

Forward KL Can Be Better: Normalizing Flow Boltzmann Sampling with Log-Ratio Variation

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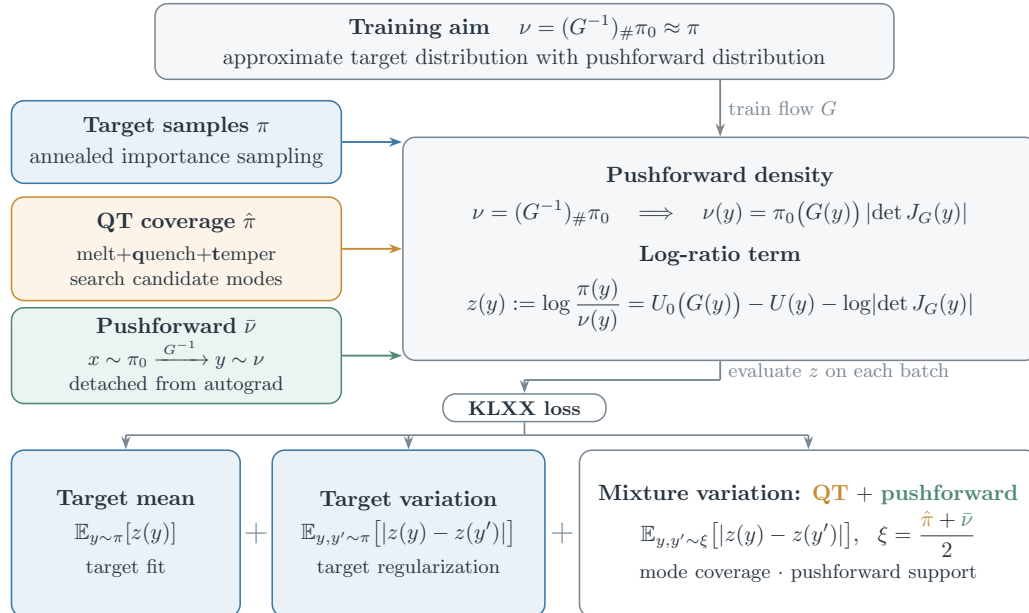
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Abstract

Normalizing flow Boltzmann generators retain a tractable pushforward density, but training with forward KL depends on target samples that may be biased or omit modes. A flow can miss target mass while its observed importance weights give a high effective sample size. We introduce the log-ratio variation X_ω , the mean absolute pairwise difference of the target-to-pushforward log-density ratio under a weighting measure ω , and use it to define KLXX, a new loss function: its two X 's are **two such variations** added to the forward KL, one weighted by the target surrogate **to improve accuracy**, the other by a mixture of quench and temper samples with pushforward samples **to search candidate modes**. We derive exponential convergence in Fisher–Rao geometry and a fixed-surrogate error bound for KLXX, then place KLXX in an adaptive-staging Boltzmann generator with exact importance reweighting and sharpening of the regularized molecular potential. We bound that generator’s sampling error and **prove it asymptotically unbiased** in the sample size. From that bound we read a propagation factor that summarizes how error accumulates across stages. In the model and molecular tests, KLXX improves mode coverage and stagewise diagnostics, and selected observables agree with independent molecular simulations. The log-ratio variations thus supply information that a straightforward KL loss can omit.

KLXX: one log-ratio (KL), two log-ratio variations (XX)



1 Introduction

Drawing samples from a target distribution $\pi \propto \exp(-U)$ on $\mathbb{R}^d$ is a recurring problem in molecular simulation, Bayesian inference, and statistical physics (Frenkel and Smit, 2002; Robert and Casella, 2004; Landau and Binder, 2014). When π is high-dimensional and multimodal, its mass splits among basins of U separated by barriers. Local sampling kernels, including Metropolis–Hastings (Metropolis et al., 1953; Hastings, 1970), Langevin dynamics (Langevin, 1908; Leimkuhler and Matthews, 2015), Nosé–Hoover dynamics (Nosé, 1984; Hoover, 1985), and Hamiltonian Monte Carlo (Duane et al., 1987), evolve from the current configuration and can therefore remain confined to one basin over accessible simulation times. When this occurs, finite-time estimates are biased toward the visited basin and give no direct account of the mass that was never reached.

Tempering and particle methods address this problem by connecting a tractable distribution to π through intermediate distributions. Parallel tempering (PT) and replica exchange run copies of the system at several temperatures and exchange neighboring configurations, so a basin reached at high temperature can enter the low-temperature ensemble (Swendsen and Wang, 1986; Geyer, 1991; Hukushima and Nemoto, 1996). Annealed importance sampling (AIS) moves along a bridge of intermediate distributions while accumulating importance weights (Neal, 2001). Sequential Monte Carlo (SMC) uses a similar bridge but resamples and rejuvenates a weighted particle set at each level (Doucet et al., 2001; Del Moral et al., 2006). For a prescribed chemical process, umbrella sampling biases selected reaction coordinates to cross the relevant barriers and reweights the biased samples to recover a free energy profile (Torrie and Valleau, 1977; Kästner, 2011). Nested sampling instead rewrites the normalizing integral as a one-dimensional quadrature over a likelihood threshold (Skilling, 2006).

Recent work improves PT through three distinct mechanisms. Nonreversible swap schedules and tuned temperature grids accelerate round trips (Syed et al., 2022); variational-reference methods adapt the reference endpoint and its connecting path (Surjanovic et al., 2022); and neural transports improve communication between adjacent distributions with limited overlap (Zhang et al., 2026). Replica exchange has also been scaled beyond molecular simulation to stochastic-gradient samplers: replica exchange Langevin diffusion accelerates nonconvex optimization (Chen et al., 2019; Dong and Tong, 2021), its stochastic-gradient variant corrects the swap test for noisy energy estimates (Deng et al., 2020), and later variants reduce estimator variance to restore swap rates (Deng et al., 2021). These accelerations strengthen communication between neighboring distributions and can transmit a discovered basin efficiently, but two limits remain: a basin that no chain has visited cannot enter the target ensemble (Woodard et al., 2009; Chehab et al., 2025), and the mass of a visited basin is still estimated through occupation frequencies rather than reweighting. A complementary strategy is to learn a global transport from an easily sampled reference to the target, thereby generating independent, nonlocal proposals.

Boltzmann generators implement this strategy with a learned transport map (Noé et al., 2019). Let $\pi_0 \propto \exp(-U_0)$ be an easily sampled reference distribution, usually Gaussian; we train an invertible normalizing flow $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ (Rezende and Mohamed, 2015; Papamakarios et al., 2017; Kingma and Dhariwal, 2018) to fit the target-to-source transport relation $G_{\#}\pi \approx \pi_0$. The inverse map then sends independent source draws toward the target, and their pushforward distribution is $\nu := (G^{-1})_{\#}\pi_0$. Its density is

$$\nu(y) = \pi_0(G(y)) |\det J_G(y)|, \quad (1)$$

where $J_G(y)$ is the Jacobian matrix of G at y . The importance weight $w(y) := \pi(y)/\nu(y)$ is the ratio of the target density to the pushforward density. Reweighting by w makes estimates from pushforward samples *unbiased*, even when the flow is not perfectly trained. Throughout the paper we also use the *log-ratio*:

$$z(y) := \log w(y) = \log \frac{\pi(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)| + \text{const}, \quad (2)$$

where the constant comes from the two normalizers and depends on neither y nor G . The weights also define the effective sample size (ESS), a diagnostic of density agreement on the support reached by ν (Liu, 2001; Doucet et al., 2001; Elvira et al., 2022). A high ESS guarantees accuracy in the flow’s reached regions, but not full mode coverage of the potential U .

The choice of loss greatly affects the training quality. The original Boltzmann generator (Noé et al., 2019) and its variational-inference predecessor (Rezende and Mohamed, 2015) minimize the reverse Kullback–

Leibler (KL) divergence

$$\text{(reverse)} \quad \text{KL}(\nu\|\pi) = \int_{\mathbb{R}^d} \nu(y) \log \frac{\nu(y)}{\pi(y)} dy = -\mathbb{E}_{y \sim \nu} [z(y)], \quad (3)$$

which needs samples only from ν and evaluations of U . Its empirical gradient is therefore accessible, but the divergence is mode-seeking (Wainwright and Jordan, 2008): a mode absent from ν contributes no samples and hence no gradient. A flow can then collapse onto a subset of the target modes (Wu et al., 2020; Felardos et al., 2023). Existing approaches interleave Markov chain Monte Carlo (MCMC) kernels with deterministic layers, encode target symmetries in the architecture, reduce gradient variance near the optimum, or anneal from a smoother potential (Wu et al., 2020; Köhler et al., 2020; Vaitl et al., 2022; Schopmans and Friederich, 2025). These changes can improve exploration, but their success still depends on whether the resulting sampler reaches each relevant mode.

The opposing loss minimizes the forward KL,

$$\text{(forward)} \quad \text{KL}(\pi\|\nu) = \int_{\mathbb{R}^d} \pi(y) \log \frac{\pi(y)}{\nu(y)} dy = \mathbb{E}_{y \sim \pi} [z(y)], \quad (4)$$

which is mass-covering for exact distributions: it diverges if ν vanishes where π has positive mass (Wainwright and Jordan, 2008). Its empirical form, however, needs samples from the target distribution π being learned. Existing work resolves this circularity in several ways. Maximum likelihood uses configurations from a reference thermodynamic state (Wirnsberger et al., 2020); adaptive flow samplers train on a running Markov chain and use the flow as a global proposal (Gabri   et al., 2022); molecular flows can fit a molecular-dynamics trajectory (Klein et al., 2023); and transferable architectures amortize coverage across related molecules (Klein and No  , 2024). Persistent chains and coupled-target regression provide other approximations to the missing target expectation (Naesseth et al., 2020; Rehman et al., 2026). Because an empirical loss reflects only the regions represented by its sample source, the mass-covering property of the exact divergence does not prevent empirical mode omission. This tension motivates a loss that retains forward KL while using samples from more than one distribution.

The sampled forward KL leaves three distinct gaps. First, a mode absent from its target surrogate contributes no training points. If the flow also misses that mode, the observed weights can be nearly uniform on the reached support, so the ESS can be close to one even though target mass is absent; we call this a *fake ESS* (No   et al., 2019; Midgley et al., 2023). Second, a finite or biased target surrogate distorts the sampled loss, and the fitted flow can inherit that error. Third, target samples give little information where the pushforward distribution places excess mass, allowing leakage between modes or into target-poor regions. Addressing one of these failures does not by itself resolve the other two.

1.1 Main contributions in this work

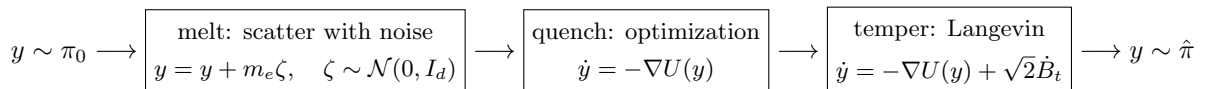
To address these gaps, we propose KLXX, a forward KL loss augmented by *log-ratio variations*:

$$X_\omega(\pi\|\nu) = \mathbb{E}_{y, y' \sim \omega} [|z(y) - z(y')|] = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(y) \omega(y') \left| \log \frac{\pi(y)}{\nu(y)} - \log \frac{\pi(y')}{\nu(y')} \right| dy dy'. \quad (5)$$

The log-ratio variation is the mean absolute pairwise difference of the log-ratio $z(y)$ in (2), with both points drawn independently from a given distribution ω . The principal idea in X_ω is that the log-ratio $z(y)$ should be *almost surely zero on the support of ω* , while the difference allows the cancellation of the unknown normalizers. As the name KLXX implies, two variation terms of the form (5) are appended:

Target variation: X_ω with $\omega = \pi$, the target distribution. It regularizes the forward KL loss at no additional cost, since it reuses the target samples the forward KL already draws, and it improves the accuracy, and hence the ESS, of the pushforward samples.

Mixture variation: X_ω with $\omega = (\hat{\pi} + \bar{\nu})/2$, the equal mixture of the quench and temper (QT) distribution $\hat{\pi}$ and the detached pushforward distribution $\bar{\nu}$. Here the QT distribution $\hat{\pi}$ is our core weapon against mode collapse, and it is built in three steps:



Each sample y from the source distribution π_0 is first scattered by Gaussian noise at a melt scale $m_e > 0$, which carries it beyond the region the source covers. Gradient descent on U then drives it into a local basin. A short Langevin run finally spreads the samples around that basin. A positive m_e is what lets the QT distribution $\hat{\pi}$ find candidate modes of U that the source distribution π_0 does not reach. The QT distribution is then paired with the detached pushforward distribution, identical to ν but detached from the differentiation graph of G , in order to restrain the intermodal leakage between the modes QT has found.

The graphical abstract summarizes the forward KL, the target variation, and the mixture variation, together with the distribution each of them uses; formally, in terms of the flow map G , the KLXX loss reads

$$\begin{aligned} \text{KLXX}[G] &= \text{KL}(\pi \parallel \nu) + \text{X}_\pi(\pi \parallel \nu) + \text{X}_{(\hat{\pi} + \bar{\nu})/2}(\pi \parallel \nu) \\ &= \mathbb{E}_{y \sim \pi} [z(y)] + \mathbb{E}_{y, y' \sim \pi} [|z(y) - z(y')|] + \mathbb{E}_{y, y' \sim (\hat{\pi} + \bar{\nu})/2} [|z(y) - z(y')|], \end{aligned} \quad (6)$$

where the log-ratio $z(y)$ depends explicitly on G , while the detached pushforward distribution $\bar{\nu}$ requires G^{-1} evaluations that do not enter the differentiation graph. Our contributions are as follows:

1. We introduce a new KLXX loss (6), whose components mix QT and pushforward distributions. Pairwise differences remove the unknown target normalizer, and the log-ratio variation retains $\nu = \pi$ as a minimizer. The target-weighted variation reuses the forward KL batch, while the mixture supplies candidate-mode and pushforward support information absent from that batch.
2. We derive the Fisher–Rao gradient flow of the generalized KLXX loss under a general choice of parameters, and prove that the forward KL decays exponentially in time. For a biased target surrogate, the two log-ratio variations improve the KL bound at the stationary distribution of the biased KLXX loss, and Theorem 1 exhibits that improvement under a mild condition.
3. We demonstrate that QT can reach target basins outside the regions covered by the source and annealed samples. In the Himmelblau and Sparse tests, the two losses trained without QT miss remote wells, while both QT-based losses recover every mode; mixing QT and pushforward samples also reduces visible intermodal leakage (Figures 4 and 5). We therefore report ESS together with coverage, since uniform weights on reached modes do not imply complete coverage.
4. We build an adaptive-staging Boltzmann generator based on KLXX without any external data or replay buffer, using only oracle access to the target potential $U(x)$ and its gradient $\nabla U(x)$; annealing and QT generate the required samples on the fly. For molecular potentials, an additional continuation sharpens a regularized interaction core toward the desired endpoint. We also bound the error of the sample set it returns: Theorem 2 controls the error at rate $N^{-1/2}$ for every bounded observable with a constant independent of N , **so the scheme is asymptotically unbiased**. Reading that constant stagewise defines the propagation factors, which quantify how the error accumulates across the accepted stages; the experiments report the computable factor $\hat{F}_\Sigma$, assembled from the per-stage ESS.

In addition, we develop general Python packages `jflows` and `jflows-md` for normalizing flow Boltzmann sampling with molecular potentials. With a JAX backend and OpenXLA compilation, the reported 22-atom alanine dipeptide run completes all ten accepted training stages in 72.7 minutes on one NVIDIA GeForce RTX 5090, without an external target dataset or a prescribed conformational landscape.

1.2 Comparison to related works

What is new in the KLXX loss (6) is mainly the mixture of the QT distribution $\hat{\pi}$ with the detached pushforward distribution $\bar{\nu}$, which is what remedies mode collapse. Its remaining ingredients draw on several ideas already present in normalizing flow training, and we compare them here.

The closest related loss is that of flow annealed importance sampling bootstrap (FAB) (Midgley et al., 2023), which couples AIS and flow training but replaces the forward KL by the α -divergence at $\alpha = 2$. Its integrand is proportional to π^2/ν , and its exact minimizer also minimizes importance-weight variance. FAB targets π^2/ν with AIS and reuses those samples through a prioritized replay buffer, which must be maintained across training, and its MCMC transitions differentiate the flow density. KLXX instead retains the forward KL and adds nonnegative log-ratio variations, so its additional cost is the QT construction and

the variation evaluations under the mixture measure; it uses no replay buffer. The 2-divergence gives the stronger tail penalty, whereas KLXX obtains its coverage information from QT.

Centered log-ratio penalties also arise in log-variance losses for learned diffusion samplers (Richter and Berner, 2024) and in LDR losses for Boltzmann generators (Schopmans et al., 2026), which places the present construction within a broader family of log-ratio regularizers. The target-weighted comparison is quantitative: LDR-L1 uses the centered penalty $\mathbb{E}_\pi|z - \mathbb{E}_\pi z|$, and Jensen’s inequality together with the triangle inequality gives $\mathbb{E}_\pi|z - \mathbb{E}_\pi z| \leq X_\pi(\pi||\nu) \leq 2\mathbb{E}_\pi|z - \mathbb{E}_\pi z|$, so at the same coefficient $\text{KL}+X_\pi$ and LDR-L1 are comparable up to a factor of two. This suggests similar training performance, although the losses are not equivalent. The two design differences are that X_π pairs the forward KL integrand at two independent target points, which yields the double integral in (5), and that KLXX evaluates a second variation under a mixture of QT and pushforward samples rather than under fixed energy-labeled data.

A second family trains one flow per annealing step inside an SMC sweep. Annealed flow transport (AFT) (Arbel et al., 2021) trains each flow by the KL divergence between the transported distribution and the next annealed target. That KL is estimated on the current weighted particles, which are then transported through the flow and resampled. Its continual repeated variant CRAFT (Matthews et al., 2022) repeats the training across successive sweeps instead of fitting each flow once. These are the closest prior works to the staged construction of Section 4, whose stage loss carries the same forward KL term; what KLXX adds at each stage are the two log-ratio variations, and in particular the mixture of QT and pushforward samples that a per-stage KL does not supply. Constrained mass transport instead constrains the KL divergence and the entropy decay between successive intermediate distributions, which keeps their overlap high and avoids the mass teleportation that a fixed geometric schedule can produce (von Klitzing et al., 2026); the adaptive stage selection of Section 4 addresses the same difficulty through the SMC screen and the ESS gate. Sequential Boltzmann generators keep a single flow, fit it to biased molecular dynamics samples by forward KL, and anneal those samples toward the target at inference time by SMC (Tan et al., 2025); that transport acts after training, whereas the annealing of Section 3.3 builds the target surrogate during it, and KLXX uses no molecular dynamics data.

A separate family replaces the loss altogether. Energy-only methods such as iDEM alternate a sampling loop with a score- or velocity-fitting loop and a replay buffer (Akhound-Sadegh et al., 2024; Woo and Ahn, 2024), adjoint sampling learns a diffusion sampler from the energy alone by stochastic optimal control (Havens et al., 2025), and controlled stochastic differential equations, denoising diffusions, and flow-matching models learn a drift or velocity rather than a bijection (Zhang and Chen, 2022; Vargas et al., 2023; Berner et al., 2024; Nüsken and Richter, 2021; Richter and Berner, 2024). Annealing enters this family as well: NETS augments AIS with a learned drift (Albergo and Vanden-Eijnden, 2025), and a temperature sequence of diffusion models can be annealed at inference time (Akhound-Sadegh et al., 2025). These stochastic transports can be more expressive than the invertible architectures used here, but their terminal density often requires per-sample ODE integration (He et al., 2025). We retain a bijection because the exact pushforward density, importance ratio, ESS, and reweighting are central to every term of (6) and to the final estimator. Table 1 summarizes these distinctions.

method	loss	training mechanism
reverse KL	$\text{KL}(\nu \pi)$	pushforward samples; no target sampler
forward KL	$\text{KL}(\pi \nu)$	annealed target surrogate in this work
FAB	2-divergence (π^2/ν)	AIS on π^2/ν ; MCMC differentiates ν ; replay buffer
LDR	$\text{KL}(\pi \nu) + \mathbb{E}_\pi[z - \mathbb{E}_\pi z ^p]$	fixed energy-labeled data; no extra pushforward samples
AFT, CRAFT	KL per annealing step	one flow per step, trained on the weighted particles
iDEM	energy matching	iterative sampler; stochastic transport; replay buffer
KLXX	$\text{KL}(\pi \nu) + X_\pi + X_{(\hat{\pi} + \bar{\nu})/2}$	target annealing, QT, and detached pushforward samples no MCMC derivative of ν or replay buffer

Table 1: Losses and training mechanisms for the methods discussed above. Here $z = \log(\pi/\nu)$; the LDR row gives its target-weighted form with $p = 1$ for LDR-L1 and $p = 2$ for LDR-L2. The forward KL row uses the annealed baseline in this paper but may instead use external target data. The table compares how samples enter each loss; it does not assume equal computational budgets.

The rest of the paper is organized as follows: [Section 2](#) analyzes the generalized KLXX loss in Fisher–Rao geometry, deriving its gradient flow and the accuracy bound it reaches under a biased target surrogate; [Appendices A and B](#) give the proofs. [Section 3](#) gives the training algorithm. [Section 4](#) proposes the adaptive-staging Boltzmann generator and bounds the error of the sample set it returns; [Appendix C](#) gives the proof. [Sections 5 and 6](#) report the model and molecular results, respectively, and [Section 7](#) concludes.

2 Analysis of the generalized KLXX model

In this section we consider a generalized form of the KLXX loss, determined by the parameters $(\lambda, \vartheta, \alpha, \beta)$. For convenience, we write the loss $\mathcal{L}$ as a functional of the pushforward distribution ν rather than of the flow map G :

$$\begin{aligned}\mathcal{L}[\nu] &= \text{KL}(\pi\|\nu) + \lambda X_\pi(\pi\|\nu) + \vartheta X_{\alpha\hat{\pi}+\beta\bar{\nu}}(\pi\|\nu) \\ &= \mathbb{E}_{y\sim\pi}[z(y)] + \lambda \mathbb{E}_{y,y'\sim\pi}[|z(y) - z(y')|] + \vartheta \mathbb{E}_{y,y'\sim\alpha\hat{\pi}+\beta\bar{\nu}}[|z(y) - z(y')|],\end{aligned}\tag{7}$$

where the parameters satisfy $\lambda, \vartheta, \alpha, \beta \geq 0$ and $\alpha + \beta = 1$, so that the mixture $\xi = \alpha\hat{\pi} + \beta\bar{\nu}$ is a probability distribution on $\mathbb{R}^d$, and the log-ratio $z = \log(\pi/\nu)$ in [\(2\)](#) carries the entire dependence on ν . Throughout the paper, we let one Greek symbol (π, ν, μ , and so on) carry both a probability distribution and its density with respect to Lebesgue measure on $\mathbb{R}^d$.

The three subsections below take the generalized loss [\(7\)](#) in turn: a formal account of how the mixture variation contributes to mode discovery, a derivation of exponential convergence for the Fisher–Rao gradient flow, and an accuracy bound under a biased target surrogate.

2.1 Principle of mixture variation

The mixture distribution $\xi = \alpha\hat{\pi} + \beta\bar{\nu}$ mainly relies on the QT distribution $\hat{\pi}$ to find the candidate modes of the potential U , as elaborated in [Section 1](#). The QT distribution $\hat{\pi}$ is generated from the source distribution π_0 in three steps: for each $y \sim \pi_0$,

1. **Melt**: scatter with Gaussian noise, $y = y + m_e\zeta$, with $\zeta \sim \mathcal{N}(0, I_d)$, at a melt scale $m_e > 0$ that carries y beyond the region the source covers;
2. **Quench**: descend the potential, $\dot{y} = -\nabla U(y)$, until y settles at a local minimizer of U , that is, at the centre of whichever basin the melted particle landed in;
3. **Temper**: run a short Langevin trajectory, $\dot{y} = -\nabla U(y) + \sqrt{2}\dot{B}_t$, which spreads y around that centre at the target temperature.

The three names are taken directly from industrial steelmaking, where a workpiece is melted, quenched, and then tempered in that order.

Only melting carries the coverage: a large $m_e > 0$ allows the particles to explore the region and cover all the missing modes, and quenching then uses a local gradient descent to drive these particles into the potential basins. Tempering finally restores a spread around each centre, so that $\hat{\pi}$ is a distribution over configurations rather than point masses at the minimizers.

We use an intuitive example to illustrate the construction. Define the 1D Rastrigin potential

$$U(x) = \frac{1}{2}x^2 + 4\cos(2\pi x), \quad x \in \mathbb{R},$$

take the source distribution $\pi_0 = \mathcal{N}(0, 1)$ and the melt scale $m_e = 2.0$, and temper for a Langevin time of 0.2 to obtain the QT distribution $\hat{\pi}$. [Figure 1](#) shows $\hat{\pi}$ against the target $\pi \propto \exp(-U)$.

Every mode of π is covered by $\hat{\pi}$, including the outer wells the source never reaches, but the probability assigned to each differs from the target one: $\hat{\pi}$ weights a well by the melted mass falling into it, so its outer wells carry far more than π gives them. This is exactly complementary to the forward KL, which is accurate where its target samples land and blind elsewhere, whereas $\hat{\pi}$ is inaccurate everywhere and blind nowhere. The difference form of the log-ratio variation [\(5\)](#) is what lets $\hat{\pi}$ be used without corrupting the fit to π : a

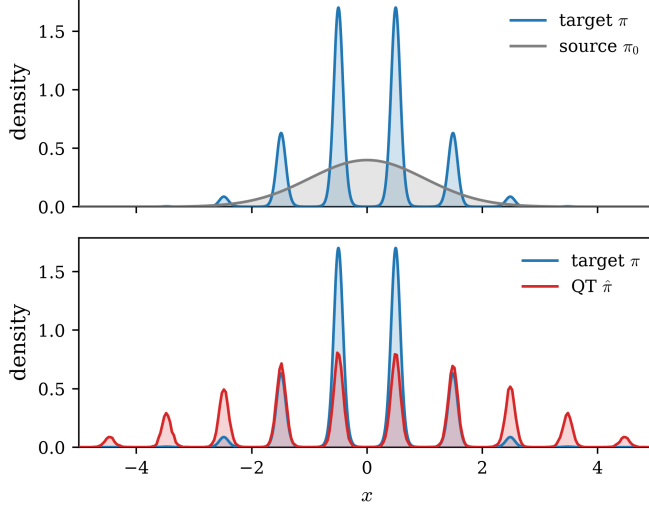


Figure 1: Quench and temper on the 1D Rastrigin target $\pi \propto \exp(-U)$ with $U(x) = \frac{1}{2}x^2 + 4\cos(2\pi x)$. Top: the Gaussian source $\pi_0 = \mathcal{N}(0, 1)$. Bottom: the QT distribution $\hat{\pi}$, built from 10^5 source samples at melt scale $m_e = 2.0$ and tempered over a Langevin time of 0.2.

variation asks only that the log-ratio be flat under its weighting measure, so $\hat{\pi}$ enters as the points at which flatness is demanded, never as an estimate of the mass they carry.

However, using the QT distribution $\hat{\pi}$ alone in the log-ratio variation may cause *intermodal leakage*: the trained flow places spurious mass between the target modes. The Sparse target shows this directly, where $\text{KL} + \text{X}_\pi + \text{X}_{\hat{\pi}}$ recovers all four modes but leaves visible mass in the intermodal regions (Figure 5). The reason is that $\text{X}_{\hat{\pi}}$ weights only the points $\hat{\pi}$ produces, and those sit in the basins of U ; it therefore says nothing about the regions the pushforward ν occupies and $\hat{\pi}$ does not, which are exactly the intermodal ones. Combining it with the detached pushforward distribution $\bar{\nu}$ is exactly the treatment. In the log-ratio variation with the mixture distribution $\xi = \alpha\hat{\pi} + \beta\bar{\nu}$, the log-ratio $z = \log \frac{\pi}{\nu}$ is penalized to be a constant under both $\hat{\pi}$ and $\bar{\nu}$, hence during the training we have approximately

$$z(y) = \log \frac{\pi(y)}{\nu(y)} \approx z(y') = \log \frac{\pi(y')}{\nu(y')}, \quad y \sim \hat{\pi}, \quad y' \sim \nu. \quad (8)$$

The relation (8) reflects a fundamental collision in the training. On the one hand, y lies in a target mode, which $\hat{\pi}$ supplies for every mode, and secures a large density $\pi(y)$, so $z(y)$ grows to $+\infty$ if the pushforward ν does not reach that mode. On the other hand, y' lies wherever the flow sends mass and secures a positive density $\nu(y')$, so $z(y')$ decreases to $-\infty$ if the pushforward reaches the intermodal regions, where π is negligible. Demanding that the two agree sets the divergences against each other, and the flow can satisfy both only by reaching every mode and staying out of the intermodal regions. The log-ratio variation with the mixture $\xi = \alpha\hat{\pi} + \beta\bar{\nu}$ resolves the collision in a clean way: $\hat{\pi}$ supplies the points at which z must not run high, $\bar{\nu}$ supplies the points at which it must not run low, and the variation balances the log-ratio across both, thus reducing the intermodal leakage.

Remark 1 We do not train with the log-ratio variation alone. A variation constrains only the flatness of z pointwise, while the forward KL (4) drives the flow globally, since the convexity in Jensen’s inequality makes it nonnegative and zero only at $\nu = \pi$. Appendix D.2 of the LDR study (Schopmans et al., 2026) reports the analogous failure: training with LDR-L1 or LDR-L2 alone, without a KL term, produces qualitatively incorrect distributions.

2.2 Convergence as Fisher–Rao gradient flow

For an arbitrary parameter choice $(\lambda, \vartheta, \alpha, \beta)$, the generalized KLXX loss (7) attains its minimum 0 at the exact target π , so it is natural to ask whether the gradient flow it induces converges to π exponentially. In the Wasserstein-2 geometry, the forward KL is not displacement convex, so the Otto–Villani and Bakry–Émery arguments used for the reverse KL do not apply (Otto and Villani, 2000; Bakry et al., 2014). We therefore work in the Fisher–Rao (FR) geometry, where the gradient flow has an explicit form. Assume all densities are positive and regular enough for differentiation.

Let ω be a weighting distribution that is fixed under the variation, and define the *nonlocal correlation* of the importance weight

$$S^\omega(x) := \mathbb{E}_{y \sim \omega} [\text{sgn}(w(x) - w(y))] \in [-1, 1], \quad (9)$$

which reports where $w(x)$ sits among the weights ω carries, and $\text{sgn}(\cdot) \in \{0, \pm 1\}$ represents the sign of a real number. The definition of S^ω naturally applies to the target distribution π and to the mixture distribution $\xi = \alpha\hat{\pi} + \beta\bar{\nu}$, written as S^π and S^ξ .

The FR gradient flow of the generalized KLXX loss (7) is a differential equation for a time-dependent pushforward density ν_t , reading

$$\partial_t \nu_t(x) = -\text{grad}^{\text{FR}} \mathcal{L}(\nu_t) = \pi(x) - \nu_t(x) + 2\lambda \pi(x) S_t^\pi(x) + 2\vartheta \xi(x) S_t^\xi(x), \quad (10)$$

where the nonlocal correlations are evaluated with the importance weight $w_t = \pi/\nu_t$,

$$S_t^\pi(x) = \mathbb{E}_{y \sim \pi} [\text{sgn}(w_t(x) - w_t(y))], \quad S_t^\xi(x) = \mathbb{E}_{y \sim \xi} [\text{sgn}(w_t(x) - w_t(y))]. \quad (11)$$

The mixture $\xi = \alpha\hat{\pi} + \beta\bar{\nu}$ depends on ν_t through $\bar{\nu}$, but $\bar{\nu}$ is detached and takes no part in the gradient calculation, so ξ is held fixed when the variation is taken. Differentiating $\text{KL}(\pi\|\nu_t)$ along (10) gives the dissipation relation of the forward KL

$$\frac{d}{dt} \text{KL}(\pi\|\nu_t) = -\chi^2(\pi\|\nu_t) - \lambda \mathbb{E}_{x,y \sim \pi} [|w_t(x) - w_t(y)|] - \vartheta \mathbb{E}_{x,y \sim \xi} [|w_t(x) - w_t(y)|], \quad (12)$$

where $\chi^2(\pi\|\nu) := \mathbb{E}_\nu [(w-1)^2]$ is the chi-squared divergence. Since both variations enter (12) as nonpositive terms and $\chi^2(\pi\|\nu_t) \geq \text{KL}(\pi\|\nu_t)$, Grönwall’s inequality yields

$$\text{KL}(\pi\|\nu_t) \leq e^{-t} \text{KL}(\pi\|\nu_0), \quad t \geq 0, \quad (13)$$

without assuming log-concavity, a spectral gap, or a growth condition on π . This is a statement in density space: the rate is the one forward KL already carries, and what the variations change is the structure of the flow rather than its speed. Appendix A gives the derivation of the gradient flow (10) and the proof details.

Remark 2 *The uniform rate (13) does not say how fast training converges. It describes a flow in density space. Training instead descends in the parameters of the flow map G , and that descent moves the density along a different path. The flow time t is also not a step count. How far one step advances depends on the magnitude of the parameter gradient, and (13) does not control it.*

2.3 Accuracy under biased target surrogate

In the training of the generalized KLXX loss (7), the target distribution π needs to be approximated by AIS (see Section 3.3 for details), so the effective target surrogate used in training is a biased distribution $\tilde{\pi}$, and the corresponding biased loss is given by

$$\tilde{\mathcal{L}}[\nu] = \mathbb{E}_{y \sim \tilde{\pi}} [z(y)] + \lambda \mathbb{E}_{y,y' \sim \tilde{\pi}} [|z(y) - z(y')|] + \vartheta \mathbb{E}_{y,y' \sim \alpha\hat{\pi} + \beta\bar{\nu}} [|z(y) - z(y')|], \quad (14)$$

where the log-ratio $z = \log(\pi/\nu)$ is still evaluated against the exact target, and only the outer measures of the first two terms are replaced by the biased $\tilde{\pi}$. The mixture term carries no surrogate bias: $\xi = \alpha\hat{\pi} + \beta\bar{\nu}$ approximates nothing, $\hat{\pi}$ being supplied by QT and $\bar{\nu}$ being the detached copy of ν .

Following the derivation of (10) for the unbiased loss, the FR gradient of the biased KLXX loss (14) with respect to ν has a similar form,

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\pi}(x) - 2\lambda \tilde{\pi}(x) S^{\tilde{\pi}}(x) - 2\vartheta \xi(x) S^\xi(x), \quad (15)$$

where $S^{\tilde{\pi}}$ and S^{ξ} are the nonlocal correlations defined in (9). Let ν_* be a stationary density of $\tilde{\mathcal{L}}$, at which the gradient (15) vanishes. Collecting the two $\tilde{\pi}$ terms gives

$$\nu_*(x) = \tilde{\pi}(x)(1 + 2\lambda S^{\tilde{\pi}}_*(x)) + 2\vartheta \xi(x) S^{\xi}_*(x), \quad (16)$$

where the correlation terms are evaluated with the importance weight $w_* = \pi/\nu_*$,

$$S^{\tilde{\pi}}_*(x) = \mathbb{E}_{y \sim \tilde{\pi}}[\text{sgn}(w_*(x) - w_*(y))], \quad S^{\xi}_*(x) = \mathbb{E}_{y \sim \xi}[\text{sgn}(w_*(x) - w_*(y))]. \quad (17)$$

Since $\bar{\nu}$ tracks ν as training updates it, (16) characterizes the densities that annihilate the gradient rather than the minimizers of a functional. At $\lambda = \vartheta = 0$ it reduces to $\nu_* = \tilde{\pi}$, whose error is $\text{KL}(\pi \|\tilde{\pi})$; the following theorem bounds how far the two log-ratio variations move that error.

Theorem 1 *Fix the coefficients $(\lambda, \vartheta, \alpha, \beta)$ and let $\tilde{\pi}$ be a positive probability density. Let ν_* be a positive density solving (16), and $w_* = \pi/\nu_*$ be the importance weight, with $\mathbb{E}_{\tilde{\pi}}[w_*] < \infty$ and $\mathbb{E}_{\xi}[w_*] < \infty$. Then*

$$\text{KL}(\pi \|\nu_*) \leq \text{KL}(\pi \|\tilde{\pi}) - \lambda \mathbb{E}_{x, y \sim \tilde{\pi}}[|w_*(x) - w_*(y)|] - \vartheta \mathbb{E}_{x, y \sim \xi}[|w_*(x) - w_*(y)|]. \quad (18)$$

Both subtracted discrepancies are nonnegative, so the two log-ratio variations improve the error that a fixed biased surrogate leaves behind, under two different weightings of the same ratio w_* .

3 Flow training with generalized KLXX loss

In this section we give the training algorithm for the generalized KLXX loss (7). Recall that the source and target distributions are $\pi_0 \propto \exp(-U_0)$ and $\pi \propto \exp(-U)$, and that the flow map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is trained to fit the transport relation $\nu = (G^{-1})_{\#} \pi_0 \approx \pi$. Hence the generalized KLXX loss in terms of G reads

$$\mathcal{L}[G] = \mathbb{E}_{y \sim \pi}[z(y)] + \lambda \mathbb{E}_{y, y' \sim \pi}[|z(y) - z(y')|] + \vartheta \mathbb{E}_{y, y' \sim \alpha \hat{\pi} + \beta \bar{\nu}}[|z(y) - z(y')|], \quad (19)$$

where the log-ratio $z(y)$ defined in (2) takes the explicit form

$$z(y) = \log \frac{\pi(y)}{\nu(y)} = U_0(G(y)) - U(y) - \log |\det J_G(y)|,$$

the normalizing constants being omitted, since they depend on neither y nor G .

The training algorithm uses two parameters throughout: the sample size N , the size of the sample set that enters and leaves the training and carries the diagnostics; and the batch size B , the number of samples per gradient step.

The generalized KLXX loss (19) draws on three sample sets: one from the target distribution π , one from the QT distribution $\hat{\pi}$, and one from the detached pushforward distribution $\bar{\nu}$. The QT samples number N and are constructed before training. The pushforward samples number B and are drawn at each gradient step, where the target surrogate also approximates π with a batch of B samples.

This section is written in a top-down hierarchy. Section 3.1 gives the KLXX training algorithm, which alternates between preparing samples and taking gradient steps; Sections 3.2 and 3.3 then construct the QT and AIS samples it consumes; and Section 3.4 gives two complementary diagnostics for the training quality.

3.1 KLXX training algorithm

We begin from a sample set $\mathcal{Y}_0$ of size N approximating the source distribution π_0 , initialize the flow G as an identity map, and run QT (Algorithm 2) once to generate a QT sample set $\hat{\mathcal{Y}}$ of size N . Each gradient step then builds the target surrogate and mixture samples of size B as follows:

- (i) Draw a random batch $\mathcal{X}_0 \subset \mathcal{Y}_0$ of size B and apply the pushforward to obtain $\mathcal{X}_{\nu} = G^{-1}(\mathcal{X}_0)$;
- (ii) Apply AIS (Algorithm 3) to $\mathcal{X}_{\nu}$ to obtain the target surrogate samples $\mathcal{X}_{\pi}$;
- (iii) Draw B subsamples $\mathcal{X}_{\hat{\pi}} \subset \hat{\mathcal{Y}}$ from the QT samples, and mix $\mathcal{X}_{\hat{\pi}}$ and $\mathcal{X}_{\nu}$ with probabilities α and β to obtain the mixture set $\mathcal{X}_{\xi}$;

(iv) Evaluate the empirical loss on the target surrogate samples $\mathcal{X}_\pi$ and the mixture samples $\mathcal{X}_\xi$,

$$\widehat{\mathcal{L}}[G] = \underbrace{\widehat{\mathbb{E}}_{y \in \mathcal{X}_\pi} [z(y)]}_{\text{forward KL}} + \lambda \underbrace{\widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\pi} [|z(y) - z(y')|]}_{\text{target variation}} + \vartheta \underbrace{\widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\xi} [|z(y) - z(y')|]}_{\text{mixture variation}}, \quad (20)$$

and take one gradient step on G , in which only $z(y)$ is differentiated.

By convention $\mathcal{X}$ always has size B and $\mathcal{Y}$ has size N . The sets $\mathcal{Y}_0$ and $\hat{\mathcal{Y}}$ are determined before training, and $\mathcal{X}_\pi$ and $\mathcal{X}_\xi$ are updated at each gradient step since their construction depends on G . For a batch sample set $\mathcal{X} = \{y_1, \dots, y_B\}$, the empirical averages of $z(y)$ and $|z(y) - z(y')|$ appearing in (20) are defined by

$$\widehat{\mathbb{E}}_{y \in \mathcal{X}} [z(y)] := \frac{1}{B} \sum_{i=1}^B z(y_i), \quad \widehat{\mathbb{E}}_{y, y' \in \mathcal{X}} [|z(y) - z(y')|] := \frac{1}{B} \sum_{i=1}^B |z(y_i) - z(y_{\sigma(i)})|, \quad (21)$$

where σ is a random permutation of $\{1, \dots, B\}$, so both variation terms carry its randomness. The target variation therefore costs no more than the forward KL, and the additional cost of KLXX comes from constructing the QT samples and evaluating the mixture variation. Algorithm 1 collects steps (i)–(iv).

Algorithm 1: KLXX training algorithm

Input: source and QT sample sets $\mathcal{Y}_0, \hat{\mathcal{Y}}$; source and target potentials U_0, U ; sizes N, B ; loss coefficients $(\lambda, \vartheta, \alpha, \beta)$

Output: trained flow map G

1 Initialize G to the identity map;

2 **for** each gradient step **do**

// (i) pushforward

3 Draw a random batch $\mathcal{X}_0 \subset \mathcal{Y}_0$ of size B and set $\mathcal{X}_\nu \leftarrow G^{-1}(\mathcal{X}_0)$;

// (ii) target surrogate

4 Apply AIS (Algorithm 3) to $\mathcal{X}_\nu$ to obtain $\mathcal{X}_\pi$;

// (iii) mixture

5 Draw B subsamples $\mathcal{X}_{\hat{\pi}} \subset \hat{\mathcal{Y}}$, and mix $\mathcal{X}_{\hat{\pi}}$ and $\mathcal{X}_\nu$ with probabilities α and β to form $\mathcal{X}_\xi$;

// (iv) gradient step

6 Take one gradient step on G for the empirical loss

$$\widehat{\mathcal{L}}[G] = \widehat{\mathbb{E}}_{y \in \mathcal{X}_\pi} [z(y)] + \lambda \widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\pi} [|z(y) - z(y')|] + \vartheta \widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\xi} [|z(y) - z(y')|],$$

where the log-ratio $z(y) = U_0(G(y)) - U(y) - \log |\det J_G(y)|$.

7 **end**

Remark 3 When the sample set $\mathcal{X}_\pi$ of size B is drawn from the target π , the variation term $\widehat{\mathbb{E}}_{y, y' \in \mathcal{X}_\pi} [|z(y) - z(y')|]$ has expectation $\frac{B-1}{B} \mathbb{E}_{y, y' \sim \pi} [|z(y) - z(y')|]$: the permutation in (21) fixes i with probability $\frac{1}{B}$, and the term $|z(y_i) - z(y_{\sigma(i)})|$ then vanishes. The implementation leaves this factor uncorrected for simplicity.

3.2 Constructing $\hat{\pi}$ by quench and temper

The QT distribution $\hat{\pi}$ is the wide-coverage measure for the target distribution π , and its melt, quench and temper construction was described in Section 2.1. In the discrete setting we implement the quench step with L-BFGS (Liu and Nocedal, 1989) and the temper step with the Metropolis-adjusted Langevin algorithm (MALA). Algorithm 2 writes the construction as the map

$$\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}_0, U)$$

from the source samples $\mathcal{Y}_0 = \{y_1, \dots, y_N\}$ and the target potential U to the QT samples $\hat{\mathcal{Y}} = \{\hat{y}_1, \dots, \hat{y}_N\}$.

When the potential function U is possibly singular, we prefer L-BFGS and MALA over plain gradient descent and unadjusted Langevin dynamics, which are less stable near the singularity. The QT samples can also be built from a subset of $\mathcal{Y}_0$, at lower cost.

Remark 4 *L-BFGS is a quasi-Newton method (Liu and Nocedal, 1989) and is not a discretization of the quench flow $\dot{y} = -\nabla U(y)$; its iterates do not follow that trajectory. The quench asks only for the local minimizer at the end of it, and in as few iterations as possible, so the better performance of L-BFGS over gradient descent is worth the departure from the flow.*

Algorithm 2: Quench and temper (QT): wide-coverage construction

Input: source sample set $\mathcal{Y}_0 = \{y_1, \dots, y_N\}$, target potential U , melt scale m_e

Output: wide-coverage set $\hat{\mathcal{Y}} = \{\hat{y}_1, \dots, \hat{y}_N\}$ forming $\hat{\pi}$, written $\hat{\mathcal{Y}} = \text{QT}(\mathcal{Y}_0, U)$

// melt

1 Scatter the samples $y_i \leftarrow y_i + m_e \zeta_i$ with $\zeta_i \sim \mathcal{N}(0, I_d)$ i.i.d. and melt scale $m_e > 0$;

// quench

2 Drive each sample toward a mode center of π by L-BFGS on U , updating $y_1, \dots, y_N$;

// temper

3 Spread the samples around each mode, $\hat{y}_{1:N} \leftarrow \text{MALA}(y_{1:N}, U)$;

3.3 Approximating π by annealed importance sampling

Step (ii) of the KLXX training algorithm (Algorithm 1) builds the target surrogate samples $\mathcal{X}_\pi$ on the fly, at every gradient step, since samples from π are not directly available. We form them by reweighting the pushforward samples $y_1, \dots, y_B$ drawn from $\nu = (G^{-1})_\# \pi_0$. The importance weight is

$$w(y) = \exp \left(U_0(G(y)) - U(y) - \log |\det J_G(y)| \right) = \frac{\pi(y)}{\nu(y)}, \quad y \in \mathbb{R}^d. \quad (22)$$

In high dimensions, however, π and ν can differ greatly, so the weights can collapse onto a few outliers. We temper the reweighting through $M + 1$ bridging distributions

$$\mu_m(y) \propto \nu(y)^{1-\frac{m}{M}} \pi(y)^{\frac{m}{M}} \propto ((G^{-1})_\# \pi_0)(y) w(y)^{\frac{m}{M}}, \quad m = 0, 1, \dots, M, \quad (23)$$

so that $\mu_0 = \nu$ and $\mu_M = \pi$. Each level reweights by only $w(y)^{1/M}$, reducing the degeneracy of direct importance sampling. After each reweighting, we resample and diffuse the duplicates before the next level with MALA (Roberts and Tweedie, 1996), which is unbiased for π .

A standard SMC sampler targeting the path (23) would rejuvenate at level m with a Markov kernel invariant for μ_m . For $m < M$, a MALA kernel for μ_m would require $\nabla_y \log \nu(y)$ and hence positional derivatives of the flow density, including $\nabla_y \log |\det J_G(y)|$; at $m = M$, this term disappears because $\mu_M = \pi$. AIS (Algorithm 3) instead applies MALA directly under π , using only $-\nabla U$ in its proposal. MALA preserves π but not the intermediate μ_m , so the output is a biased target surrogate: the construction trades intermediate invariance for a kernel that needs only $-\nabla U$.

The AIS construction in Algorithm 3 differs from FAB (Midgley et al., 2023). FAB places the flow inside the chains, and its MCMC transitions target the surrogate π^2/ν , whose log-density gradient is

$$\nabla_y \log \frac{\pi^2(y)}{\nu(y)} = -2\nabla U(y) - \nabla_y \log \nu(y),$$

and therefore contains the positional gradient of the pushforward density at every MCMC step. FAB trains from a prioritized replay buffer of weighted AIS samples with a staleness correction. Reusing this buffer

Algorithm 3: Annealed importance sampling (AIS) for a biased target surrogate

Input: pushforward samples $y_1, \dots, y_B$ from $\nu = (G^{-1})_{\#}\pi_0$, target distribution π , ladder length M
Output: biased target-surrogate batch $y_1, \dots, y_B$ approximating π

```
1 for  $m = 1, \dots, M$  do
2   Compute the incremental weights  $w(y_i)^{1/M} = \exp(z(y_i)/M)$  on  $y_1, \dots, y_B$ ;
   // resample
3   Resample  $y'_1, \dots, y'_B$  from  $y_{1:B}$  with weights  $w(y_i)^{1/M}$ ;
   // rejuvenate under the target
4   Diffuse the resampled duplicates by a short MALA chain on  $\pi$ ,  $y_{1:B} \leftarrow \text{MALA}(y'_{1:B}, U)$ ;
5 end
```

reduces repeated AIS generation but adds storage, prioritized sampling, and correction work. Here every MALA chain instead targets π and uses only $-\nabla U$; the flow enters through the initial pushforward samples and evaluations of z . Each gradient step starts from a new source batch, so no replay buffer is kept.

The exploration costs also differ. Relative to forward KL, KLXX adds the QT construction and the pairwise log-ratio evaluations, of which X_π reuses the target batch and the mixture term evaluates one more. Coverage then comes from the fixed N -sample QT set of Section 3.2, whereas FAB relies on its AIS budget for exploration and on the stronger tail incentive of its 2-divergence.

Remark 5 *In principle, AIS or another SMC sampler can propagate from the source π_0 to the target π on its own, with no flow at all. Two drawbacks argue against that choice. First, such a sampler explores only where its own trajectories go, so nothing guarantees coverage of the target modes; the QT set of Section 3.2 is what supplies candidate modes in KLXX. Second, with no flow to deform the proposal toward the next distribution, each level reweights and resamples at a much lower ESS, and repeated resampling of a degenerate weight set raises the Monte Carlo error of the returned samples.*

The Boltzmann generator of Section 4 compares each trained flow with the identity proposal and keeps whichever has the larger sample ESS, so an accepted stage is never worse than the identity in that diagnostic, up to the finite sample set. A trained Boltzmann generator therefore has no reason to perform systematically worse than the identity, which propagates by reweighting, resampling, and MALA alone. Section 6.3 reports the empirical side: the KLXX Boltzmann generator has the smallest propagation factor in all ten recorded alkane settings, while forward KL approaches the identity result as the molecule grows.

3.4 Evaluation: ESS, coverage, and importance sampling

We assess the quality of the pushforward samples through two complementary diagnostics. The ESS measures density agreement through the observed importance weights on support reached by the flow. Coverage instead checks whether the pushforward samples reach the regions represented by a fixed QT reference set. A high ESS can coexist with a missing mode, while high coverage does not establish the correct relative density within the reached regions. We therefore report the two diagnostics together.

At evaluation time the trained flow G supplies the pushforward distribution $\nu = (G^{-1})_{\#}\pi_0$ and its tractable density $\nu(y) = \pi_0(G(y))|\det J_G(y)|$. The importance weight $w(y)$ in (22) can therefore be evaluated on pushforward samples without fitting another density model. Let $x_1, \dots, x_N$ be samples from the source π_0 and let $y_i = G^{-1}(x_i)$, so that $\mathcal{Y} = \{y_1, \dots, y_N\}$ are pushforward samples with weights $w_i = w(y_i)$.

The first diagnostic is the ESS, normalized to $(0, 1]$ by dividing the classical ESS by the sample count N . For the importance weights $(w_1, \dots, w_N)$, define

$$\text{ESS}(w) = \frac{(\sum_{i=1}^N w_i)^2}{N \sum_{i=1}^N w_i^2} \in (0, 1]. \quad (24)$$

The upper bound follows from Cauchy–Schwarz and is attained when all weights are equal. The factor N distinguishes this normalized ESS from the classical count $(\sum_i w_i)^2 / \sum_i w_i^2 \in [1, N]$ (Elvira et al., 2022). A

high ESS indicates uniform observed weights only on support reached by ν ; a target mode absent from the sample cannot lower it.

To detect such missing modes, we pair the ESS with the coverage diagnostic of Naeem et al. (2020). Let $\hat{\mathcal{Y}} = \{\hat{y}_1, \dots, \hat{y}_N\}$ be the QT sample set produced by Algorithm 2. For each $1 \leq i \leq N$, the κ -nearest-neighbor radius of $\hat{y}_i$ (with $\kappa = 5$) is

$$\text{NND}_\kappa(\hat{y}_i; \hat{\mathcal{Y}}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq N : 0 < |\hat{y}_i - \hat{y}_j| < r\} \geq \kappa \right\}, \quad (25)$$

i.e., the distance from $\hat{y}_i$ to its κ -th nearest neighbor in $\hat{\mathcal{Y}}$. The coverage of $\mathcal{Y}$ in $\hat{\mathcal{Y}}$ is then

$$\text{Coverage}_\kappa(\mathcal{Y}; \hat{\mathcal{Y}}) = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - \hat{y}_i| \leq \text{NND}_\kappa(\hat{y}_i; \hat{\mathcal{Y}}) \right\}, \quad (26)$$

i.e., the fraction of points in $\hat{\mathcal{Y}}$ whose κ -nearest-neighbor ball contains a point of $\mathcal{Y}$. A coverage close to 1 indicates that the finite QT reference set is well covered by the flow, while a coverage well below 1 exposes missing regions that the ESS alone cannot see. Because the reference set is itself produced by QT, the diagnostic checks only modes found by that reference construction.

After these diagnostics assess the pushforward samples, the same weights define the resampling step. Normalize them as $p_i = w_i / \sum_{j=1}^N w_j$, draw N samples by multinomial resampling from the importance-weighted pushforward samples $(y_i, w_i)_{i=1}^N$ with probabilities p_i , and apply a short MALA chain under π to rejuvenate the duplicates that resampling produces. The resampled samples form an unweighted approximation of the target-weighted empirical measure, while MALA produces fresh configurations and retains π as its invariant distribution.

Remark 6 *MALA or other local Monte Carlo methods are necessary to keep the samples distinct after resampling, because each resampling possibly introduces duplicates even when the importance weights are completely uniform, and all the more so in the general case, where the ESS can be low. The same rejuvenation can in principle be applied to the batch subsamples drawn in steps (i) and (iii) of the KLXX training algorithm (Algorithm 1); we omit it there for simplicity.*

Under [standard regularity consistency conditions](#), the target-weighted empirical measure and its resampled particle approximation converge to π as $N \rightarrow \infty$ (Liu, 2001; Doucet et al., 2001). The π -invariant MALA refresh preserves this target limit. We call samples with this property *asymptotically unbiased*: their bias against the target vanishes as $N \rightarrow \infty$.

Remark 7 *Asymptotic unbiasedness is stated here formally. Theorem 2 proves it rigorously for the Boltzmann generator, with MALA as the rejuvenation kernel.*

4 Adaptive-staging Boltzmann generator

For high-dimensional distributions, a single flow from the source π_0 to the target π can be inadequate when the source-to-target deformation is too large, so we divide the training into stages that interpolate between the two potentials. Each stage changes the potential by a controlled amount and trains one flow with the generalized KLXX loss (19) at a fixed choice of $(\lambda, \vartheta, \alpha, \beta)$; the KLXX Boltzmann generator is the instance at the default $(\lambda, \vartheta, \alpha, \beta) = (1, 1, \frac{1}{2}, \frac{1}{2})$. Sampling diagnostics determine the stage schedule.

4.1 Why staging must be adaptive

Two obstacles prevent a single flow from carrying π_0 to π in one map.

- *Curse of dimensionality.* In low dimension a unimodal source can often be reshaped into a multimodal target by one map, but as d grows the target landscape develops more basins, separated by higher barriers and occupying a smaller fraction of the source support. A normalizing flow of fixed depth and width can represent only a bounded change of shape, so the deformation it realizes in one training run

is limited by its capacity rather than by the training budget. Requesting the whole source-to-target change from a single flow therefore leaves part of the target unfitted in exactly the high-dimensional, multimodal problems of interest. Splitting the interpolation into stages keeps each requested deformation within the capacity of one flow, and the number of stages needed to do so grows with dimension.

- *Structural limitation of a diffeomorphism.* A trained flow is a diffeomorphism of $\mathbb{R}^d$, so the pushforward ν inherits the connected support of π_0 : the map deforms that support continuously and cannot split it into disconnected components. Mass therefore remains on the paths joining the target basins even when the basins themselves are well fitted, an effect visible between the isolated modes of the Sparse target in Figure 5. Reweighting and MALA rejuvenation at the interpolating distributions remove the leaked samples at every stage, so the support changes shape gradually across the schedule.

The stage schedule cannot be fixed in advance, because we assume no external information about the target. Training uses no target dataset, no reference trajectory, and no prior knowledge of the number, location, or depth of the potential wells. Under this restriction the difficulty of a candidate stage cannot be read off the potential before attempting it. The algorithm therefore probes the interpolation parameter: it proposes a stage increment, trains the stage flow, accepts the increment only when the resulting proposal passes a sampling gate on the sample set, and otherwise shrinks the increment and retries. Adaptive staging is thus more than a convenience. It is the mechanism that turns oracle access to $U(x)$ and $\nabla U(x)$ into a stage schedule matched to the difficulty the target actually presents, and it lets easy regions advance in few large stages while hard regions receive as many small stages as they require.

4.2 Construction of the adaptive stage schedule

To interpolate from the source $\pi_0 \propto \exp(-U_0)$ to the target $\pi \propto \exp(-U)$, we seek an increasing sequence

$$0 = t_0 < t_1 < \cdots < t_K = 1$$

defining the interpolating potential and measure

$$U_t(x) = (1-t)U_0(x) + tU(x), \quad \pi_t(x) \propto \exp(-U_t(x)), \quad t \in [0, 1].$$

Here t is a dimensionless interpolation parameter. The discrete values t_k are stage points, and stage k is the transition from t_{k-1} to t_k . At the accepted stage points, write $\pi_k := \pi_{t_k}$. Stage k trains a target-to-source map G_k , whose inverse induces the pushforward distribution $\nu_k = (G_k^{-1})_{\#}\pi_{k-1}$ for π_k . The number of stages K is selected during the run. The central scheduling choice is the stage increment $\Delta t_k = t_k - t_{k-1}$: a large increment can be hard to fit, whereas a small increment adds stages.

Training G_k requires the target surrogate for π_k defined in Section 3.3. Before training, the adaptive SMC screen (Algorithm 4) can screen the proposed stage increment Δt_k with an optional classical SMC calculation. Its smallest per-level ESS measures overlap along that candidate stage for the current sample set. This is a difficulty diagnostic, not a mode-coverage guarantee. The screen floor τ_{SMC} lies in $[0, 1)$, and setting it to zero disables the screen. Throughout this section, the sample set is carried across stages and used for training batches, QT, and stage selection; QT or the SMC screen may act on the whole set or on a separately sized subset.

Throughout, the adaptive SMC screen (Algorithm 4) and AIS (Algorithm 3) take the same ladder length M : one value sets how finely both the screen and the training surrogate resolve a stage, and the Boltzmann generator (Algorithm 5) passes it to both. These SMC ladders involve only the potentials U_0 and U . They use no QT samples, so their per-level ESS can overlook a mode the SMC samples never reach. We therefore use this ESS only to screen a candidate stage; it is a heuristic measure of stage difficulty, not a coverage certificate. The stage ESS in step (v) below supplies the acceptance gate, while QT supplies the separate wide-coverage set. By default $\tau_{\text{SMC}} = 0$, so the screen returns t_{init} without running SMC. For a positive floor, it returns the first candidate stage along the shrink sequence whose estimated ESS clears that floor. The shrink factor is $\gamma = 0.7$. As t_k approaches t_{k-1} , the stage becomes trivial and its ESS approaches 1, so a floor below 1 is attainable under repeated shrinking in exact arithmetic.

Algorithm 4: Adaptive selection of the stage increment Δt_k

Input: potentials U_0, U , stage point t_{k-1} , sample set $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\}$ approximating π_{k-1} , ladder length M , SMC ESS floor $\tau_{\text{SMC}} = 0$, shrink factor $\gamma = 0.7$, initial guess t_{init}

Output: next stage point t_k , giving the stage increment $\Delta t_k = t_k - t_{k-1}$

```
1  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U;$   
   // start from the supplied initial guess  
2  $t_k \leftarrow t_{\text{init}};$   
3 if  $\tau_{\text{SMC}} = 0$  then  
4   | return  $t_k;$   
5 end  
6 while true do  
   // run a classical SMC ladder across the candidate stage  
7   Run an  $M$ -level SMC ladder on the set  $\mathcal{Y}_{k-1}$ , using the linear potential bridge from  $U_{t_{k-1}}$  to  $U_{t_k}$ ;  
   // evaluate the ladder by its worst per-level ESS  
8   Record  $\text{ESS}_{\min}$ , the smallest per-level ESS over the  $M$  ladder levels;  
9   if  $\text{ESS}_{\min} \geq \tau_{\text{SMC}}$  then  
10  | return  $t_k;$   
11  else  
    // shrink the stage increment toward  $t_{k-1}$   
12  |  $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1});$   
13  end  
14 end
```

4.3 The full training procedure

Algorithm 5 combines the stage-selection rule with the training procedure. It carries across stages a fixed-size sample set $\mathcal{Y}_k$, whose empirical measure approximates the per-stage target π_k . Each candidate stage builds its QT set, trains a flow with the selected loss weights, and compares that flow with the identity proposal. If neither proposal reaches the sample ESS floor $\tau_v \in [0, 1)$, the algorithm shrinks the stage increment and retries. The displayed loop ends when $t_K = 1$. Practical implementations use an endpoint tolerance and finite caps on retries and stages; reaching a cap returns an incomplete stage schedule.

- (i) *Sample set.* Stage k uses the *sample size* N and the *batch size* B , taken at least 2 so that the permuted-pair estimator has a pair to form. The sample set $\mathcal{Y}_{k-1} = \{y_1, \dots, y_N\} \approx \pi_{k-1}$ is carried over from the previous stage. Training batches are drawn at run time, B samples per gradient step, from $\mathcal{Y}_{k-1}$; no separate training set is built.
- (ii) *Adaptive stage selection.* The adaptive SMC screen (Algorithm 4) selects the next stage point t_k . The first stage starts from $t_{\text{init}} = t_{\text{safe}} = 0.2$. Later stages extrapolate the last accepted stage increment, $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$, with enlarge factor $\Gamma = 1.5$. When a positive floor is supplied, the algorithm shrinks the stage increment until the classical SMC ESS on all N samples clears that floor. The selected stage target is $\pi_k \propto \exp(-U_{t_k})$, with $U_{t_k} = (1 - t_k)U_0 + t_k U$.
- (iii) *QT set construction.* QT (Algorithm 2) acts on the sample set $\mathcal{Y}_{k-1}$ under U_{t_k} . The resulting N samples $\hat{\mathcal{Y}}_k = \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$ are held fixed for the stage.
- (iv) *Flow training.* Each attempt initializes G_k to the identity map and trains it by the KLXX training algorithm (Algorithm 1), applied to the stage $\pi_{k-1} \rightarrow \pi_k$: the batches of size B are drawn from $\mathcal{Y}_{k-1}$, the target surrogate samples come from AIS (Algorithm 3) with $G = G_k$, the QT sample set is $\hat{\mathcal{Y}}_k$, and

the stage log-ratio is

$$z_k(y) = U_{t_{k-1}}(G_k(y)) - U_{t_k}(y) - \log |\det J_{G_k}(y)|.$$

The coefficients $(\lambda, \vartheta, \alpha, \beta)$ are fixed for the run. Taking $\lambda = \vartheta = 0$ gives forward KL, $(\lambda, \vartheta) = (1, 0)$ gives KL+X $_{\pi}$, and $(\lambda, \vartheta, \alpha, \beta) = (1, 1, 1, 0)$ gives KL+X $_{\pi}$ +X $_{\tilde{\pi}}$; the default gives KLXX.

- (v) *Sample set update.* Once the candidate G_k is trained, each $y \in \mathcal{Y}_{k-1}$ is pushed through its inverse to $\tilde{y} = G_k^{-1}(y)$. This gives the trained set of pushforward samples $\tilde{\mathcal{Y}}_k^{\text{tr}} = G_k^{-1}(\mathcal{Y}_{k-1})$ from $\nu_k = (G_k^{-1})_{\#} \pi_{k-1}$, together with the importance weights

$$w_k^{\text{tr}}(\tilde{y}) := \exp(z_k(\tilde{y})), \quad \tilde{y} \in \tilde{\mathcal{Y}}_k^{\text{tr}}.$$

The identity proposal keeps $\mathcal{Y}_{k-1}$ unchanged and has weights $w_k^{\text{id}}(y) = \exp(U_{t_{k-1}}(y) - U_{t_k}(y))$. We select the pair with the larger ESS over the sample set and denote it by $(\tilde{\mathcal{Y}}_k, w_k)$, taking the stage map to be G_k or the identity accordingly. If $\text{ESS}(w_k) < \tau_v$, with default $\tau_v = 0.6$, we discard the candidate, shrink to $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$, and repeat steps (iii)–(v) at the reduced stage increment. Otherwise we draw N samples from $\tilde{\mathcal{Y}}_k$ with weights w_k and rejuvenate them by MALA under U_{t_k} , giving the next sample set $\mathcal{Y}_k \approx \pi_k$.

The accepted stage schedule defines a Boltzmann generator for π . The process starts with samples from π_0 . At stage k , it applies the selected inverse map or identity, computes w_k , resamples, and rejuvenates under U_{t_k} . The resulting sample set supplies stage $k+1$.

4.4 Propagation factor analysis

Algorithm 5 traverses the interpolation distributions $\{\pi_k\}_{k=0}^K$, training a flow map G_k for each stage from π_{k-1} to π_k . A natural question is how the Monte Carlo sampling error accumulates across these stages. Let $w_k = \pi_k/\nu_k$ be the importance weight at stage k , written as the ratio of the target density π_k and the pushforward density ν_k . Assume w_k is well defined so that importance sampling is available at every stage.

To generate N samples for the target distribution π_K , draw $X_0^1, \dots, X_0^N$ independently from the source distribution π_0 and repeat, for $k = 1, \dots, K$:

- *Flow pushforward.* Apply the inverse stage map to each particle, $Y_k^i = G_k^{-1}(X_{k-1}^i)$.
- *Importance sampling.* Evaluate the stage weight at each pushforward particle, $W_k^i = w_k(Y_k^i)$, and draw independent $Z_k^1, \dots, Z_k^N$ by multinomial resampling from $\sum_{i=1}^N (W_k^i / \sum_{l=1}^N W_k^l) \delta_{Y_k^i}$.
- *Rejuvenation.* Rejuvenate each resampled particle, $X_k^i \sim Q_k(Z_k^i, \cdot)$, independently across i .

Here Q_k is any Markov kernel leaving π_k invariant; the MALA of step (v) in Algorithm 5 is one such kernel, and nothing below uses more than that invariance.

After stage K the scheme returns the empirical measure

$$\pi_K^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_K^i} \quad (27)$$

as the estimate of π_K . For a general distribution μ and a function φ , we write $\mu\langle\varphi\rangle := \int_{\mathbb{R}^d} \varphi d\mu$ for the integral. For a bounded observable φ we write $\text{osc}(\varphi) = \text{ess sup } \varphi - \text{ess inf } \varphi$, and for a function h write $\|h\|_{\infty}$ for the essential supremum of $|h|$ on $\mathbb{R}^d$, taken for φ with respect to π_K and for w_k with respect to ν_k .

Theorem 2 Suppose that, for every stage $k = 1, \dots, K$, $w_k > 0$, the ν_k -essential supremum $\|w_k\|_{\infty}$ is finite, and Q_k is a Markov kernel leaving π_k invariant. Then for every bounded observable φ and every $N \geq 1$,

$$\mathbb{E} \left[(\pi_K^N \langle \varphi \rangle - \pi_K \langle \varphi \rangle)^2 \right] \leq \frac{C}{N}, \quad C = \frac{\text{osc}(\varphi)^2}{4} \left(\sum_{k=0}^K \prod_{j=k+1}^K 3 \|w_j\|_{\infty} \right)^2, \quad (28)$$

with the empty product equal to one.

Algorithm 5: Adaptive-staging Boltzmann generator training

Input: potentials U_0, U ; sizes $N, B \geq 2$; loss coefficients $(\lambda, \vartheta, \alpha, \beta)$; ladder length M ; ESS floors $\tau_{\text{SMC}} = 0$, $\tau_v = 0.6$; factors $\gamma = 0.7$, $\Gamma = 1.5$; safe start $t_{\text{safe}} = 0.2$

Output: completed stage schedule $0 = t_0 < \dots < t_K = 1$ and selected stage maps $\{G_k\}_{k=1}^K$

// sample set, $\mathcal{Y}_k \approx \pi_k$

```
1  $\mathcal{Y}_0 \leftarrow N$  samples from  $\pi_0$ ,  $t_0 \leftarrow 0$ ,  $k \leftarrow 0$ ;  
2 while  $t_k < 1$  do  
3    $k \leftarrow k + 1$ ;  
4   Set  $U_{t_{k-1}} \leftarrow (1 - t_{k-1})U_0 + t_{k-1}U$ ;  
   // (i) sample set  
5   Use  $\mathcal{Y}_{k-1}$  for stage selection, the ESS gate, and QT;  
   // (ii) adaptive stage selection  
6   if  $k = 1$  then  
7     Set  $t_{\text{init}} = t_{\text{safe}}$ ;  
8   else  
9     Set  $t_{\text{init}} = \min(1, t_{k-1} + \Gamma(t_{k-1} - t_{k-2}))$ ;  
10  end  
11  Choose  $t_k$  by the adaptive SMC screen (Algorithm 4) on  $\mathcal{Y}_{k-1}$  from  $t_{\text{init}}$ ;  
12  Set  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU$ ;  
13  repeat  
14    Initialize  $G_k$  to the identity map;  
    // (iii) QT sample set  
15    Set the QT sample set  $\hat{\mathcal{Y}}_k \leftarrow \text{QT}(\mathcal{Y}_{k-1}, U_{t_k})$  by Algorithm 2;  
    // (iv) flow training  
16    Train  $G_k$  by the KLXX training algorithm (Algorithm 1) on the sample set  $\mathcal{Y}_{k-1}$  and the QT sample  
    set  $\hat{\mathcal{Y}}_k$ , from  $U_{t_{k-1}}$  to  $U_{t_k}$ , at the coefficients  $(\lambda, \vartheta, \alpha, \beta)$ ;  
    // (v) sample set update  
17    Form the trained and identity pushforward sets with their weights;  
18    Keep the larger-ESS pair as  $(\tilde{\mathcal{Y}}_k, w_k)$  and set  $G_k$  to the corresponding map;  
19    if  $\text{ESS}(w_k) < \tau_v$  then  
20      // reject the stage and shrink toward  $t_{k-1}$   
       $t_k \leftarrow t_{k-1} + \gamma(t_k - t_{k-1})$ , then update  $U_{t_k} \leftarrow (1 - t_k)U_0 + t_kU$ ;  
21    else  
22      Resample  $N$  samples from  $\tilde{\mathcal{Y}}_k$  with weights  $w_k$ ;  
23      Rejuvenate the resampled set by MALA under  $U_{t_k}$  and set  $\mathcal{Y}_k$  to the result;  
24    end  
25  until  $\text{ESS}(w_k) \geq \tau_v$ ;  
26 end  
27  $K \leftarrow k$ ;  
28 return  $\{t_k\}_{k=0}^K$  and  $\{G_k\}_{k=1}^K$ ;
```

Appendix C gives the proof of Theorem 2. The constant is explicit and independent of N , and depends on the stage maps only through the weight suprema; it is a product over stages, summed over the stage at which the error enters. Rejuvenation does not enlarge the constant. Given the pushforward particles, the resampling and the kernel draw amount to a single draw per particle. A Markov kernel also cannot widen the range of a bounded function.

Theorem 2 also settles how the scheme behaves in N . Writing $\|\cdot\|_2$ for the L^2 norm over all the randomness of the scheme, the bound gives $\|\pi_K^N\langle\varphi\rangle - \pi_K\langle\varphi\rangle\|_2 \leq \sqrt{C/N}$ for every bounded observable, which also bounds the bias and vanishes as N grows: the inference is *asymptotically unbiased*, and its error is removed by increasing the sample size alone. Classical Monte Carlo sampling of π instead requires the simulation time to grow and, for unadjusted integrators, the time step to vanish; a Metropolis-adjusted kernel such as MALA removes the second requirement but not the first. The generator of Algorithm 5 is covered directly, its rejuvenation being MALA, whose Metropolis adjustment leaves each stage target invariant exactly (Metropolis et al., 1953; Hastings, 1970; Roberts and Tweedie, 1996; Del Moral, 2004). The rejuvenation also keeps the particles distinct: resampling only copies and the stage maps are bijections, so without it the number of distinct configurations could only fall from stage to stage, and the independent noise scatters the copies apart again at no cost in the bound.

The constant C in Theorem 2 is a square up to the observable, and its square root is defined as the *propagation factor* of the training schedule,

$$F_{\Sigma+} := \sum_{k=0}^K \prod_{j=k+1}^K 3\|w_j\|_{\infty}, \quad (29)$$

so that the inequality (28) equivalently reads

$$\|\pi_K^N\langle\varphi\rangle - \pi_K\langle\varphi\rangle\|_2 \leq \frac{\text{osc}(\varphi)}{2} \frac{F_{\Sigma+}}{\sqrt{N}}.$$

The propagation factor multiplies the error itself rather than its square.

$F_{\Sigma+}$ is built from the essential supremum of each stage weight, and that is what makes it impractical. One rare configuration of large weight fixes the supremum, so $\|w_j\|_{\infty}$ can be enormous at a stage that otherwise behaves well; for a singular potential it is difficult to estimate accurately, and need not be finite at all. We therefore estimate the training difficulty by a second propagation factor, assembled from the ESS of each accepted stage, a quantity the run computes anyway. For the stage weight, the ESS of (24) reads

$$\text{ESS}(w_k) := \frac{1}{\int_{\mathbb{R}^d} w_k(x)^2 \nu_k(x) dx} \in (0, 1], \quad (30)$$

and replacing each supremum by the reciprocal square root of the stage ESS, and dropping the factors 3, which arise from the recurrence in the proof and carry no information about the problem, gives

$$F_{\Sigma} := \sum_{k=0}^K \prod_{j=k+1}^K \frac{1}{\sqrt{\text{ESS}(w_j)}}. \quad (31)$$

The substitution is ordered stagewise. Since $\|w_j\|_{\infty} \geq \int w_j d\nu_j = 1$,

$$\frac{1}{\text{ESS}(w_j)} = \int_{\mathbb{R}^d} w_j^2 d\nu_j \leq \|w_j\|_{\infty} \leq \|w_j\|_{\infty}^2 \quad \text{for every accepted stage } j \implies F_{\Sigma} \leq F_{\Sigma+}.$$

The ordering of the inequality is reversed, so F_{Σ} is not a bound on the constant of (28). Every summand in (31) is at least one, so $F_{\Sigma} \geq K+1$, even when every stage proposal equals its target.

The calculation of F_{Σ} requires the ν_j -integral $\int w_j^2 d\nu_j$. A Boltzmann generator run already evaluates the stage weight vectors $W_j = (W_j^1, \dots, W_j^N)$ with $W_j^i = w_j(Y_j^i)$, and the sample ESS of each vector is read off them at no further cost, giving the propagation factor that the experiments report,

$$\hat{F}_{\Sigma} := \sum_{k=0}^K \prod_{j=k+1}^K \frac{1}{\sqrt{\text{ESS}(W_j)}}, \quad (32)$$

invariant to the scale of the weights and therefore available when the target is known only up to a constant. It is not, however, a bound on F_Σ at finite N . Appendix C.3 takes up how far F_Σ can sit from $F_{\Sigma+}$ and why it is nevertheless the quantity to report. F_Σ summarizes weight degeneracy along one accepted schedule, and does not measure the difficulty of the target. The sample size required by importance sampling grows exponentially in the KL divergence between target and proposal (Chatterjee and Diaconis, 2018), and with the intrinsic dimension of the problem (Agapiou et al., 2017); neither dependence appears in F_Σ .

Our propagation factor analysis is closely related to the asymptotic study of AFT (Arbel et al., 2021) and CRAFT (Matthews et al., 2022). Under uniformly bounded incremental weights, AFT establishes convergence in probability. Under further assumptions, among them convergence of the flows, it also establishes a central limit theorem whose asymptotic variance satisfies a recurrence: the variance injected at one annealing step, plus the previous step’s variance carried through that step. The constant of (28) is the unrolled form of such a recurrence, with the sum over the stage at which the error enters and the product over the stages it must still traverse, but bounded rather than exact. That is what makes the rate $N^{-1/2}$ and the constant explicit at finite N for the Boltzmann generator of Algorithm 5. The same reading favors F_Σ over $F_{\Sigma+}$, since the injected term of that recurrence is a second moment of the incremental weights, the quantity the stage ESS of (30) measures, whereas an essential supremum is not.

5 Numerical experiments on model problems

We test forward KL against three instances of the generalized KLXX loss (19): $\text{KL} + \text{X}_\pi$, $\text{KL} + \text{X}_\pi + \text{X}_{\hat{\pi}}$, and $\text{KL} + \text{X}_\pi + \text{X}_{(\hat{\pi} + \bar{\nu})/2}$, corresponding to different choices of the coefficients $(\lambda, \vartheta, \alpha, \beta)$. The last is KLXX, whose mixture variation weights the QT distribution $\hat{\pi}$ and the pushforward $\bar{\nu}$ equally. The tests are the 2D benchmark distributions, the high-dimensional product multi-well, the lattice ϕ^4 field theory, and the p -state clock model. They vary in dimension and in how the modes are distributed. Where a fake ESS can be an issue, we report ESS together with the coverage diagnostic.

5.1 2D benchmark distributions

For every target, we initialize the same neural spline flow (NSF) (Durkan et al., 2019) at the identity and compare the four losses above. The flow has 32 spline bins, 6 transforms, and conditioner widths (128, 128), and each training step constructs a target surrogate through the current flow on a ladder of length $M = 1$, followed by MALA rejuvenation. QT acts on a fixed sample set of 5×10^4 to 8×10^4 points. Coverage, with $\kappa = 5$ against an independently generated QT reference set, measures the fraction of reference configurations represented by the pushforward samples through the nearest-neighbor test of Section 3.4. Because coverage tests occupancy of the reference balls, a collapsed flow can have ESS near one but cannot have full coverage. Each loss–target pair is one fixed-seed training realization, so the numerical differences below describe these matched runs rather than variation across independent trainings.

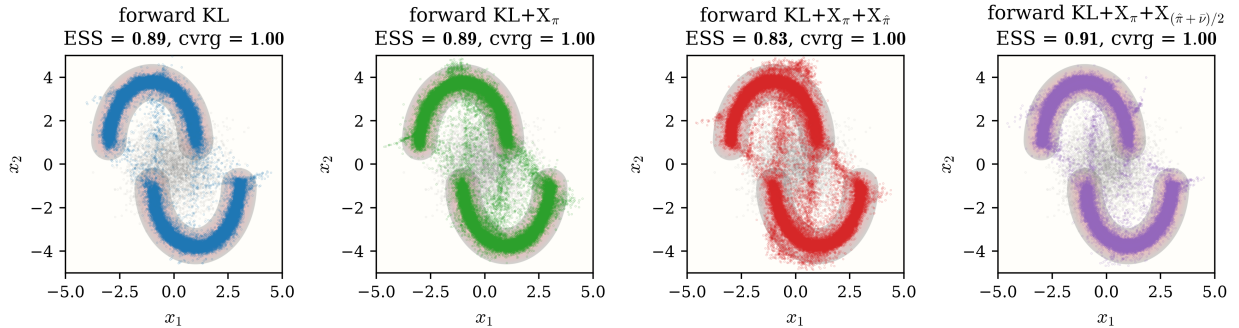


Figure 2: Two-Moon target. Columns, from left to right, are forward KL, $\text{KL} + \text{X}_\pi$, $\text{KL} + \text{X}_\pi + \text{X}_{\hat{\pi}}$, and $\text{KL} + \text{X}_\pi + \text{X}_{(\hat{\pi} + \bar{\nu})/2}$. Each panel overlays the pushforward samples on the target energy and reports final ESS and coverage. Gray points show the source.

Two-Moon. The source is $\mathcal{N}(0, I_2)$ and both crescents lie within its reach, so all four losses attain coverage 1 (Figure 2). With batch size 500, 200 Adam steps, and learning rate 10^{-3} , forward KL reaches ESS 0.8934, adding X_π leaves it at 0.8929, and the equal-weight mixture is best at 0.9087. The loss $\text{KL}+X_\pi+X_{\hat{\pi}}$ falls to 0.8313, since QT supplies no mode that the target surrogate is missing.

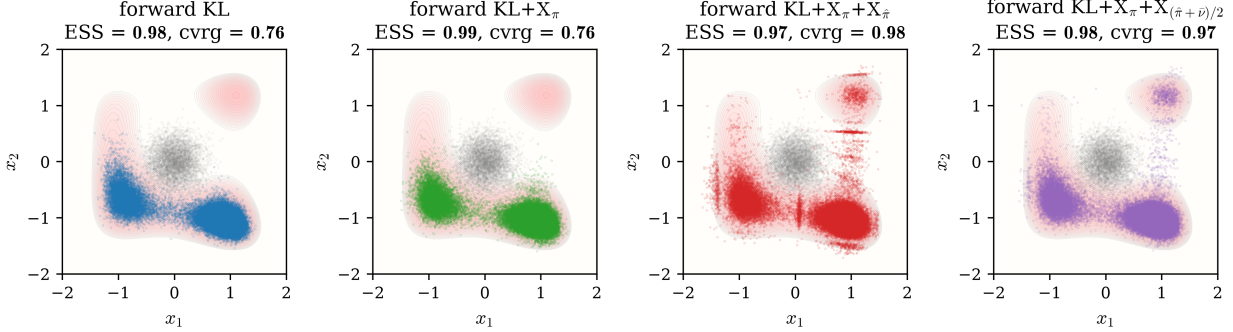


Figure 3: Three-Well target. From left to right, the panels show forward KL, $\text{KL}+X_\pi$, $\text{KL}+X_\pi+X_{\hat{\pi}}$, and $\text{KL}+X_\pi+X_{(\hat{\pi}+\bar{\nu})/2}$. Each panel overlays the pushforward samples on the target energy and reports final ESS and coverage; gray points show the source.

Three-Well. The source is $\mathcal{N}(0, 0.25^2 I_2)$, narrower than the well separation and initially covering no basin (Figure 3). Training uses batch size 1000, 500 steps, and learning rate 10^{-3} . Forward KL and $\text{KL}+X_\pi$ fill the two favored wells and report ESS 0.9769 and 0.9939, respectively, while their coverage remains 0.7595. The QT set also represents the suppressed third well, and both losses that use it cover all three wells at comparable ESS.

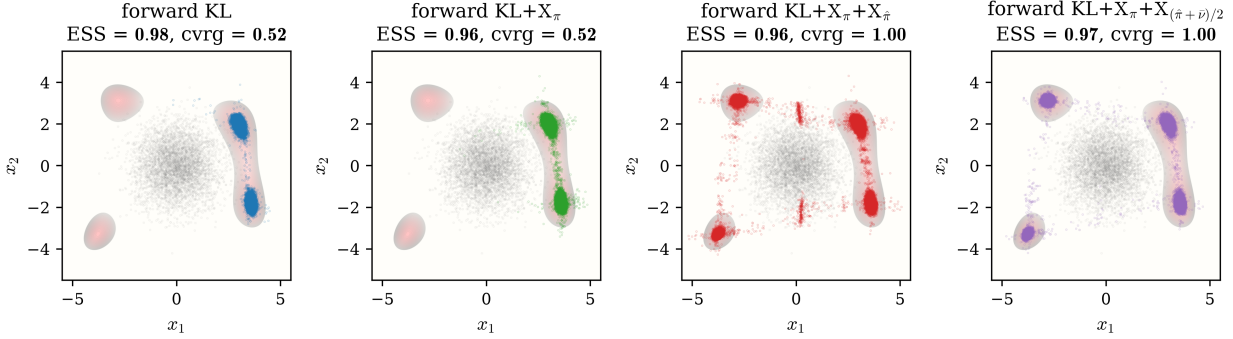


Figure 4: Himmelblau target. Columns show forward KL, $\text{KL}+X_\pi$, $\text{KL}+X_\pi+X_{\hat{\pi}}$, and $\text{KL}+X_\pi+X_{(\hat{\pi}+\bar{\nu})/2}$; each panel shows the pushforward samples over the target energy with final ESS and coverage, and gray points show the source.

Himmelblau. The source is $\mathcal{N}(0, I_2)$, whereas the four wells lie at radius approximately 3.6–5.0. Training uses batch size 1000, 1000 steps, and learning rate 10^{-3} . Forward KL and $\text{KL}+X_\pi$ capture only two wells, giving coverage 0.5160 despite ESS 0.9814 and 0.9596. Both losses using $\hat{\pi}$ attain coverage 1, and relative to $\text{KL}+X_\pi+X_{\hat{\pi}}$, KLXX shows less intermodal mass and raises ESS from 0.9618 to 0.9726.

Sparse. The target is a four-component Gaussian mixture with component standard deviation 0.08: two modes lie near the source center and two lie in opposite corners, more than five source standard deviations away. With source standard deviation 0.6, batch size 1000, 1000 steps, and learning rate 5×10^{-3} , forward KL and $\text{KL}+X_\pi$ cover only the central pair, at coverage 0.5335 and ESS 0.9031 and 0.9567. The loss

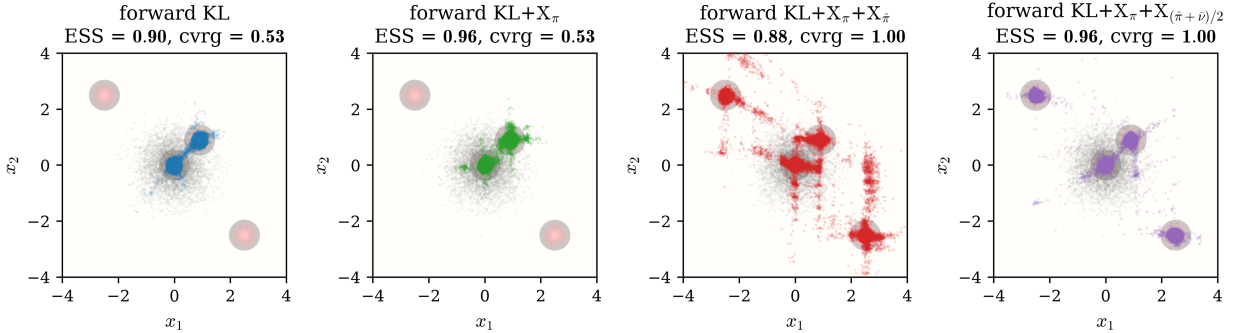


Figure 5: Sparse target. Columns show forward KL, $\text{KL}+X_\pi$, $\text{KL}+X_\pi+X_{\hat{\pi}}$, and $\text{KL}+X_\pi+X_{(\hat{\pi}+\bar{\nu})/2}$, with the pushforward samples overlaid on the target energy; panel titles report final ESS and coverage, and gray points show the source.

$\text{KL}+X_\pi+X_{\hat{\pi}}$ restores all four modes but places visible mass between them, at ESS 0.8783, while KLXX attains coverage 1 with less intermodal mass and ESS 0.9559.

target	KL	$\text{KL}+X_\pi$	$\text{KL}+X_\pi+X_{\hat{\pi}}$	$\text{KL}+X_\pi+X_{(\hat{\pi}+\bar{\nu})/2}$
Two-Moon	0.893 (1.000)	0.893 (1.000)	0.831 (1.000)	0.909 (1.000)
Three-Well	0.977 (0.760)	0.994 (0.760)	0.968 (0.977)	0.982 (0.971)
Himmelblau	0.981 (0.516)	0.960 (0.516)	0.962 (1.000)	0.973 (1.000)
Sparse	0.903 (0.534)	0.957 (0.534)	0.878 (1.000)	0.956 (1.000)

Table 2: Final ESS, with $\kappa = 5$ coverage in parentheses, on the four 2D targets. Bold marks the highest ESS among methods whose sample panel covers every target mode. The coverage diagnostic slightly undercounts the visibly occupied third mode on Three-Well.

The four targets support the intended division of labor. Adding X_π leaves ESS at the forward KL level on Two-Moon, where coverage is already complete. QT samples supply information about modes absent from the target surrogate. Adding the pushforward samples to that mixture gives cleaner intermodal regions than the loss $\text{KL}+X_\pi+X_{\hat{\pi}}$. KLXX has the highest mode-complete ESS on every target in Table 2.

5.2 High-dimensional product multi-well

We next isolate accuracy scaling over the dimensions $d = 16, 32, 48, \dots, 256$. Letting $k = \lfloor \log_2 d \rfloor$, we use

$$U(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|^2 + 12 \sum_{i=1}^k \exp(-x_i^2). \quad (33)$$

The first k coordinates are symmetric double wells with minima at $\pm\sqrt{\ln 24}$ and barrier approximately 9.9, while the remaining $d - k$ coordinates are standard Gaussian. The target therefore has exactly 2^k equal-weight sign-pattern modes, where $2^k \leq d < 2^{k+1}$. Multimodality occupies only a k -dimensional subspace, but the log-weight error accumulates across all d coordinates.

All four losses use the same identity-initialized NSF on $[-4, 4]^d$, with 16 bins, 6 transforms, and conditioner widths (256, 256). At every dimension, training uses batch size 250, 2000 Adam steps at learning rate 10^{-3} , and a fixed sample set of size $5000d$. The target surrogate uses a ladder of length $M = 1$ followed by 50 MALA steps of size 2×10^{-3} . QT uses $1000d$ samples, melt scale 2, and an L-BFGS quench with initial step 0.5 and 200 iterations. Final sample ESS is evaluated on 80000 fresh source samples. Each loss-dimension pair is one fixed-seed training realization.

In each of the 64 evaluated pairs of loss and dimension, all 2^k sign patterns of the double-well coordinates occur among the fresh evaluation samples, including all 256 patterns for each of the four losses at $d = 256$.

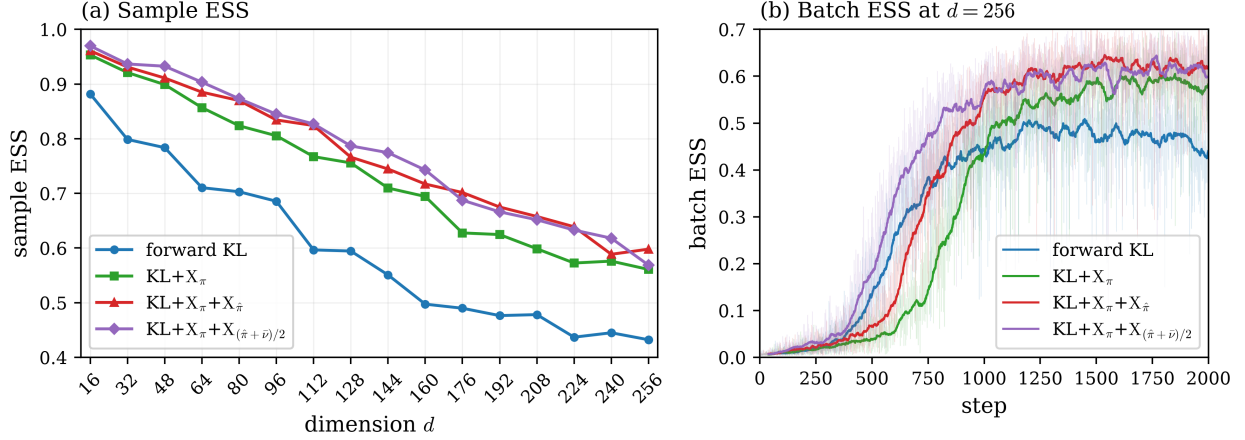


Figure 6: ESS on the product multi-well target (33). (a) Final sample ESS over sixteen tested dimensions. (b) Batch ESS during the $d = 256$ training; bold curves are 40-step moving averages of the raw values.

Figure 6 shows that each of the three losses with log-ratio variation has higher final sample ESS than forward KL at every one of the sixteen dimensions, so forward KL is the lowest of the four at each dimension. At $d = 256$, the final values are 0.4322 for forward KL, 0.5608 for $\text{KL}+X_\pi$, 0.5979 for $\text{KL}+X_\pi+X_{\hat{\pi}}$, and 0.5687 for $\text{KL}+X_\pi+X_{(\hat{\pi}+\bar{\nu})/2}$. The two losses containing $\hat{\pi}$ behave similarly across dimensions. During the $d = 256$ training realization, the batch ESS of $\text{KL}+X_\pi$, evaluated on the training batch after each gradient step, rises more slowly than that of forward KL over the first several hundred steps and overtakes it only near step 800. Adding $X_{\hat{\pi}}$ removes that delay: both losses containing $\hat{\pi}$ cross forward KL earlier, the mixture term first, so the wide-coverage term recovers the initial convergence speed and improves on it. All three losses with log-ratio variation then continue to improve after forward KL begins to plateau.

5.3 Lattice ϕ^4 field theory: fake ESS at a collective barrier

On a periodic $L \times L$ lattice with $d = L^2$, the tilted scalar ϕ^4 action is

$$U(\phi) = -0.8 \sum_{\langle jl \rangle} \phi_j \phi_l + \sum_{j=1}^d \left[\phi_j^2 + \frac{1}{2}(\phi_j^2 - 1)^2 \right] + h \sum_{j=1}^d \phi_j.$$

The on-site term alone has a single minimum at 0; the nearest-neighbor coupling is what makes a uniform configuration bistable, with effective per-site potential $-0.6\phi^2 + \frac{1}{2}(\phi^2 - 1)^2$ and wells near ± 1.26 . The magnetization of a configuration is

$$m(\phi) = \frac{1}{d} \sum_{j=1}^d \phi_j.$$

We take $h = 0.0257$ at $L = 6$ and $h = 0.0144$ at $L = 8$, matching the scaling $hL^2 \approx 0.92$ to produce comparable well imbalance in the targets of the two settings. A collective domain-wall barrier separates the two wells, and the minority weight $p_+ = \mathbb{P}(m > 0)$ must be learned rather than fixed by symmetry.

PT provides the references $p_+ = 0.1251 \pm 0.0009$ for $L = 6$ and 0.1284 ± 0.0014 for $L = 8$, with 3 independent runs for both L . Each loss is run at seeds 0, 1, 2 using an identity-initialized NSF on $[-3, 3]^d$, with 16 bins, 6 transforms, and conditioner widths (256, 256). The source is $\mathcal{N}(0, 0.5^2 I_d)$, centered on the barrier. Training uses a fixed sample set of 10^5 source samples, batch size 500, and 2000 Adam steps at learning rate 10^{-3} . The target surrogate uses a ladder of length $M = 1$ followed by 50 MALA steps of size 2×10^{-3} . QT acts on the sample set, with melt scale 2 and an L-BFGS quench using initial step 0.1 and 200 iterations.

Forward KL and $\text{KL}+X_\pi$ collapse at every size and seed (Table 3). Their target surrogate starts from the pushforward samples and does not cross the collective barrier within the run, so the second well remains

	$L = 6$		$L = 8$	
	ESS	p_+	ESS	p_+
forward KL	0.794/0.840/0.826	1/1/0	0.622/0.686/0.684	1/0/1
KL+ X_π	0.935/0.933/0.945	1/1/0	0.804/0.824/0.821	1/0/0
KL+ $X_\pi+X_{\hat{\pi}}$	0.875/0.873/0.871	0.127/0.128/0.127	0.636/0.645/0.654	0.125/0.124/0.124
KL+ $X_\pi+X_{(\hat{\pi}+\bar{\nu})/2}$	0.896/0.877/0.894	0.127/0.128/0.127	0.690/0.676/0.681	0.131/0.130/0.129
PT reference	—	0.1251 ± 0.0009	—	0.1284 ± 0.0014

Table 3: Final ESS and reweighted minority-well weight p_+ for three seeds on the two tilted ϕ^4 targets, $(L, h) = (6, 0.0257)$ and $(8, 0.0144)$. Integer values 0 and 1 indicate collapse onto one well; bold marks the higher ESS among the losses that recover both wells.

absent and reweighting gives $p_+ = 0$ or 1. Nevertheless, KL+ X_π has the highest ESS at both sizes, and it is a fake ESS: it is earned on the single well the flow reaches, and the missing one does not enter it. KLXX has both a trustworthy ESS and samples that populate both wells.

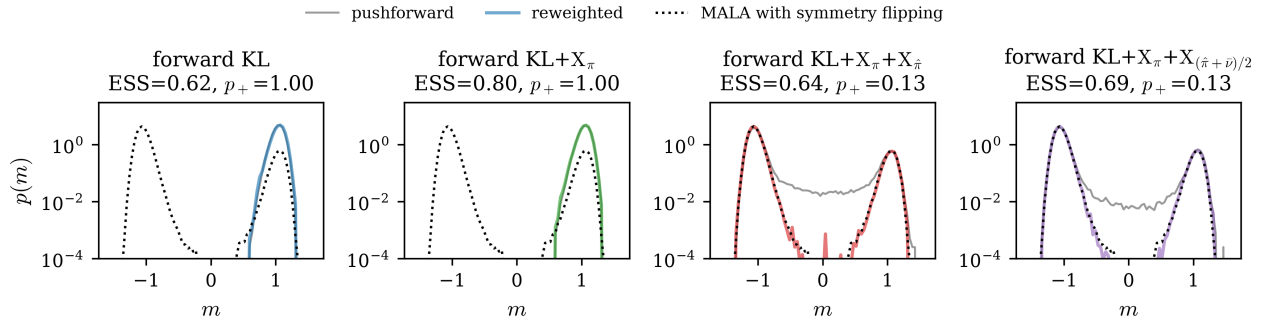


Figure 7: Magnetization density on the $L = 8$ lattice at seed 0. Each panel shows the pushforward density (gray), the reweighted density (color), and the resampled reference density from MALA with symmetry flipping (dotted); the title reports ESS and the reweighted p_+ .

The latter two losses using the QT samples $\hat{\pi}$ populate both wells at every seed, with recovered p_+ between 0.1238 and 0.1312, within 0.005 of the reference at the corresponding lattice size. In these finite-capacity fits, recovering both wells is accompanied by some mass in the suppressed region between them and lower ESS. At $L = 8$, KLXX has the higher ESS and less visible mass in that region (Figure 7), consistent with the intended role of the pushforward samples.

5.4 Boltzmann generator on p -state clock model

The preceding benchmarks use a single flow. We now test the full adaptive-staging construction on the $p = 6$ clock model on a periodic 8×8 lattice,

$$U(\theta) = - \sum_{\langle jl \rangle} \cos(\theta_j - \theta_l) - \frac{1}{2} \sum_{j=1}^d \cos(6\theta_j), \quad \theta \in [-\pi, \pi)^d, \quad d = 64, \quad (34)$$

where $\langle jl \rangle$ runs over nearest-neighbor pairs. The source is uniform on the torus and the interpolation is $U_t = tU$. The target distribution $\pi \propto \exp(-U)$ separates into six symmetry-related clock sectors.

Each stage uses an identity-initialized neural circular spline flow with 16 bins, 6 autoregressive transforms, and conditioner widths (256, 256). The sample set contains 4×10^5 samples, and both SMC and QT act directly on this set. Every stage uses 3000 Adam steps at learning rate 10^{-3} and ladder length $M = 4$ for both the adaptive SMC screen and the target surrogate, with 100 MALA steps of size 10^{-3} . For KLXX, QT uses melt scale 2π and an L-BFGS quench with initial step 10^{-2} and 100 iterations. The clock stage

schedule uses the nondefault settings $t_{\text{safe}} = 0.25$, enlarge factor $\Gamma = 2$, and gates $\tau_{\text{SMC}} = 0.7$ and $\tau_v = 0.4$. We report batch sizes $B = 2000, 1000, 500, 250$.

KLXX uses the same stage schedule as forward KL for fair comparison. That schedule is constructed with forward KL under the adaptive SMC screen and the ESS gate, and KLXX is then trained on the identical accepted values of t without a second gate. Every configuration is a single run, so individual stage ESS values carry single-run fluctuation.

The principal diagnostic is the sample ESS of each accepted stage. The propagation factor $\hat{F}_\Sigma$ of (32) summarizes stagewise weight degeneracy across the stage sequence; smaller is better. It is not a full-chain ESS or an endpoint error estimate. Table 4 reports every accepted stage and the resulting factor.

stage k	1	2	3	4	5	6	7	8	$\hat{F}_\Sigma$
t_k ($B = 2000$)	0.250	0.495	0.663	0.779	0.887	1.000	—	—	
forward KL	0.751	0.570	0.471	0.474	0.511	0.602	—	—	21.4
KL+ X_π + $X_{(\hat{\pi}+\bar{\nu})/2}$	0.921	0.759	0.608	0.562	0.558	0.649	—	—	15.6
t_k ($B = 1000$)	0.250	0.495	0.613	0.728	0.841	0.952	1.000	—	
forward KL	0.672	0.479	0.580	0.463	0.427	0.520	0.830	—	28.3
KL+ X_π + $X_{(\hat{\pi}+\bar{\nu})/2}$	0.866	0.642	0.700	0.545	0.462	0.541	0.859	—	21.2
t_k ($B = 500$)	0.250	0.421	0.590	0.705	0.784	0.895	1.000	—	
forward KL	0.580	0.549	0.408	0.432	0.525	0.417	0.530	—	40.7
KL+ X_π + $X_{(\hat{\pi}+\bar{\nu})/2}$	0.778	0.695	0.506	0.500	0.595	0.429	0.566	—	29.3
t_k ($B = 250$)	0.250	0.421	0.539	0.654	0.734	0.811	0.887	1.000	
forward KL	0.468	0.448	0.495	0.428	0.506	0.484	0.514	0.424	62.7
KL+ X_π + $X_{(\hat{\pi}+\bar{\nu})/2}$	0.659	0.586	0.601	0.480	0.563	0.536	0.564	0.464	41.7

Table 4: Stage ESS, the sample ESS of the map selected at a stage, on every accepted stage of the schedule-matched clock-model comparison between forward KL and KL+ X_π + $X_{(\hat{\pi}+\bar{\nu})/2}$. Within each batch size block, both losses traverse the listed stage schedule. The final column is the propagation factor $\hat{F}_\Sigma$ in (32); bold marks the better loss at each stage and the smaller propagation factor.

KLXX has higher sample ESS on all 28 shared stages, raising the geometric mean stage ESS from 0.556 to 0.665 at $B = 2000$, from 0.553 to 0.643 at $B = 1000$, from 0.487 to 0.571 at $B = 500$, and from 0.470 to 0.553 at $B = 250$. It reduces the propagation factor $\hat{F}_\Sigma$ at every reported batch size, by ratios between 1.33 and 1.50: the factor falls from 21.4 to 15.6 at the largest batch and from 62.7 to 41.7 at the smallest. As the batch shrinks, the schedule lengthens and both propagation factors grow.

The single-site marginal exhibits all six clock wells with nearly equal peak heights. Pair differences are most concentrated at zero for nearest neighbors; as the lattice separation increases, the alignment peak falls and the shoulders near $\pm\pi/3$ gain mass. This is the expected decay of local angular order with distance.

To quantify whether staged resampling preserves the sixfold symmetry, we perform two fresh-rebuild tests through the trained stage sequences. The batch size test fixes $N = 6.4 \times 10^5$, uses $B \in \{2000, 1000, 500, 250\}$, and averages over four matched sampling seeds for each method and batch size. The particle-count test fixes $B = 2000$ and uses $N = 10^4 2^k$, $k = 0, \dots, 6$, with 2^{8-k} independent rebuilds at each size so every particle-count value has equal total work of 2.56×10^6 particles. For each sample, let s be the clock sector nearest to the argument of the complex magnetization $d^{-1} \sum_{j=1}^d e^{i\theta_j}$, and let p_s be the empirical fraction assigned to sector s . We define

$$\text{occupancy bias} := \frac{1}{6} \sum_{s=0}^5 \left| p_s - \frac{1}{6} \right|.$$

At fixed N , panel (a) of Figure 9 shows that the mean occupancy bias increases as the training batch shrinks for both methods. KLXX has the lower mean at $B = 2000, 1000$, and 250 . At those batch sizes, the forward KL means are 0.00573, 0.00888, and 0.02760, while the KLXX means are 0.00235, 0.00517, and 0.01506. At $B = 500$ the ordering reverses slightly, from 0.00963 for forward KL to 0.01082 for KLXX, and the intervals defined by ± 2 standard errors overlap.

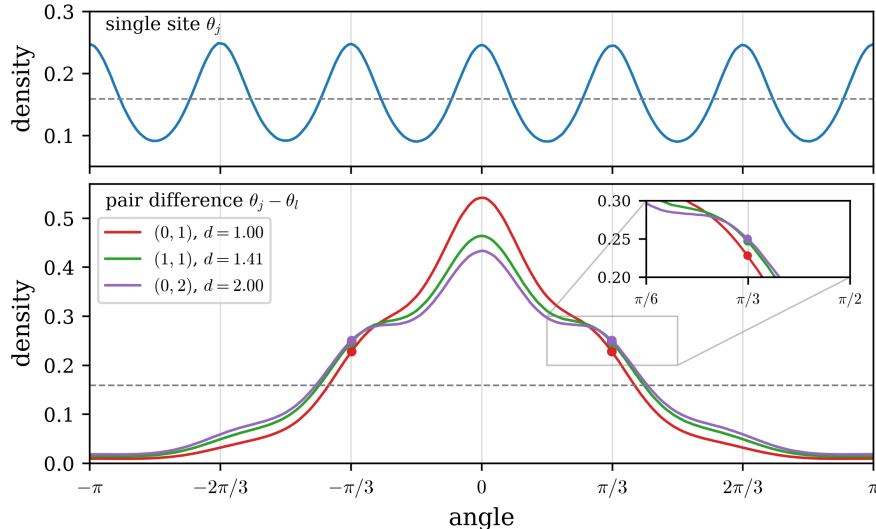


Figure 8: Angle densities from 2×10^6 fresh samples advanced through the trained $B = 2000$ $\text{KL} + \text{X}_\pi + \text{X}_{(\hat{\pi} + \bar{\nu})/2}$ stage sequence by map, reweighting, resampling, and MALA. The upper panel is the single-site marginal. The lower panel gives pair differences at offsets $(0, 1)$, $(1, 1)$, and $(0, 2)$; dots and the inset resolve the one-clock-step angle $\pi/3$. Dashed lines show the uniform density.

At fixed $B = 2000$, panel (b) follows the Monte Carlo rate: the log-log slopes of occupancy bias against N are -0.453 for forward KL and -0.545 for $\text{KL} + \text{X}_\pi + \text{X}_{(\hat{\pi} + \bar{\nu})/2}$. The absence of a visible bias floor is consistent with sampling fluctuation rather than a particle-count-independent sector imbalance over the tested range. The KLXX curve is lower at all seven reported sizes, reducing the mean bias from 0.03735 to 0.02397 at $N = 10^4$ and from 0.00573 to 0.00235 at $N = 6.4 \times 10^5$. Together with the six peaks in Figure 8, these tests show that the staged samples are not confined to one global sector. The rebuild seeds measure sampling variability; each stage schedule for a given method and batch size is one training run, so the between-method differences do not estimate variation over independently trained models.

6 Boltzmann generators on molecular potential energies

The model problems above separate the target and QT terms on explicit targets and probe the term built from pushforward samples. Molecular systems add a singular nonbonded core and more conformational degrees of freedom as molecular size grows, and their whitened coordinates (Noé et al., 2019) bring a mixed Euclidean-periodic domain. We first define the molecular measure and its regularized potential, then add sharpening to the staged generator and report the completed alkane family, three additional achiral molecules, and three chiral KLXX runs with fixed stereochemical configurations. The reported molecular wall times were recorded on a single NVIDIA GeForce RTX 5090 GPU.

6.1 Molecular potential models

For Cartesian coordinates x , let $E(x)$ denote the complete molecular-mechanics energy

$$E(x) = E_{\text{bond}} + E_{\text{angle}} + E_{\text{torsion}} + E_{\text{LJ}} + E_{\text{Coul}} + E_{\text{GB}}^{\text{OBC1}} + E_{\text{ACE}}. \quad (35)$$

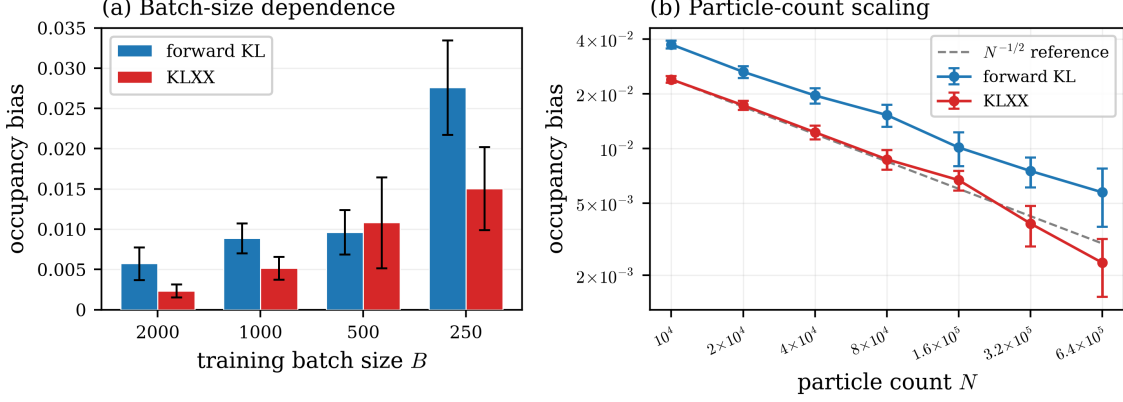


Figure 9: Clock-sector occupancy bias under two fresh-rebuild tests. (a) Mean bias at fixed $N = 6.4 \times 10^5$ across training batch sizes; each bar averages four matched sampling seeds. (b) Mean bias at fixed $B = 2000$ versus particle count; at $N = 2^k \times 10^4$, each mean uses 2^{8-k} independent rebuilds, keeping the total work per point at 2.56×10^6 particles. Error bars in both panels are ± 2 standard errors. The dashed line in (b) is an $N^{-1/2}$ reference. The two panels’ numbers align at $B = 2000$, $N = 6.4 \times 10^5$.

The bonded and vacuum nonbonded part has the Amber form (Cornell et al., 1995)

$$\begin{aligned}
 E_{\text{vac}}(x) = & \sum_b k_b (r_b - r_b^0)^2 + \sum_a k_a (\theta_a - \theta_a^0)^2 + \sum_\tau \frac{V_\tau}{2} [1 + \cos(n_\tau \varphi_\tau - \gamma_\tau)] \\
 & + \sum_{\substack{i < j \\ \text{nonbonded}}} \left\{ 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + k_e \frac{q_i q_j}{r_{ij}} \right\}.
 \end{aligned} \tag{36}$$

Here b , a , and τ index bonds, angles, and proper or improper torsions, while i, j index nonbonded atom pairs after topology-defined exclusions and scaling. The symbols r_b^0 , θ_a^0 , k_b , k_a , V_τ , n_τ , and γ_τ are force-field equilibrium values and coefficients; σ_{ij} , ϵ_{ij} , and q_i are the nonbonded parameters. OBC1 supplies the generalized-Born electrostatic solvation term (Onufriev et al., 2004), and ACE supplies the nonpolar term (Schaefer and Karplus, 1996). All terms in (35) belong to the sampled potential.

A nonlinear molecule with A atoms has $d = 3A - 6$ internal degrees of freedom after rigid translation and rotation are removed. A fixed internal-coordinate tree reconstructs $x = x(q)$ from whitened bond and angle coordinates and periodic torsions. Write d_E and d_T for the numbers of nonperiodic Euclidean coordinates and periodic torsions, respectively, and let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. Consequently,

$$\mathcal{Q} = \mathbb{R}^{d_E} \times \mathbb{T}^{d_T}, \quad d_E + d_T = 3A - 6,$$

and the physical reduced potential at inverse temperature $\beta_T = (k_B T)^{-1}$ is

$$U(q) = \beta_T E(x(q)) - \log J_x(q), \tag{37}$$

where $J_x(q)$ is the Cartesian volume Jacobian of the reconstruction, including the whitening maps. The source is Gaussian on the Euclidean block and uniform on the torus. The mixed spline flow uses ordinary rational-quadratic splines on the Euclidean coordinates and circular splines on the torsions.

Table 5 gives the force field and implicit-solvent model of each target. GAFF and AM1-BCC are described in Wang et al. (2004) and Jakalian et al. (2002), the bundles are prepared with AmberTools (Case et al., 2023), and the independent references are propagated with OpenMM (Eastman et al., 2017). We abbreviate N -acetyl-L-proline N -methylamide as Ac-Pro-NHMe. The chiral targets retain only the named configurations: L-alanine for alanine dipeptide, (2*R*, 3*R*) for 2,3-butanediol, and L-proline for Ac-Pro-NHMe. The achiral targets fix no stereocenter, and every conformational basin within a selected configuration belongs to the target, so the restriction defines the target rather than filtering its samples.

target	atoms / d	model
<i>Achiral targets</i>		
alkanes, C ₁ –C ₆	5–20/9–54	GAFF2/AM1-BCC; OBC1/ACE
<i>N</i> -methylacetamide (NMA)	12/30	ff96; OBC1/ACE
glycerol	14/36	GAFF2/AM1-BCC; OBC1/ACE
neutral diethanolamine	18/48	GAFF2/AM1-BCC; OBC1/ACE
<i>Chiral targets: one fixed stereochemical configuration</i>		
alanine dipeptide	22/60	ff96; OBC1/ACE
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol	16/42	GAFF2/AM1-BCC; OBC1/ACE
Ac-Pro-NHMe	26/72	ff96; OBC1/ACE

Table 5: Molecular targets in mixed internal coordinates, grouped by configurational stereochemistry. All systems are at 300 K, use no nonbonded cutoff or bond constraints, and include the coordinate Jacobian in (37). For each chiral target, the generator and reference are restricted to one specified stereochemical configuration, while all conformations within that configuration remain part of the target.

The Lennard–Jones core in (36) grows as r^{-12} , so early flow proposals can place a few samples at energies and gradients far beyond the scale of the thermally occupied region. We therefore use a two-parameter regularized potential. Let x_* be the bundle reference geometry. After flooring only the regular and exception Coulomb/Lennard–Jones pair distances at r , denote the complete force-field energy by $E_r(x)$, its reference value by $E_r^* := E_r(x_*)$, and its reference-relative value by $\Delta E_r(x) := E_r(x) - E_r^*$. For an energy threshold e , define the soft cap

$$\mathcal{C}_e(\Delta E) = \begin{cases} \Delta E, & \Delta E \leq e, \\ e[1 + \log(\Delta E/e)], & \Delta E > e, \end{cases} \quad (38)$$

and collect the two regularization parameters in $\rho = (e, r)$. The corresponding reduced potential is

$$U^{\{\rho\}}(q) = \beta_T [E_r^* + \mathcal{C}_e(\Delta E_r(x(q)))] - \log J_x(q). \quad (39)$$

The threshold e is measured in kJ mol^{−1}, and the pair floor r is measured in nm. The bonded and OBC1/ACE terms keep their component formulas, and the scalar cap is applied to the complete energy. The coordinate Jacobian and domain are not capped. The physical potential U in (37) is recovered in the formal limit $e \rightarrow \infty, r \rightarrow 0$.

A run ends at a finite regularization setting ρ rather than at that formal limit. For endpoint samples $q_1, \dots, q_N$, define the endpoint-to-physical weights and the corresponding regularization ESS (RESS) by

$$w_i^{\text{phys}} := \exp[U^{\{\rho\}}(q_i) - U(q_i)], \quad \text{RESS} := \text{ESS}(w^{\text{phys}}). \quad (40)$$

6.2 Boltzmann generators with sharpening

A softer molecular potential can make early stages easier in a given adaptive-staging run. After each flow stage, *sharpening* reweights the sample set toward a stricter regularization setting, resamples it, and applies MALA under the stricter potential. To extend the source-to-target interpolation of Section 4, let $\rho_0 = (e_0, r_0)$ and $\rho_1 = (e_1, r_1)$ be the soft and final settings and interpolate them by

$$\rho_s = (1 - s)\rho_0 + s\rho_1, \quad s \in [0, 1].$$

A fixed-regularization run has $\rho_0 = \rho_1$. Extending that interpolation by the single index s , set

$$U_{t,s}(q) = (1 - t)U_0(q) + tU^{\{\rho_s\}}(q), \quad \pi_{t,s} \propto \exp(-U_{t,s}).$$

Thus t remains the source-to-target interpolation parameter defined in Section 4, while s changes only the regularization setting. If s is fixed, suppressing the second index recovers the notation U_t, π_t used there.

Accepted sample sets lie on the diagonal $s = t$, but one stage advances the two coordinates in sequence:

$$\mathcal{Y}_{k-1} \approx \pi_{t_{k-1}, t_{k-1}} \xrightarrow[\text{selected } (\tilde{\mathcal{Y}}_k, w_k)]{\text{flow or identity}} \mathcal{Y}_k^- \approx \pi_{t_k, t_{k-1}} \xrightarrow[w_k^{\text{sharp}}]{\text{sharpen}} \mathcal{Y}_k \approx \pi_{t_k, t_k}. \quad (41)$$

The first arrow of (41) holds $s = t_{k-1}$ fixed and advances t from t_{k-1} to t_k . Potential-space SMC and mixed-domain MALA carry a copy of all N particles along this arrow, and in KLXX the additional QT samples are built under $U_{t_k, t_{k-1}}$. After training, the candidate flow and identity proposals are evaluated on the carried set $\mathcal{Y}_{k-1}$; the pair with larger sample ESS is denoted by $(\tilde{\mathcal{Y}}_k, w_k)$, ties retaining the identity, and reweighting, resampling, and MALA under $U_{t_k, t_{k-1}}$ produce $\mathcal{Y}_k^-$. Both ends of this arrow carry the same regularization parameters, so the map has only the interpolation coordinate to absorb.

The second arrow of (41) holds $t = t_k$ fixed and advances the regularization coordinate from t_{k-1} to t_k . Its exact log weight is

$$\log w_k^{\text{sharp}}(q) = U_{t_k, t_{k-1}}(q) - U_{t_k, t_k}(q) = t_k \left[U^{\{\rho_{t_{k-1}}\}}(q) - U^{\{\rho_{t_k}\}}(q) \right]. \quad (42)$$

Sharpening is accepted only when its ESS over the sample set clears the stage threshold, after which the samples are resampled and rejuvenated by MALA under U_{t_k, t_k} ; a proposal or a sharpening step that fails its gate shrinks $t_k - t_{k-1}$ and retries. This arrow therefore carries the change of regularization by exact reweighting rather than by the map, and it raises barriers the map is not asked to raise: a softer core flattens the short-range repulsion and the high-energy tail, which can leave passages between modes that the endpoint potential does not have, and advancing s toward t_k removes those spurious connections.

The propagation factor of (32) is evaluated on both arrows of (41). An accepted stage on a sharpening path reweights and resamples twice, so it supplies two of the weight vectors W_j : the weight w_k of the map selected at that stage, from the first arrow, and the sharpening weight w_k^{sharp} of (42), each evaluated on the N samples. A run of K accepted stages therefore supplies $2K$ weight vectors, K from the proposals and K from sharpening, and the sum and products of $\hat{F}_\Sigma$ run over them in the order the generator performs them. On a fixed regularization path the sharpening weight is identically one, and the step still enters the sequence because the generator resamples there. As before, $\hat{F}_\Sigma$ summarizes stagewise weight degeneracy; it is not a fresh source-to-endpoint ESS or an endpoint error estimator.

Temperature-annealed Boltzmann generators (TA-BG) (Schopmans and Friederich, 2025) address the same barrier problem through a different continuation coordinate: TA-BG lowers the temperature toward the physical target, whereas sharpening holds the temperature fixed and anneals the regularization ρ_s . Sharpening is the stronger intervention at the singular core, since changing temperature rescales the physical energy but preserves its divergent short-range form. It is also the less interpretable one: before the endpoint the sharpening states are algorithmic ensembles rather than physical molecular systems, whereas the TA-BG intermediates remain canonical distributions of the same potential at explicitly defined elevated temperatures. We have no matched numerical comparison with TA-BG and leave it to future work.

The generators below differ only in how each stage proposal is obtained; the SMC selection, reweighting, resampling, and MALA steps are the same throughout.

6.3 Alkane family results

We test methane (CH_4) through n -hexane (C_6H_{14} , straight chain). The regularization paths are listed in Table 6: fixed-regularization runs start and remain at the common endpoint, while sharpening runs begin at a softer setting and advance linearly to it. The three methods use method-specific adaptive stage schedules, so stage counts and recorded wall times are descriptive rather than fixed-compute comparisons.

All molecular flows use 6 mixed spline transforms, 32 bins, conditioner widths (256, 256), spline bound 8, and minimum slope 10^{-3} , and are initialized at the identity. The sample and batch sizes (N, B) are (100000, 5000) for methane, (120000, 6000) for ethane, (160000, 8000) for propane, and (200000, 10000) for butane through hexane. Each accepted flow stage uses 500 Adam steps for methane through propane, 400 for butane, and 300 for pentane and hexane, with learning rate 10^{-3} and 25 warmup steps. Every molecular run uses ladder length $M = 8$, so the SMC screen and the training surrogate both traverse 8 annealing levels; mixed-domain MALA uses 100 steps of size 10^{-3} . QT uses melt scale 1, 200 L-BFGS steps with initial step 10^{-2} , and the complete molecular QT set. The safe start is 0.25 for methane and ethane and 0.1 for the

larger molecules. The ESS gate is 0.4 throughout; the SMC gate is 0.3 for methane and ethane, 0.5 for propane, and 0.8 for butane through hexane.

molecule (d)	start ρ_0	identity			forward KL			KL+X $_{\pi}$ +X $_{(\hat{\pi}+\bar{\nu})/2}$		
		$\hat{F}_{\Sigma}$	stages	time	$\hat{F}_{\Sigma}$	stages	time	$\hat{F}_{\Sigma}$	stages	time
methane (9)	(100,0.15)	26.13	8	1.4	7.464	3	1.2	7.033	3	2.4
ethane (18)	(100,0.15)	50.02	9	3.5	10.36	3	1.8	7.305	3	4.1
propane (27)	(100,0.15)	93.24	11	7.7	92.97	11	13.4	20.01	7	23.0
	(50,0.15)	65.01	10	6.9	64.99	10	12.1	19.41	6	18.3
butane (36)	(100,0.15)	138.6	12	13.3	138.6	12	20.3	37.49	10	49.0
	(50,0.20)	86.75	10	10.8	86.68	10	14.9	36.19	10	46.1
pentane (45)	(100,0.15)	217.0	13	19.2	208.0	13	24.6	65.43	12	60.4
	(50,0.20)	117.0	10	13.9	117.6	10	17.2	54.89	9	38.5
hexane (54)	(100,0.15)	334.8	13	23.3	336.5	13	29.0	190.3	12	69.9
	(50,0.25)	171.3	10	17.4	170.1	10	22.7	114.5	10	60.1

Table 6: Alkane Boltzmann generator training results, including propagation factor $\hat{F}_{\Sigma}$, the number of accepted stages, and total wall time in minutes. Identity applies the identity map at every stage, whereas forward KL and KLXX may select a trained flow when it improves sample ESS. Each row reports one seed and one regularization path; bold marks the lowest $\hat{F}_{\Sigma}$ among the three methods for that path. Every path ends at (100,0.15), and rows that start at (100,0.15) use no sharpening.

Within the single-seed runs, KLXX has the smallest factor in all ten settings. Forward KL selects trained maps through methane and part of ethane. From propane onward, every accepted forward KL stage selects the identity proposal, and its factor is close to the identity result. KLXX selects the trained map at every accepted stage through pentane. At hexane, it selects trained maps in 6 of 12 fixed-path stages and 8 of 10 sharpening stages.

The difference between the fixed and sharpening paths increases with molecule size in these runs. The KLXX fixed-path factor divided by the sharpening-path factor is 1.03, 1.04, 1.19, and 1.66 for propane through hexane. The butane, pentane, and hexane KLXX sharpening runs take 3.10, 2.24, and 2.65 times the recorded forward KL wall time. The KLXX sharpening paths have the lowest factor in the larger-molecule comparisons, with additional QT, reweighting, and flow-training cost.

The endpoint-to-physical weights of (40) are nearly uniform on the final KLXX sample sets: $1 - \text{RESS} = 0$ numerically from methane through butane, 9.5×10^{-6} for pentane, and 2.3×10^{-5} for hexane. These values measure overlap on the sampled sets; they do not prove global equality between $U^{\{\rho_1\}}$ and U .

The KLXX curves of Figure 10 are unweighted summaries of the finite-endpoint sample sets; the physical energy E is evaluated only as an observable, without endpoint-to-physical resampling. The physical-potential reference uses two seeds and a 0.25 fs timestep. Methane through propane use 300 K Langevin trajectories; butane through hexane use 300–800 K PT to mix the backbone rotamers. Across the six molecules, the maximum relative difference between KLXX and reference physical-energy means is 0.644%. The largest relative difference among the nonzero carbon radius-of-gyration and terminal-distance means is 1.058%, and the largest absolute rotamer-fraction difference is 0.051. These results show agreement for the selected energy, size, and rotamer summaries across the C₁–C₆ series.

All thirty identity, forward KL, and KLXX runs in Table 6 are complete, but each configuration has one training seed. The reference frames are correlated, and the two-seed ranges are descriptive rather than framewise confidence intervals. The table therefore compares the recorded runs; it does not estimate run-to-run variation or distance between the complete equilibrium measures.

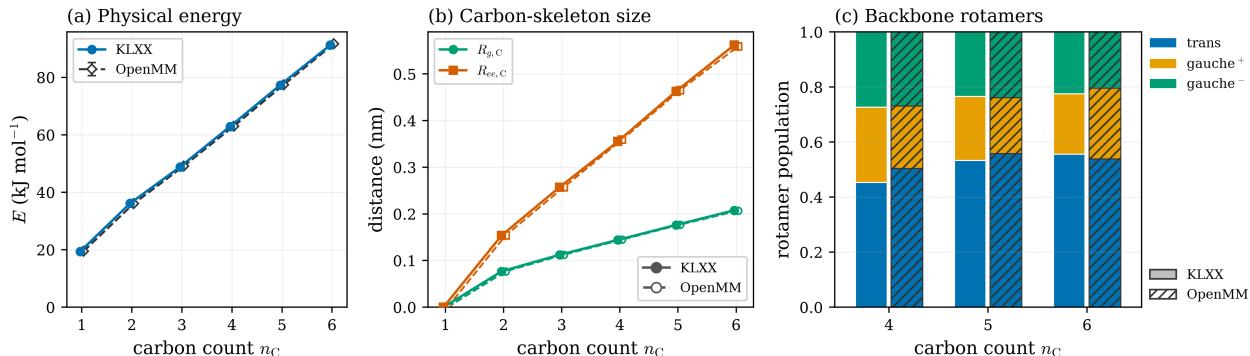


Figure 10: Macroscopic observables from the smallest-factor KLXX sample set for each alkane and an independent physical-potential reference. Fixed-path KLXX is used for methane through propane and sharpening KLXX for butane through hexane. (a) Physical energy evaluated under the exact, unregularized potential. (b) Carbon-skeleton radius of gyration $R_{g,C}$ and terminal-carbon distance $R_{e,C}$. (c) Pooled backbone trans and gauche fractions. Filled solid curves and unhatched bars denote KLXX; open dashed curves and hatched bars denote the reference.

6.4 Additional achiral molecule results

We next test NMA, glycerol, and neutral diethanolamine. Each method uses one seed, a sample set of size $N = 200000$, and batch size $B = 10000$. The flow, SMC, MALA, and QT settings match the larger-alkane configuration above. NMA and glycerol use 400 Adam steps per accepted flow stage, while neutral diethanolamine uses 300. The three sharpening paths are listed in Table 7. The two losses use method-specific adaptive stage schedules, so the stage counts and recorded wall times are descriptive rather than fixed-compute comparisons.

molecule (d)	start ρ_0	end ρ_1	KL+X π			KL+X π +X $_{(\hat{\pi}+\hat{\nu})/2}$		
			$\hat{F}_\Sigma$	stages	time	$\hat{F}_\Sigma$	stages	time
NMA (30)	(50, 0.20)	(100, 0.15)	20.19	6	6.0	15.12	5	12.1
glycerol (36)	(50, 0.20)	(100, 0.10)	146.2	9	16.5	54.16	8	35.8
neutral diethanolamine (48)	(50, 0.20)	(100, 0.10)	303.1	12	23.1	140.9	9	43.1

Table 7: Boltzmann generator training results for the additional achiral molecules, including propagation factor $\hat{F}_\Sigma$, the number of accepted stages, and total wall time in minutes. The two method groups use the full losses KL+X π and KL+X π +X $_{(\hat{\pi}+\hat{\nu})/2}$, respectively. Each row reports one seed and one sharpening path; bold marks the smaller $\hat{F}_\Sigma$ for that molecule.

Comparing KL+X π with KLXX, the latter has the smaller propagation factor for all three molecules. Relative to KL+X π , it reduces $\hat{F}_\Sigma$ by factors of 1.34, 2.70, and 2.15 for NMA, glycerol, and neutral diethanolamine, respectively, and reaches $t = 1$ in fewer accepted stages in every case. The corresponding KLXX runs require 2.02, 2.16, and 1.86 times the recorded wall time. Thus the factor reduction is accompanied by the additional QT and optimization work in KLXX.

Figure 11 resolves each factor into its accepted-stage contributions. Because the two losses use method-specific adaptive stage schedules, the individual arcs are descriptive rather than transition-matched comparisons. Sharpening can be the tighter operation at intermediate stages, so the total factor includes both ESS values reported on each arc.

The final regularized samples are compared with OpenMM PT references, taken at 300 K, in Figure 12. The dashed blue curves are plotted above the solid red curves so that close agreement between samples from the two losses remains visible. The selected torsions resolve the NMA amide trans basin and the trans/gauche wells of glycerol and neutral diethanolamine.

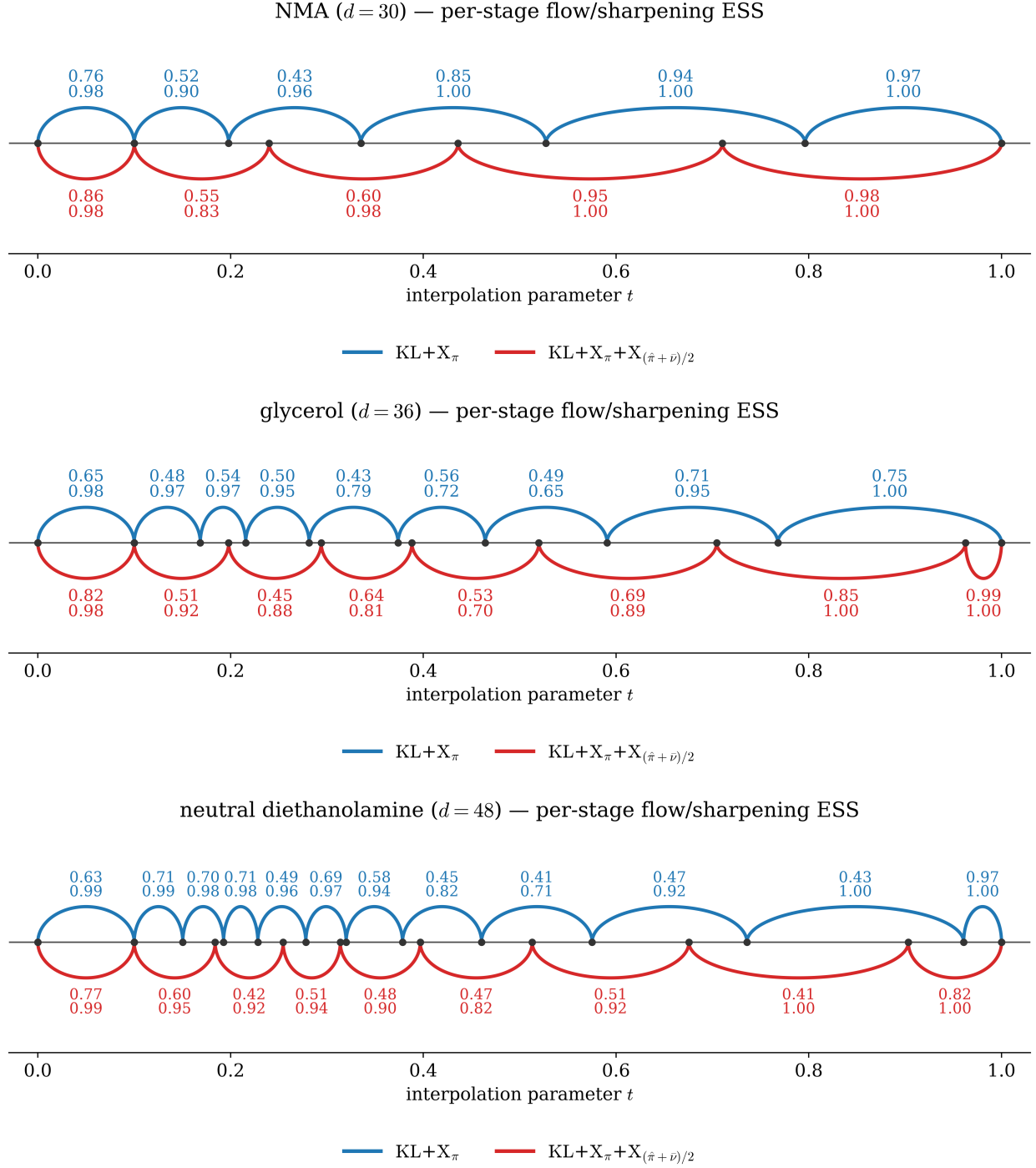


Figure 11: Per-stage flow and sharpening ESS for NMA (top), glycerol (middle), and neutral diethanolamine (bottom). Each arc spans one accepted transition $t_{k-1} \rightarrow t_k$. Within each stacked annotation, the first row is flow ESS and the second row is sharpening ESS. Blue upper arcs show $\text{KL} + \text{X}_\pi$; red lower arcs show $\text{KL} + \text{X}_\pi + \text{X}_{(\hat{\pi} + \bar{\nu})/2}$. The horizontal coordinate is the source-to-target interpolation parameter $t \in [0, 1]$.

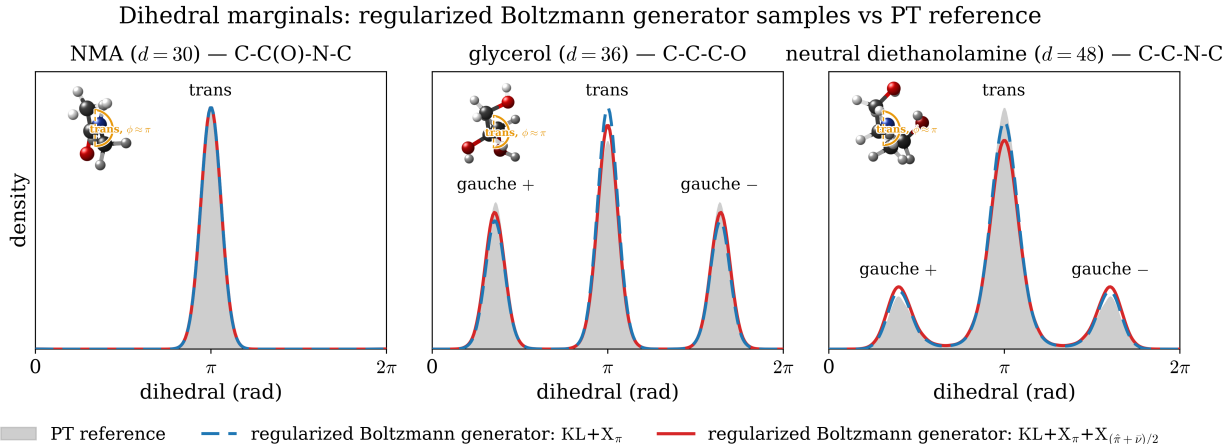


Figure 12: Selected dihedral marginals from regularized Boltzmann generator samples and OpenMM PT references for NMA, glycerol, and neutral diethanolamine. Gray denotes the reference; dashed blue denotes $KL+X_\pi$, and solid red denotes $KL+X_\pi+X_{(\hat{\pi}+\bar{\nu})/2}$. The upper-left insets show representative trans conformers. The NMA reference is a PT trajectory initialized in the trans basin, whereas the glycerol and neutral diethanolamine references pool two seeds. The NMA panel is therefore a trans-basin comparison rather than an estimate of the equilibrium cis/trans populations.

The curves overlap closely on these selected one-dimensional marginals, but this diagnostic does not determine the complete molecular distributions. In particular, it does not resolve the NMA cis/trans weights, joint glycerol or neutral diethanolamine rotamers, or unrestricted signed-volume statistics. The reference frames are also correlated, and every generator configuration has one training seed.

6.5 Chiral molecule results

Molecular setup. We report completed runs for alanine dipeptide in its L form, (2*R*, 3*R*)-2,3-butanediol, and Ac-Pro-NHMe. Only the KLXX loss is evaluated, so this subsection contains no within-target loss comparison. Every run uses seed 0, 250 Adam steps per accepted flow stage, an 8-level SMC ladder, and 100 mixed-domain MALA steps. The sample set and batch size pairs are $3.0 \times 10^5/1.5 \times 10^4$, $2.4 \times 10^5/1.2 \times 10^4$, and $3.2 \times 10^5/1.6 \times 10^4$, respectively. After training, each stage flow is replayed on a larger particle set, giving 10^7 inference samples for alanine dipeptide and 2,3-butanediol and 1.5×10^7 for Ac-Pro-NHMe. The surfaces below use those inference sets, on which the endpoint-to-physical weights are nearly uniform, with RESS equal to 1.000000, 0.999981, and 1.000000, respectively, so the regularized endpoint and the physical potential are indistinguishable on the sampled region.

The training and replay costs are deliberately asymmetric. Training never carries more than 3.2×10^5 samples, which keeps the wall times of Table 8 modest, while the inference sets are produced afterwards by replaying frozen flows without backpropagation. Replay is nonetheless the more expensive half in absolute terms, because sampling evaluates the NSF inverse G^{-1} , the expensive direction for this flow structure. Cost therefore sits where it is easiest to pay: replay is a fixed number of independent map evaluations and parallelizes across devices, whereas the training schedule is sequential in the stage index and the reference simulations of Section 6.3 must advance their trajectories sequentially in time.

All three runs reach $t = 1$; Table 8 gives their sharpening paths and stage summaries. The propagation factor is accumulated over the flow and sharpening ESS of every accepted stage, as in the earlier molecular tables. The three rows are different molecules with different sample sizes and regularization paths, so they are not comparable with one another.

Figure 13 shows one representative final-stage conformation per target, with gold rings on the fixed stereocenters: the *S* alanine C_α , both *R* centers of 2,3-butanediol, and the *S* proline C_α .

All three chiral targets carry an independent equilibrium reference; the first is published and the other two are our own. For alanine dipeptide it is the implicit-solvent trajectory data released with the FAB

molecule (d)	start ρ_0	end ρ_1	KL+X $_{\pi}$ +X $_{(\hat{\pi}+\bar{\nu})/2}$		
			$\hat{F}_{\Sigma}$	stages	time (min)
alanine dipeptide (60)	(50, 0.25)	(125, 0.10)	68.08	10	72.7
(2 <i>R</i> , 3 <i>R</i>)-2,3-butanediol (42)	(50, 0.20)	(100, 0.10)	37.45	8	35.8
Ac-Pro-NHMe (72)	(50, 0.25)	(150, 0.10)	176.7	11	124.7

Table 8: Boltzmann generator results for the chiral molecules, including propagation factor $\hat{F}_{\Sigma}$, the number of accepted stages, and total stage time in minutes. The method uses the loss KL+X $_{\pi}$ +X $_{(\hat{\pi}+\bar{\nu})/2}$. Each row reports one seed and one sharpening path.

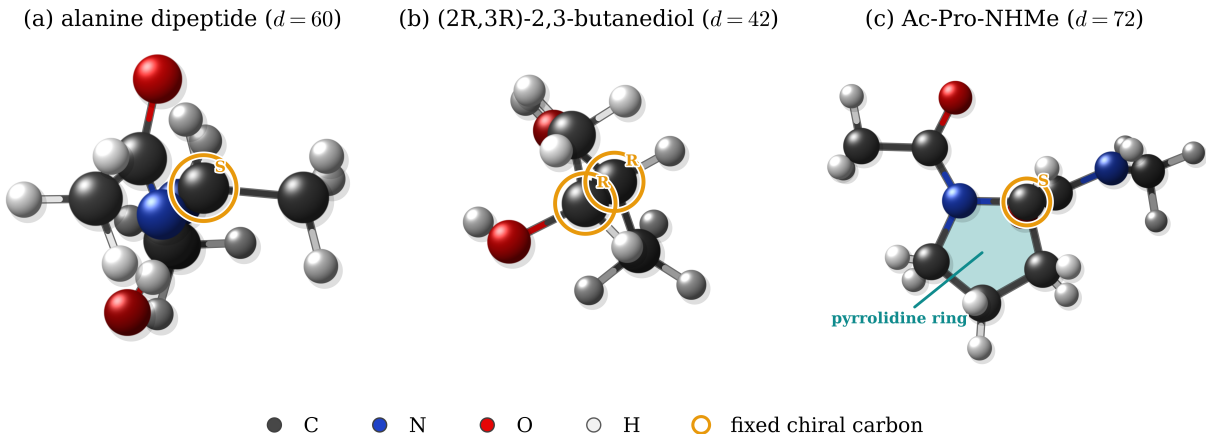


Figure 13: Representative final-stage conformations for the three completed chiral targets. From left to right: L-alanine dipeptide, (2*R*,3*R*)-2,3-butanediol, and Ac-Pro-NHMe. Gold rings and the adjacent *R/S* labels identify every fixed stereocenter; the atom colors follow the CPK convention shown in the legend. Each structure is the sample nearest the modal center of a molecule-specific two-dihedral histogram.

study (Midgley et al., 2023), generated for the same Hamiltonian and used here as ground truth, with 10^7 samples. For 2,3-butanediol it is a PT run on the unregularized physical potential at 300 K, six temperatures on a geometric 300–800 K grid, retaining 10^5 frames over 50.0 ns after a discarded 0.5 ns equilibration, with mean per-pair exchange probability 0.529 and 2.2×10^4 transitions among the three central rotamers. For Ac-Pro-NHMe the ACE-PRO amide barrier is about $26 k_B T$, so neither a plain trajectory nor PT crosses it. Its reference instead adds to the 300 K Langevin dynamics a Metropolis rigid π rotation of the proline side of the amide bond; that move is an involution with unit Jacobian, so the jump is exact and the target is never deformed. It retains 2×10^6 samples over 501.0 ns at 0.2028 jump acceptance, giving 4.06×10^5 accepted *cis/trans* jumps and a blocking standard error of 0.0003 on p_{cis} .

Results comparison. Figure 14 tracks the alanine dipeptide Ramachandran surface along four accepted sharpening stages. Table 9 compares the final KLXX stage against ground truth on the same 100×100 grid over (ϕ, ψ) with the same smoothing, each surface normalized to its own most populated bin, so only per-method values are quoted. The three populations are raw bin counts on that grid, taken before any smoothing: α_L is the half-plane $\phi > 0$, and the bins with $\phi \leq 0$ are split by the sign of ψ into β /PPII and α_R . The $\phi > 0$ half-plane is the sparse region of this target. It holds 0.33% of the ground-truth mass, a factor of about 240 below β /PPII, and the KLXX Boltzmann generator places 0.0031 of its own mass there against the ground-truth 0.0033, so a region of this weight is reproduced at its own scale rather than left empty or overfilled. The three populations differ by at most 0.0018, and the mean free energies agree to $0.03 k_B T$ in every band. A single target does not by itself establish accuracy across systems.

Figure 15 compares the accepted 2,3-butanediol stage nearest $t = 0.5$ with the final stage; both rows

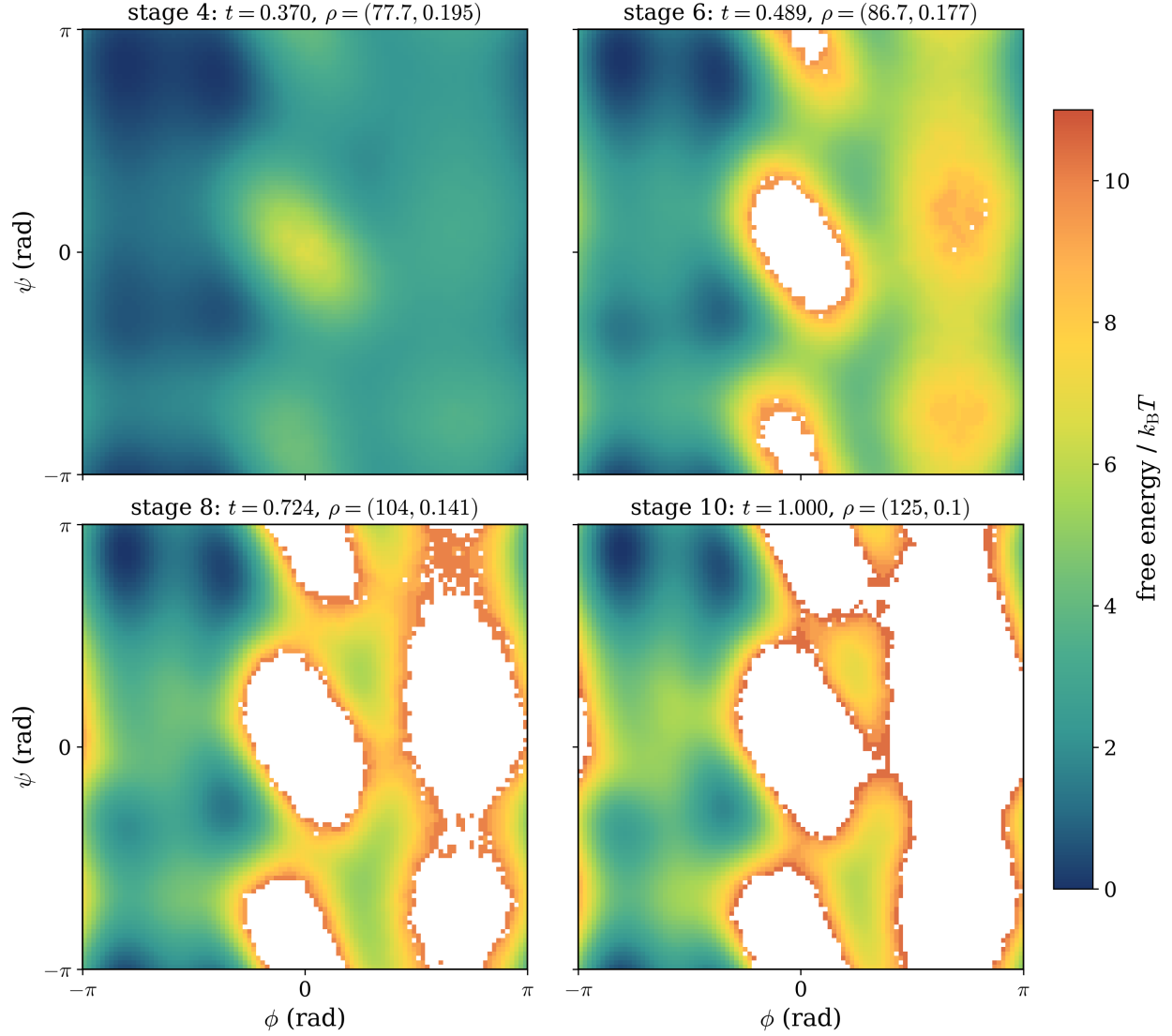


Figure 14: Alanine dipeptide Ramachandran free energy surfaces at accepted stages 4, 6, 8, and 10. Each panel contains 10^7 inference samples obtained by generating pushforward samples with the corresponding stage flow and then applying resampling, rejuvenation, sharpening, and post-sharpen rejuvenation. The panel titles give the interpolation parameter and regularization value; stage 10 reaches $t = 1$ and $\rho = (125, 0.10)$. A common colorbar reports free energy relative to the observed minimum in units of $k_B T$.

sample set	backbone basin population		
	α_L	β/PPII	α_R
KL+X π +X $_{(\hat{\pi}+\bar{\nu})/2}$	0.0031	0.7958	0.2012
ground truth	0.0033	0.7940	0.2027
sample set	mean free energy by region		
	low	high	all
KL+X π +X $_{(\hat{\pi}+\bar{\nu})/2}$	1.230	6.323	5.379
ground truth	1.230	6.293	5.355

Table 9: Alanine dipeptide at the final accepted stage against ground truth, the implicit-solvent reference data of Midgley et al. (2023), both with 10^7 samples. Basins split the (ϕ, ψ) plane by $\phi > 0$ for α_L and, for $\phi \leq 0$, by the sign of ψ . Bins are classified by the reference surface as low below $2k_B T$ and high above $3k_B T$, and restricted to bins finite in both surfaces; entries are mean free energies in units of $k_B T$.

resolve the heavy-atom rotamers and hydroxyl orientations. Table 10 compares the final KLXX stage against the 2,3-butanediol reference on the same 100×100 torsion grid with the same smoothing. The two sample sets differ in size by two orders of magnitude, and the two halves of the table are read accordingly. A basin population and a mean over occupied bins are ratios, estimated without bias at either size, so both are quoted from the full 10^7 inference set. A maximum free energy is not such a quantity: each surface is normalized to its own most populated bin, so the deepest value it can resolve is set by the bin holding a single sample. Maxima are quoted only at matched count, with the same inference set thinned to the reference’s 10^5 .

sample set	central C-C-C-C rotamer population		
	gauche−	gauche+	trans
KL+X π +X $_{(\hat{\pi}+\bar{\nu})/2}$	0.1486	0.1861	0.6652
OpenMM reference	0.1425	0.1726	0.6849
sample set	mean free energy by region		
	low	high	all
KL+X π +X $_{(\hat{\pi}+\bar{\nu})/2}$	1.229	5.126	4.137
OpenMM reference	1.259	5.353	4.309

Table 10: (2*R*,3*R*)-2,3-butanediol at the final accepted stage against an independent OpenMM PT reference, with 10^7 inference samples and 10^5 reference frames. Rotamers split the central C-C-C-C torsion at its eclipsed barriers, trans for $|\phi| \geq 2\pi/3$ and one gauche well on either side of $\phi = 0$. Bins are classified by the reference surface as low below $2k_B T$ and high above $3k_B T$, and restricted to bins finite in both surfaces; entries are mean free energies in units of $k_B T$ on the heavy-atom torsions.

KLXX and the 2,3-butanediol reference agree closely: the rotamer populations differ by at most 0.020, in the dominant trans well; the mean free energies on the heavy-atom torsions by at most $0.23k_B T$; and at matched sample count the maxima by $0.02k_B T$ on the heavy-atom torsions, 7.52 against 7.54, and $0.00k_B T$ on the hydroxyl torsions, 4.04 against 4.04. The residual trans difference is larger than the reference’s own sampling error, which the rotamer transition count places near 0.003, so it is not attributable to reference noise alone.

Ac-Pro-NHMe adds a ring degree of freedom and a cis/trans peptide torsion. Figure 16 compares the accepted stage nearest $t = 0.5$ with the final stage. Table 11 compares the final KLXX stage against the Ac-Pro-NHMe reference. The two sets agree on the qualitative picture and disagree quantitatively on the minority sector: both populate cis and trans with trans as the majority, and the two cis-to-trans free energy differences are within 0.53 kJ/mol of each other, while the cis populations differ by 0.0244.

These figures and tables diagnose the regularized KLXX sample sets and their retained stereochemical components. What this subsection does not provide is a matched loss comparison or uncertainty over

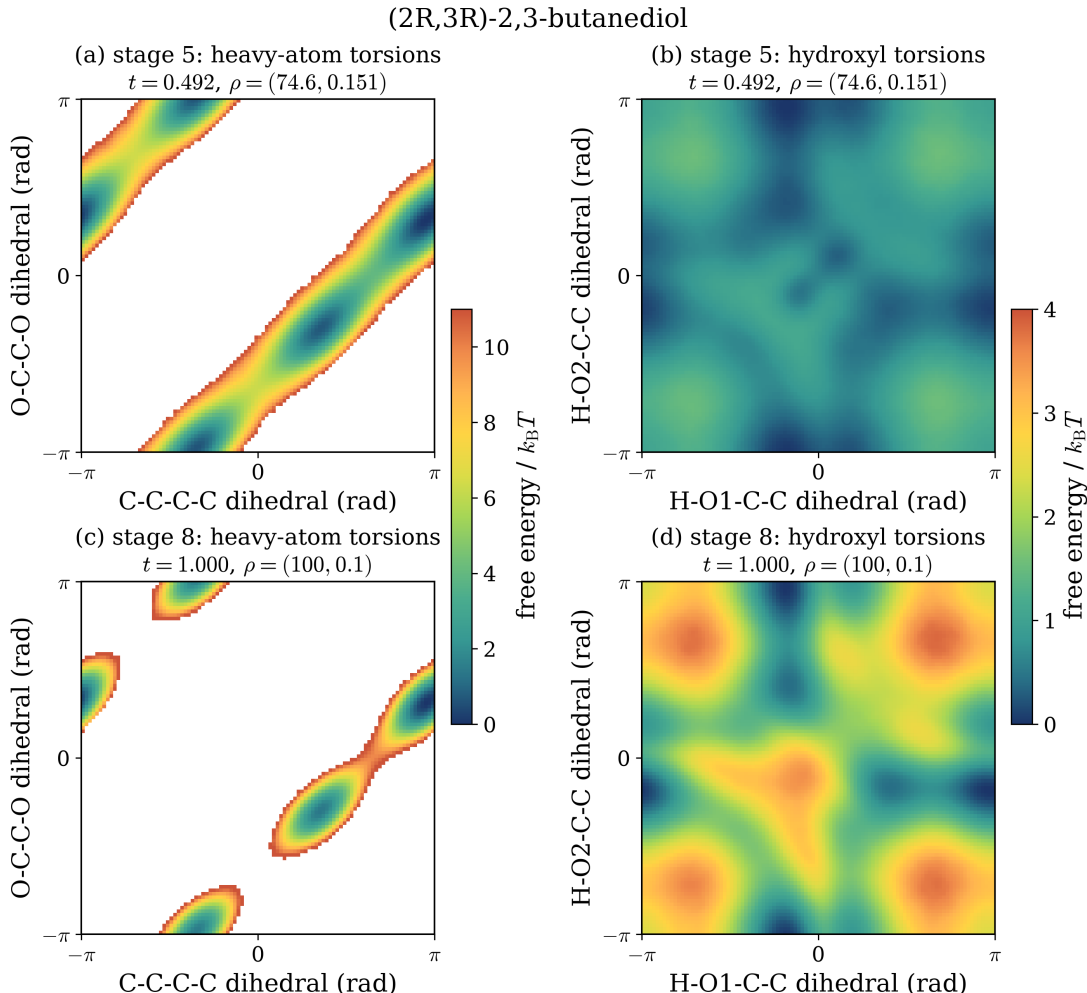


Figure 15: Conformational free energy surfaces for (2R,3R)-2,3-butanediol from 10^7 inference samples per stage. The upper row is stage 5, the accepted stage nearest $t = 0.5$, at $t = 0.492$ and $\rho = (74.6, 0.151)$; the lower row is the final stage 8 at $t = 1$ and $\rho = (100, 0.10)$. The left column shows the C-C-C-C and O-C-C-O heavy-atom torsions, and the right column shows the two hydroxyl-orientation torsions. Each column shares its scale across stages, and the two diagnostics have separate colorbars in units of $k_B T$.

independently trained generators.

Ac-Pro-NHMe residual gap. Ac-Pro-NHMe is the most difficult of the three chiral targets. Its two peptide sectors are separated by the ACE-PRO amide barrier of about $26 k_B T$, which no unbiased local move crosses, so even the reference needs a jump move built for that one bond before it can equilibrate them. The KLXX Boltzmann generator is given none of that: no reference frames, no bias potential, no elevated temperature, and no conformational prior, and it still populates both peptide sectors, keeps the S proline C_α , and orders the sectors correctly.

The residual gap is what one expects from a finite sample rather than from a systematic error in the method. By Theorem 2, the inference of Algorithm 5 is asymptotically unbiased whenever the stage weight suprema are finite: its error is then bounded at rate $N^{-1/2}$ for every bounded observable and is removed by enlarging the sample alone, so a discrepancy that survives at fixed N is the finite-sample term and should shrink as N grows. Its size is not predicted by the propagation factors of Table 8, where $\hat{F}_\Sigma$ is 176.7 for Ac-Pro-NHMe against 68.08 for alanine dipeptide, a ratio of about 2.6. Both comparisons are basin populations,

N-acetyl-L-proline N-methylamide

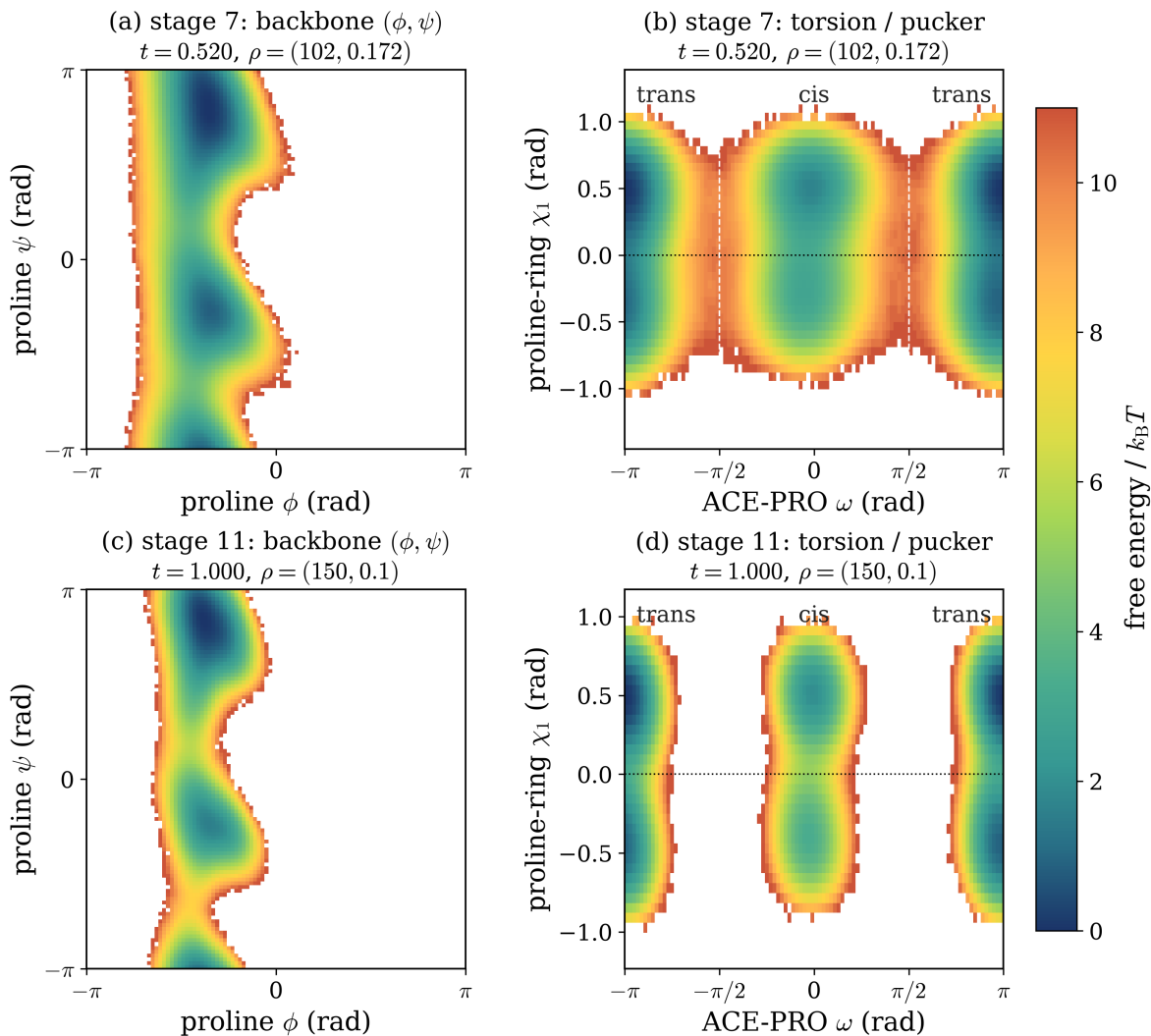


Figure 16: Conformational free energy surfaces for Ac-Pro-NHMe from 1.5×10^7 inference samples per stage. The upper row is stage 7, the accepted stage nearest $t = 0.5$, at $t = 0.520$ and $\rho = (102, 0.172)$; the lower row is the final stage 11 at $t = 1$ and $\rho = (150, 0.10)$. The left column shows the proline backbone (ϕ, ψ) surface, and the right column shows the joint ACE-PRO peptide torsion ω and proline-ring pucker χ_1 , with the cis and trans sectors marked. All four panels share one free energy scale in units of $k_B T$.

bounded observables with $\text{osc}(\varphi) = 1$, so the observable is not what separates them; what does not carry over is the factor. As Section 4.4 and Appendix C.3 state, $\hat{F}_\Sigma$ is assembled from the per-stage ESS and is not an upper bound on the constant of (28), which depends on the stage weights through their suprema. The r^{-12} Lennard-Jones core of (36) is what makes large weights available here. Every molecular run uses the regularized potential (39), in which the pair floor r softens that core and the soft cap $\mathcal{C}_e$ reduces the growth of the complete reference-relative energy to logarithmic; the weight suprema the constant depends on can nonetheless be very large, and the constant with them, while the reported factor stays small. A factor larger by 2.6 therefore does not predict a bias larger by 2.6, and the near-singular core of the potential explains the residual gap more plausibly than the dimension of the target.

Enlarging the inference set beyond 1.5×10^7 is the direct test, and it is the one we do not run. The replay cost grows in proportion to the sample count, this target is already the most expensive of the three, and pushing N far enough to resolve a difference of 0.02 in a population of 0.14 is beyond the budget of this

sample set	cis	trans	free energy
KL+X π +X $_{(\hat{\pi}+\hat{\nu})/2}$	0.1199	0.8801	4.97
OpenMM reference	0.1443	0.8557	4.44

Table 11: Ac-Pro-NHMe peptide torsion populations at the final accepted stage against an independent OpenMM reference, with 1.5×10^7 inference samples and 2×10^6 reference samples. Cis is $|\omega| < \pi/2$, counted per sample on both sides. Free energy differences are cis minus trans, in kJ/mol.

study. We leave the gap open rather than close it by tuning.

7 Conclusions

Forward KL is mass-covering for exact distributions, but its sampled loss sees only the target regions supplied to it. This gap can produce a fake ESS, transfer surrogate error to the flow, and leave target-poor pushforward regions weakly constrained. Log-ratio variation addresses these failures by comparing the exact log-density ratio at pairs of samples. The weighting measure determines which regions are compared: target samples measure density variation on reached target regions, QT samples add candidate modes, and pushforward samples expose regions occupied by the pushforward distribution. KLXX combines these comparisons while retaining the exact target as a minimizer and the flow as a tractable importance proposal.

The theory covers the generalized KLXX loss at any coefficients $(\lambda, \vartheta, \alpha, \beta)$. In FR geometry it induces a gradient flow along which the forward KL decays as e^{-t} , with no log-concavity, spectral gap, or growth assumption on π , and both log-ratio variations enter the dissipation (12) as nonpositive terms. Under a fixed biased target surrogate, Theorem 1 subtracts one nonnegative discrepancy for each variation from the stationary error bound, requiring only that two importance-weight expectations be finite. These are statements in density space, and they do not prove finite-step behavior for KLXX trained by Adam on a parametrized flow map. Theorem 2 is separate and complete: it bounds the L^2 error of the scheme of Algorithm 5 at rate $N^{-1/2}$ for every bounded observable, with a constant independent of N , so the scheme is asymptotically unbiased. The propagation factors read from that constant summarize stagewise weight degeneracy, and the reported $\hat{F}_\Sigma$ is a computable substitute rather than a bound on the error constant.

The numerical results provide the remaining evidence. The 2D and ϕ^4 ablations are consistent with the roles of the target, QT, and pushforward samples; the product test shows higher sample ESS for every loss with log-ratio variation at all sixteen tested dimensions. On the shared clock stage schedules, KLXX has higher sample ESS on all 28 shared stages and a smaller propagation factor at every reported batch size. In the single-seed alkane runs, KLXX has the smallest factor in all ten matched settings, while forward KL approaches the identity result as molecule size grows. The sharpening paths have lower KLXX factors for the larger alkanes. For NMA, glycerol, and neutral diethanolamine, KLXX also has a smaller factor than KL+X π and reaches the endpoint in fewer accepted stages. The three chiral KLXX runs reach $t = 1$, retain their specified stereochemical components, and populate multiple torsional basins in the reported conformational diagnostics. For alanine dipeptide, the final sample set matches the published equilibrium reference of Midgley et al. (2023) at 10^7 samples per side: the $\phi > 0$ half-plane, which holds 0.33% of the reference mass, is reproduced at 0.0031 against 0.0033, the three backbone basin populations differ by at most 0.0018, and the mean free energies agree to $0.03 k_B T$ in every band. For Ac-Pro-NHMe, the hardest target and the one reached with no external data and no conformational prior, the KLXX Boltzmann generator populates both peptide sectors and orders them correctly, and its cis population differs from an independent reference of our own by 0.0244. Selected alkane observables and achiral dihedral marginals agree with independent physical-potential simulations on the reported diagnostics. These findings are consistent with an explanation based on additional support information; they are not a theorem of universal dominance or a fixed-compute comparison.

The clock and molecular configurations require additional training seeds to measure run-to-run variation. The finite alkane endpoints have nearly uniform endpoint-to-physical weights on the reported sample sets, but a continuation to U would remove the remaining regularization. The additional achiral figures report selected one-dimensional marginals and do not determine joint conformational weights. Among the chiral

results, alanine dipeptide is checked against published ground truth, and 2,3-butanediol and Ac-Pro-NHMe against references of our own. The Ac-Pro-NHMe comparison agrees on the qualitative picture and leaves a quantitative gap in the cis population; the propagation bound makes that gap a finite-sample term, but resolving it would require an inference set beyond the budget of this study, so we report it open. No alternative-loss run was made for any of the three. Larger molecules and explicit solvent remain outside the completed results.

Acknowledgments

The normalizing flow computations use the JAX backend through two Python packages maintained by Xuda Ye. The packages `jflows` and `jflows-md` can be installed directly from PyPI with:

```
pip install jflows==0.5.4 jflows-md==0.5.4
```

Molecular benchmark data were generated with OpenMM and AmberTools, and the alanine dipeptide reference data were taken from the FAB study (Midgley et al., 2023). The source files are available at:

<https://github.com/xuda-ye-math/KLXX>

The GPU-accelerated runs used one MSI GeForce RTX 5090 32G Gaming Trio OC with 32 GB of VRAM. The card ran at a 600 W power limit with clock offsets of +250 MHz on the core and +2000 MHz on the memory. All reported wall times correspond to this overclocked configuration.

References

- S. Agapiou, O. Papaspiliopoulos, D. Sanz-Alonso, and A. M. Stuart. Importance sampling: intrinsic dimension and computational cost. *Statistical Science*, 32(3):405–431, 2017.
- T. Akhound-Sadegh, J. Rector-Brooks, A. J. Bose, S. Mittal, P. Lemos, C.-H. Liu, M. Sendera, S. Ravanbakhsh, G. Gidel, Y. Bengio, N. Malkin, and A. Tong. Iterated denoising energy matching for sampling from Boltzmann densities. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 760–786. PMLR, 2024.
- T. Akhound-Sadegh, J. Lee, A. J. Bose, V. De Bortoli, A. Doucet, M. M. Bronstein, D. Beaini, S. Ravanbakhsh, K. Neklyudov, and A. Tong. Progressive inference-time annealing of diffusion models for sampling from Boltzmann densities. In *Advances in Neural Information Processing Systems*, 2025.
- M. S. Albergo and E. Vanden-Eijnden. NETS: A non-equilibrium transport sampler. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 1026–1055. PMLR, 2025.
- L. Ambrosio, N. Gigli, and G. Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, 2 edition, 2008.
- M. Arbel, A. Matthews, and A. Doucet. Annealed flow transport Monte Carlo. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 318–330. PMLR, 2021.
- D. Bakry, I. Gentil, and M. Ledoux. *Analysis and Geometry of Markov Diffusion Operators*, volume 348 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2014.
- J. Berner, L. Richter, and K. Ullrich. An optimal control perspective on diffusion-based generative modeling. *Transactions on Machine Learning Research*, 2024.

- D. A. Case, H. M. Aktulga, K. Belfon, D. S. Cerutti, G. A. Cisneros, V. W. D. Cruzeiro, N. Forouzes, T. J. Giese, A. W. Götz, H. Gohlke, S. Izadi, K. Kasavajhala, M. C. Kaymak, E. King, T. Kurtzman, T. S. Lee, P. Li, J. Liu, T. Luchko, R. Luo, M. Manathunga, M. R. Machado, H. M. Nguyen, K. A. O’Hearn, A. V. Onufriev, F. Pan, S. Pantano, R. Qi, A. Rahnamoun, D. R. Roe, A. Roitberg, C. Sagui, S. Schott-Verdugo, J. Shen, C. L. Simmerling, N. R. Skrynnikov, J. Smith, J. Swails, R. C. Walker, J. Wang, H. Wei, R. M. Wolf, X. Wu, Y. Xue, D. M. York, S. Zhao, and P. A. Kollman. AmberTools. *Journal of Chemical Information and Modeling*, 63(20):6183–6191, 2023. doi: 10.1021/acs.jcim.3c01153.
- S. Chatterjee and P. Diaconis. The sample size required in importance sampling. *The Annals of Applied Probability*, 28(2):1099–1135, 2018.
- O. Chehab, A. Korba, A. Stromme, and A. Vacher. Provable convergence and limitations of geometric tempering for Langevin dynamics. In *International Conference on Learning Representations*, volume 2025, pages 5451–5493, 2025.
- Y. Chen, J. Chen, J. Dong, J. Peng, and Z. Wang. Accelerating nonconvex learning via replica exchange Langevin diffusion. In *International Conference on Learning Representations*, 2019.
- W. D. Cornell, P. Cieplak, C. I. Bayly, I. R. Gould, K. M. Merz, D. M. Ferguson, D. C. Spellmeyer, T. Fox, J. W. Caldwell, and P. A. Kollman. A second generation force field for the simulation of proteins, nucleic acids, and organic molecules. *Journal of the American Chemical Society*, 117(19):5179–5197, 1995.
- P. Del Moral. *Feynman–Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Springer, 2004.
- P. Del Moral, A. Doucet, and A. Jasra. Sequential Monte Carlo samplers. *Journal of the Royal Statistical Society: Series B*, 68(3):411–436, 2006.
- W. Deng, Q. Feng, L. Gao, F. Liang, and G. Lin. Non-convex learning via replica exchange stochastic gradient MCMC. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 2474–2483. PMLR, 2020.
- W. Deng, Q. Feng, G. Karagiannis, G. Lin, and F. Liang. Accelerating convergence of replica exchange stochastic gradient MCMC via variance reduction. In *International Conference on Learning Representations*, 2021.
- J. Dong and X. T. Tong. Replica exchange for non-convex optimization. *Journal of Machine Learning Research*, 22(173):1–59, 2021.
- A. Doucet, N. De Freitas, and N. Gordon. An introduction to sequential Monte Carlo methods. In *Sequential Monte Carlo methods in practice*, pages 3–14. Springer, 2001.
- S. Duane, A. D. Kennedy, B. J. Pendleton, and D. Roweth. Hybrid Monte Carlo. *Physics Letters B*, 195(2):216–222, 1987. doi: 10.1016/0370-2693(87)91197-X.
- C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios. Neural spline flows. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- P. Eastman, J. Swails, J. D. Chodera, R. T. McGibbon, Y. Zhao, K. A. Beauchamp, L.-P. Wang, A. C. Simmonett, M. P. Harrigan, C. D. Stern, R. P. Wiewiora, B. R. Brooks, and V. S. Pande. OpenMM 7: rapid development of high performance algorithms for molecular dynamics. *PLOS Computational Biology*, 13(7):e1005659, 2017. doi: 10.1371/journal.pcbi.1005659.
- V. Elvira, L. Martino, and C. P. Robert. Rethinking the effective sample size. *International Statistical Review*, 90(3):525–550, 2022.
- L. Felardos, J. Hénin, and G. Charpiat. Designing losses for data-free training of normalizing flows on Boltzmann distributions. *arXiv preprint arXiv:2301.05475*, 2023.

- D. Frenkel and B. Smit. *Understanding Molecular Simulation: From Algorithms to Applications*, volume 1 of *Computational Science Series*. Academic Press, 2 edition, 2002.
- M. Gabrié, G. M. Rotskoff, and E. Vanden-Eijnden. Adaptive Monte Carlo augmented with normalizing flows. *Proceedings of the National Academy of Sciences*, 119(10):e2109420119, 2022.
- C. J. Geyer. Markov chain Monte Carlo maximum likelihood. In *Computing Science and Statistics, Proceedings of the 23rd Symposium on the Interface*, pages 156–163. Interface Foundation of North America, 1991.
- W. K. Hastings. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57(1):97–109, 1970.
- A. J. Havens, B. K. Miller, B. Yan, C. Domingo-Enrich, A. Sriram, D. S. Levine, B. M. Wood, B. Hu, B. Amos, B. Karrer, X. Fu, G.-H. Liu, and R. T. Q. Chen. Adjoint sampling: highly scalable diffusion samplers via adjoint matching. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 22204–22237. PMLR, 2025.
- J. He, Y. Du, F. Vargas, D. Zhang, S. Padhy, R. OuYang, C. Gomes, and J. M. Hernández-Lobato. No trick, no treat: pursuits and challenges towards simulation-free training of neural samplers. In *ICLR 2025 Workshop on Frontiers in Probabilistic Inference: Learning meets Sampling*, 2025.
- W. G. Hoover. Canonical dynamics: Equilibrium phase-space distributions. *Physical Review A*, 31(3):1695–1697, 1985. doi: 10.1103/PhysRevA.31.1695.
- K. Hukushima and K. Nemoto. Exchange Monte Carlo method and application to spin glass simulations. *Journal of the Physical Society of Japan*, 65(6):1604–1608, 1996.
- A. Jakalian, D. B. Jack, and C. I. Bayly. Fast, efficient generation of high-quality atomic charges. AM1-BCC model: II. parameterization and validation. *Journal of Computational Chemistry*, 23(16):1623–1641, 2002. doi: 10.1002/jcc.10128.
- J. Kästner. Umbrella sampling. *WIREs Computational Molecular Science*, 1(6):932–942, 2011. doi: 10.1002/wcms.66.
- D. P. Kingma and P. Dhariwal. Glow: Generative flow with invertible 1x1 convolutions. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- L. Klein and F. Noé. Transferable Boltzmann generators. In *Advances in Neural Information Processing Systems*, volume 37, pages 45281–45314, 2024.
- L. Klein, A. Krämer, and F. Noé. Equivariant flow matching. In *Advances in Neural Information Processing Systems*, volume 36, pages 59886–59910, 2023.
- J. Köhler, L. Klein, and F. Noé. Equivariant flows: exact likelihood generative learning for symmetric densities. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 5361–5370. PMLR, 2020.
- D. P. Landau and K. Binder. *A Guide to Monte Carlo Simulations in Statistical Physics*. Cambridge University Press, 4 edition, 2014. doi: 10.1017/CBO9781139696463.
- P. Langevin. Sur la théorie du mouvement brownien. *Comptes rendus hebdomadaires des séances de l’Académie des sciences*, 146:530–533, 1908.
- B. Leimkuhler and C. Matthews. *Molecular Dynamics: With Deterministic and Stochastic Numerical Methods*, volume 39 of *Interdisciplinary Applied Mathematics*. Springer, 2015. doi: 10.1007/978-3-319-16375-8.
- D. C. Liu and J. Nocedal. On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45:503–528, 1989. doi: 10.1007/BF01589116.
- J. S. Liu. *Monte Carlo strategies in scientific computing*, volume 10. Springer, 2001.

- A. Matthews, M. Arbel, D. J. Rezende, and A. Doucet. Continual repeated annealed flow transport monte carlo. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 15196–15219. PMLR, 2022.
- N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953.
- L. I. Midgley, V. Stimper, G. N. Simm, B. Schölkopf, and J. M. Hernández-Lobato. Flow annealed importance sampling bootstrap. In *International Conference on Learning Representations*, 2023.
- M. F. Naeem, S. J. Oh, Y. Uh, Y. Choi, and J. Yoo. Reliable fidelity and diversity metrics for generative models. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7176–7185. PMLR, 2020.
- C. A. Naesseth, F. Lindsten, and D. M. Blei. Markovian score climbing: variational inference with $\text{KL}(p||q)$. In *Advances in Neural Information Processing Systems*, volume 33, pages 15499–15510, 2020.
- R. M. Neal. Annealed importance sampling. *Statistics and Computing*, 11(2):125–139, 2001.
- F. Noé, S. Olsson, J. Köhler, and H. Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.
- S. Nosé. A unified formulation of the constant temperature molecular dynamics methods. *The Journal of Chemical Physics*, 81(1):511–519, 1984. doi: 10.1063/1.447334.
- N. Nüsken and L. Richter. Solving high-dimensional Hamilton–Jacobi–Bellman PDEs using neural networks: perspectives from controlled diffusions and measures on path space. *Partial Differential Equations and Applications*, 2:48, 2021.
- A. V. Onufriev, D. Bashford, and D. A. Case. Exploring protein native states and large-scale conformational changes with a modified generalized Born model. *Proteins: Structure, Function, and Bioinformatics*, 55(2):383–394, 2004. doi: 10.1002/prot.20033.
- F. Otto and C. Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis*, 173(2):361–400, 2000.
- G. Papamakarios, T. Pavlakou, and I. Murray. Masked autoregressive flow for density estimation. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- T. Popoviciu. Sur les équations algébriques ayant toutes leurs racines réelles. *Mathematica*, 9:129–145, 1935.
- D. Rehman, O. Davis, J. Lu, J. Tang, M. M. Bronstein, Y. Bengio, A. Tong, and A. J. Bose. Efficient regression-based training of normalizing flows for Boltzmann generators. In *International Conference on Learning Representations*, 2026.
- D. J. Rezende and S. Mohamed. Variational inference with normalizing flows. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1530–1538. PMLR, 2015.
- L. Richter and J. Berner. Improved sampling via learned diffusions. In *International Conference on Learning Representations*, 2024.
- C. P. Robert and G. Casella. *Monte Carlo Statistical Methods*. Springer Texts in Statistics. Springer, 2 edition, 2004. doi: 10.1007/978-1-4757-4145-2.
- G. O. Roberts and R. L. Tweedie. Exponential convergence of Langevin distributions and their discrete approximations. *Bernoulli*, 2(4):341–363, 1996. doi: 10.2307/3318418.
- M. Schaefer and M. Karplus. A comprehensive analytical treatment of continuum electrostatics. *The Journal of Physical Chemistry*, 100(5):1578–1599, 1996. doi: 10.1021/jp9521621.

- H. Schopmans and P. Friederich. Temperature-annealed Boltzmann generators. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 53467–53500. PMLR, 2025. arXiv:2501.19077.
- H. Schopmans, C. von Klitzing, and P. Friederich. Efficient training of Boltzmann generators using off-policy log-dispersion regularization. *arXiv preprint arXiv:2602.03729*, 2026.
- J. Skilling. Nested sampling for general Bayesian computation. *Bayesian Analysis*, 1(4):833–859, 2006.
- N. Surjanovic, S. Syed, A. Bouchard-Côté, and T. Campbell. Parallel tempering with a variational reference. In *Advances in Neural Information Processing Systems*, volume 35, pages 565–577, 2022.
- R. H. Swendsen and J.-S. Wang. Replica Monte Carlo simulation of spin-glasses. *Physical Review Letters*, 57(21):2607–2609, 1986.
- S. Syed, A. Bouchard-Côté, G. Deligiannidis, and A. Doucet. Non-reversible parallel tempering: a scalable highly parallel MCMC scheme. *Journal of the Royal Statistical Society Series B*, 84(2):321–350, 2022.
- C. B. Tan, A. J. Bose, C. Lin, L. Klein, M. M. Bronstein, and A. Tong. Scalable equilibrium sampling with sequential Boltzmann generators. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 58467–58498. PMLR, 2025.
- G. M. Torrie and J. P. Valleau. Nonphysical sampling distributions in Monte Carlo free-energy estimation: Umbrella sampling. *Journal of Computational Physics*, 23(2):187–199, 1977. doi: 10.1016/0021-9991(77)90121-8.
- L. Vaitl, K. A. Nicoli, S. Nakajima, and P. Kessel. Gradients should stay on path: better estimators of the reverse- and forward-KL divergence for normalizing flows. *Machine Learning: Science and Technology*, 2022.
- F. Vargas, W. Grathwohl, and A. Doucet. Denoising diffusion samplers. In *International Conference on Learning Representations*, 2023.
- C. Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, 2009.
- C. von Klitzing, D. Blessing, H. Schopmans, P. Friederich, and G. Neumann. Learning Boltzmann generators via constrained mass transport. In *International Conference on Learning Representations*, 2026.
- M. J. Wainwright and M. I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008.
- J. Wang, R. M. Wolf, J. W. Caldwell, P. A. Kollman, and D. A. Case. Development and testing of a general Amber force field. *Journal of Computational Chemistry*, 25(9):1157–1174, 2004. doi: 10.1002/jcc.20035.
- P. Wirnsberger, A. J. Ballard, G. Papamakarios, S. Abercrombie, S. Racanière, A. Pritzel, D. J. Rezende, and C. Blundell. Targeted free energy estimation via learned mappings. *The Journal of Chemical Physics*, 153(14), 2020.
- D. Woo and S. Ahn. Iterated energy-based flow matching for sampling from Boltzmann densities. *arXiv preprint arXiv:2408.16249*, 2024.
- D. B. Woodard, S. C. Schmidler, and M. Huber. Conditions for rapid mixing of parallel and simulated tempering on multimodal distributions. *The Annals of Applied Probability*, 19(2):617–640, 2009.
- H. Wu, J. Köhler, and F. Noé. Stochastic normalizing flows. In *Advances in Neural Information Processing Systems*, volume 33, pages 5933–5944, 2020.
- L. Zhang, P. Potapchik, J. He, Y. Du, A. Doucet, F. Vargas, H.-D. Dau, and S. Syed. Accelerated parallel tempering via neural transports. In *International Conference on Learning Representations*, 2026.
- Q. Zhang and Y. Chen. Path integral sampler: A stochastic control approach for sampling. In *International Conference on Learning Representations*, 2022.

A Proof of Fisher–Rao gradient flow convergence

This appendix derives the Fisher–Rao gradient flow (10) of the generalized KLXX loss (19) and proves the dissipation identity (12) and the exponential decay (13) stated in Section 2.2. We view ν as a generic probability density evolving under a Riemannian metric on the space of densities (Ambrosio et al., 2008; Villani, 2009) and ask how fast ν_t approaches the target in $\text{KL}(\pi\|\nu_t)$. Throughout, we assume the densities are positive and sufficiently regular, and that the expectations displayed below are finite.

The FR gradient of a functional $\mathcal{F}[\nu]$ and the flow it generates are given by

$$\text{grad}^{\text{FR}} \mathcal{F}(\nu) = \nu \left(\frac{\delta \mathcal{F}}{\delta \nu} - \mathbb{E}_\nu \left[\frac{\delta \mathcal{F}}{\delta \nu} \right] \right), \quad \partial_t \nu_t = -\text{grad}^{\text{FR}} \mathcal{F}(\nu_t). \quad (43)$$

The mean subtraction enforces $\int_{\mathbb{R}^d} \text{grad}^{\text{FR}} \mathcal{F} dx = 0$ at unit mass, so the flow preserves probability mass. In the generalized KLXX loss (19), the forward KL contributes a local term: its first variation is $\delta \text{KL}(\pi\|\nu)/\delta \nu = -w$, and $\mathbb{E}_\nu[w] = \int_{\mathbb{R}^d} \pi dx = 1$, so (43) gives

$$\text{grad}^{\text{FR}} [\text{KL}(\pi\|\nu)](x) = \nu(x) - \pi(x). \quad (44)$$

The two log-ratio variations are handled at once, because neither weighting density is varied and nothing in the calculation uses that the weighting density and the target coincide.

Lemma 1 *Let ω be a probability density that does not depend on ν , and let S^ω be its nonlocal correlation (9). Then*

$$\frac{\delta X_\omega(\pi\|\nu)}{\delta \nu}(x) = -2 \frac{\omega(x)}{\nu(x)} S^\omega(x), \quad \text{grad}^{\text{FR}} [X_\omega(\pi\|\nu)](x) = -2 \omega(x) S^\omega(x). \quad (45)$$

Proof. Since $z = \log \pi - \log \nu$, the variation of the inner difference at an evaluation point r is

$$\frac{\delta}{\delta \nu(r)} [z(x) - z(y)] = -\frac{\delta(r-x)}{\nu(x)} + \frac{\delta(r-y)}{\nu(y)},$$

with δ the Dirac distribution, so integration against $\delta(r-x)$ evaluates the remaining integrand at $x=r$. Because log is increasing, $\text{sgn}(z(x) - z(y)) = \text{sgn}(w(x) - w(y))$. Differentiating (5) and using that ω is fixed,

$$\begin{aligned} \frac{\delta X_\omega(\pi\|\nu)}{\delta \nu}(r) &= \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x) \omega(y) \text{sgn}(w(x) - w(y)) \left[-\frac{\delta(r-x)}{\nu(x)} + \frac{\delta(r-y)}{\nu(y)} \right] dx dy \\ &= -2 \frac{\omega(r)}{\nu(r)} \int_{\mathbb{R}^d} \omega(y) \text{sgn}(w(r) - w(y)) dy, \end{aligned}$$

the two terms being equal after exchanging x and y and using the antisymmetry of the sign. That antisymmetry also gives

$$\mathbb{E}_\omega [S^\omega] = \mathbb{E}_{x,y \sim \omega} [\text{sgn}(w(x) - w(y))] = 0, \quad (46)$$

so the mean subtraction in (43) contributes $\mathbb{E}_\nu [-2(\omega/\nu)S^\omega] = -2 \int_{\mathbb{R}^d} \omega S^\omega dx = 0$, and the FR gradient is $\nu \cdot (-2(\omega/\nu)S^\omega)$. $\square$

The mixture $\xi = \alpha \hat{\pi} + \beta \bar{\nu}$ depends on ν through $\bar{\nu}$, but it is detached from the gradient calculation, so the lemma applies at $\omega = \xi$ as it does at $\omega = \pi$. Adding (44) to the two variations, the FR gradient of the generalized loss is

$$\text{grad}^{\text{FR}} [\mathcal{L}](x) = \nu(x) - \pi(x) - 2\lambda \pi(x) S^\pi(x) - 2\vartheta \xi(x) S^\xi(x),$$

which is the flow (10). It is a nonlocal differential equation: its right-hand side at x needs the entire profile of the importance weight, under π through S^π and under ξ through S^ξ . Integrating it and using (46) at both weighting densities, the two correlation terms drop out and

$$\frac{d}{dt} \int_{\mathbb{R}^d} \nu_t(x) dx = 1 - \int_{\mathbb{R}^d} \nu_t(x) dx,$$

so a unit mass at $t=0$ is preserved for all t . Mass neutrality of the correlation terms follows from antisymmetry alone and requires no normalization. The condition $\alpha + \beta = 1$ enters elsewhere: it makes ξ a probability density, hence $S^\xi \in [-1, 1]$, and it is what lets the mixture integrals below be read as expectations under ξ .

Remark 8 The integrand $|z(x) - z(y)|$ has no derivative on the tie set $\{z(x) = z(y)\}$, where its subdifferential is $[-1, 1]$ and the convention $\text{sgn}(0) = 0$ selects the element of least norm. This costs differentiability of $\nu \mapsto X_\omega(\pi \| \nu)$ only where the tie set carries positive $\omega \otimes \omega$ mass, the diagonal alone being null; it does so at $\nu = \pi$, where $w \equiv 1$ and every pair ties. Writing $u = \log \nu$ makes $z(x) - z(y)$ affine in u , so X_ω is convex in u and the second identity of (45) is a subgradient there. Accordingly (10) is a differential inclusion, and we still write gradient flow for simplicity.

We take $\text{KL}(\pi \| \nu_t)$ as the error and differentiate it along (10). The sign of the contribution each log-ratio variation makes to that derivative is settled by one identity, which the mixture term needs at a weighting density other than the target.

Lemma 2 Let ω be a probability density with $\mathbb{E}_\omega[w] < \infty$. Then

$$\mathbb{E}_{x \sim \omega} [w(x) S^\omega(x)] = \frac{1}{2} \mathbb{E}_{x, y \sim \omega} [|w(x) - w(y)|] \geq 0, \quad (47)$$

with equality if and only if w is ω -almost-everywhere constant.

Proof. By the definition (9) of S^ω ,

$$\mathbb{E}_{x \sim \omega} [w(x) S^\omega(x)] = \mathbb{E}_{x, y \sim \omega} [w(x) \text{sgn}(w(x) - w(y))],$$

which is finite because $\mathbb{E}_\omega[w] < \infty$. Symmetrizing in $x \leftrightarrow y$ and using the antisymmetry of the sign,

$$\mathbb{E}_{x \sim \omega} [w(x) S^\omega(x)] = \frac{1}{2} \mathbb{E}_{x, y \sim \omega} [(w(x) - w(y)) \text{sgn}(w(x) - w(y))] = \frac{1}{2} \mathbb{E}_{x, y \sim \omega} [|w(x) - w(y)|].$$

The last expression vanishes exactly when $w(x) = w(y)$ for $\omega \otimes \omega$ -almost every pair. $\square$

Differentiating $\text{KL}(\pi \| \nu_t)$ along (10) with $\delta \text{KL} / \delta \nu = -w_t$ and the correlations (11), we obtain the dissipation relation as in (12):

$$\begin{aligned} \frac{d}{dt} \text{KL}(\pi \| \nu_t) &= - \int_{\mathbb{R}^d} w_t [\pi - \nu_t + 2\lambda \pi S_t^\pi + 2\vartheta \xi S_t^\xi] dx \\ &= - \left[\int_{\mathbb{R}^d} \frac{\pi^2}{\nu_t} dx - 1 \right] - 2\lambda \mathbb{E}_{x \sim \pi} [w_t S_t^\pi] - 2\vartheta \mathbb{E}_{x \sim \xi} [w_t S_t^\xi] \\ &= -\chi^2(\pi \| \nu_t) - \lambda \mathbb{E}_{x, y \sim \pi} [|w_t(x) - w_t(y)|] - \vartheta \mathbb{E}_{x, y \sim \xi} [|w_t(x) - w_t(y)|], \end{aligned}$$

where the last equality holds by applying Lemma 2 at $\omega = \pi$ and $\omega = \xi$, respectively.

The elementary inequality $u \log u \leq u^2 - u$ for $u > 0$, applied at $u = w_t$ and integrated against ν_t , gives $\chi^2(\pi \| \nu_t) \geq \text{KL}(\pi \| \nu_t)$. Discarding the two nonpositive variation terms of (12) therefore leaves

$$\frac{d}{dt} \text{KL}(\pi \| \nu_t) \leq -\chi^2(\pi \| \nu_t) \leq -\text{KL}(\pi \| \nu_t),$$

and Grönwall's inequality yields the exponential decay (13), with no structural assumption on π and uniformly in $(\lambda, \vartheta, \alpha, \beta)$. This contrasts with the Wasserstein-2 geometry, where the forward KL is not displacement convex and an analogous exponential rate is unavailable without further assumptions.

B Proof of accuracy under biased target surrogate

This appendix derives the stationarity identity (16) and proves Theorem 1 on the accuracy of the biased KLXX loss $\tilde{\mathcal{L}}$ in (14) under a biased target surrogate $\tilde{\pi}$. Applying (44) with $\tilde{\pi}$ in place of π to the first term of (14), and Lemma 1 to the other two at $\omega = \tilde{\pi}$ and at $\omega = \xi$, gives

$$\text{grad}^{\text{FR}}[\tilde{\mathcal{L}}](x) = \nu(x) - \tilde{\pi}(x)(1 + 2\lambda S_\star^{\tilde{\pi}}(x)) - 2\vartheta \xi(x) S_\star^\xi(x).$$

Hence the stationary density $\nu_\star$ of $\tilde{\mathcal{L}}$ satisfies (16), that is

$$\nu_\star(x) = \tilde{\pi}(x)(1 + 2\lambda S_\star^{\tilde{\pi}}(x)) + 2\vartheta \xi(x) S_\star^\xi(x).$$

Integrating this identity and using (46) at $\omega = \tilde{\pi}$ and at $\omega = \xi$,

$$\int_{\mathbb{R}^d} \nu_\star(x) dx = \int_{\mathbb{R}^d} \tilde{\pi}(x) dx + 2\lambda \int_{\mathbb{R}^d} \tilde{\pi}(x) S_\star^{\tilde{\pi}}(x) dx + 2\vartheta \int_{\mathbb{R}^d} \xi(x) S_\star^\xi(x) dx = 1,$$

so $\nu_\star$ has unit mass.

Proof. Define the ratio $\varrho := \nu_\star/\tilde{\pi}$, positive because both densities are, so that (16) divided by $\tilde{\pi}$ reads

$$\varrho(x) = 1 + 2\lambda S_\star^{\tilde{\pi}}(x) + 2\vartheta \frac{\xi(x)}{\tilde{\pi}(x)} S_\star^\xi(x). \quad (48)$$

Multiply (48) by $\tilde{\pi} w_\star$, with $w_\star = \pi/\nu_\star$, and integrate. In the last term $\tilde{\pi} \cdot (\xi/\tilde{\pi}) = \xi$, so that integral is carried by the mixture rather than by the surrogate; subtracting $\int_{\mathbb{R}^d} \tilde{\pi} w_\star dx$ from both sides,

$$\int_{\mathbb{R}^d} \tilde{\pi} w_\star (\varrho - 1) dx = 2\lambda \int_{\mathbb{R}^d} \tilde{\pi} w_\star S_\star^{\tilde{\pi}} dx + 2\vartheta \int_{\mathbb{R}^d} \xi w_\star S_\star^\xi dx. \quad (49)$$

Both correlations are averages of signs under probability densities, so $|S_\star^{\tilde{\pi}}| \leq 1$ and $|S_\star^\xi| \leq 1$, whence

$$\int_{\mathbb{R}^d} \tilde{\pi} w_\star |S_\star^{\tilde{\pi}}| dx \leq \mathbb{E}_{\tilde{\pi}}[w_\star] < \infty, \quad \int_{\mathbb{R}^d} \xi w_\star |S_\star^\xi| dx \leq \mathbb{E}_\xi[w_\star] < \infty$$

by the two hypotheses of Theorem 1. The right side of (49) therefore converges absolutely.

It remains to relate $\text{KL}(\pi\|\tilde{\pi})$ to $\text{KL}(\pi\|\nu_\star)$. Taking logarithms in $\nu_\star = \tilde{\pi}\varrho$ gives $\log(\pi/\tilde{\pi}) = \log(\pi/\nu_\star) + \log \varrho$, and integrating against π ,

$$\text{KL}(\pi\|\tilde{\pi}) = \text{KL}(\pi\|\nu_\star) + \int_{\mathbb{R}^d} \pi \log \varrho dx.$$

In the last integral $\pi = \nu_\star w_\star = \tilde{\pi} \varrho w_\star$, so

$$\text{KL}(\pi\|\tilde{\pi}) = \text{KL}(\pi\|\nu_\star) + \int_{\mathbb{R}^d} \tilde{\pi} w_\star \varrho \log \varrho dx. \quad (50)$$

The inequality $\varrho \log \varrho \geq \varrho - 1$ for $\varrho > 0$ bounds the last integral of (50) below by the left side of (49), which is finite, so both summands of (50) are well defined in $(-\infty, +\infty]$ and

$$\text{KL}(\pi\|\nu_\star) \leq \text{KL}(\pi\|\tilde{\pi}) - \int_{\mathbb{R}^d} \tilde{\pi} w_\star (\varrho - 1) dx.$$

Substituting (48) into the remaining integral returns the right side of (49), and Lemma 2, applied at $\omega = \tilde{\pi}$ and at $\omega = \xi$ with $\nu = \nu_\star$, absorbs the factor two in each term, giving (18). $\square$

Identity (49) also settles how much (18) can subtract. Since $\tilde{\pi}\varrho = \nu_\star$ and $\nu_\star w_\star = \pi$, the left side of (49) evaluates exactly, and Lemma 2 turns its right side into the two discrepancies, so the two subtracted terms sum to $1 - \mathbb{E}_{\tilde{\pi}}[w_\star]$, the value quoted after Theorem 1. Both are nonnegative, which forces $\mathbb{E}_{\tilde{\pi}}[w_\star] \leq 1$, and $\mathbb{E}_{\tilde{\pi}}[w_\star] > 0$ then leaves the subtraction strictly below one nat. Equality holds at $\lambda = \vartheta = 0$, and when either coefficient is positive only in the degenerate case $\tilde{\pi} = \nu_\star = \pi$: the equality case of Lemma 2 makes $w_\star$ constant $\tilde{\pi}$ -almost everywhere when $\lambda > 0$ and ξ -almost everywhere when $\vartheta > 0$, both densities are positive, and $\int \pi = \int \nu_\star = 1$ then forces $w_\star \equiv 1$, whence $S_\star^{\tilde{\pi}} = S_\star^\xi = 0$ and (16) gives $\nu_\star = \tilde{\pi} = \pi$.

Remark 9 *The two finiteness hypotheses of Theorem 1 are mild, and one sufficient condition is worth recording. Suppose the mixture is dominated by the surrogate, $\xi \leq c\tilde{\pi}$ pointwise for some constant c , and $\lambda + c\vartheta < \frac{1}{2}$; note that $\xi \leq c\tilde{\pi}$ implies $c \geq 1$, since both densities integrate to one. Both correlations lie in $[-1, 1]$, being averages of signs under probability densities, so (48) gives $\varrho \geq 1 - 2\lambda - 2c\vartheta > 0$, that is $\nu_\star \geq (1 - 2\lambda - 2c\vartheta)\tilde{\pi} > 0$. Dividing π by this bound and integrating gives $\mathbb{E}_{\tilde{\pi}}[w_\star] \leq (1 - 2\lambda - 2c\vartheta)^{-1}$ and, using the domination once more, $\mathbb{E}_\xi[w_\star] \leq c(1 - 2\lambda - 2c\vartheta)^{-1}$.*

C Proof of propagation factor analysis

Section 4.4 states the error bound and the propagation factors read from it. This appendix proves Theorem 2 for the scheme stated there, with a general rejuvenation kernel; Appendix C.3 examines the factor the experiments report.

Throughout, the accepted schedule and the trained maps $G_1, \dots, G_K$ are held fixed: the theorem describes inference with a trained Boltzmann generator, replayed on a fresh particle set with no further selection or training. The proof keeps every stage in its own equivalence class and never leaves the triple (π_k, ν_k, w_k) . The identities that carry the error across a stage, (54) below, ask only how one stage relates to the next, so adding a kernel rebuilds no object in the proof. That the rejuvenation costs nothing rests on two things: the invariance of Q_k , and the inability of a Markov kernel to widen the range of a bounded function.

C.1 Notation and what one stage does

We write $\|\cdot\|_2$ for the L^2 norm over all the randomness of the scheme, namely the N initial draws and, at every stage, the resampling draws and the kernel draws. The Markov kernel Q_k acts on a bounded measurable function from one side and on a distribution from the other,

$$Q_k h(x) := \int_{\mathbb{R}^d} h(y) Q_k(x, dy), \quad (\pi_k Q_k)(A) := \int_{\mathbb{R}^d} Q_k(x, A) \pi_k(dx), \quad (51)$$

so that the invariance assumed of Q_k reads $\pi_k Q_k = \pi_k$. Each G_k is a diffeomorphism of $\mathbb{R}^d$ onto itself, so $\nu_k = (G_k^{-1})_{\#} \pi_{k-1}$ has a density whenever π_{k-1} does, and $w_k = \pi_k / \nu_k$ is a ratio of densities. That w_k is well defined and positive makes ν_k and π_k equivalent, so each stage carries an equivalence class of its own, that of π_0 at $k = 0$, and every essential quantity below is taken with respect to the class of its stage. No class is shared between stages and none is needed: (54) below is what moves such a quantity from one stage to the next. The proof follows the particles where they live, one stage at a time, so we name the two empirical measures each stage produces:

$$\pi_k^N := \frac{1}{N} \sum_{i=1}^N \delta_{X_k^i} \quad (k = 0, 1, \dots, K), \quad \nu_k^N := \frac{1}{N} \sum_{i=1}^N \delta_{Y_k^i} \quad (k = 1, \dots, K). \quad (52)$$

The particles enter stage k distributed approximately as π_{k-1} and are counted by π_{k-1}^N ; the stage map moves them, and between that move and the resampling they sit at the pushforward positions Y_k^i , distributed approximately as ν_k and counted by ν_k^N . Thus π_k^N collects the particles carried out of stage k , and ν_k^N the pushforward particles that stage k reweights. The estimate the scheme returns is π_K^N , and π_0^N is the empirical measure of the N initial draws. Each superscript- N measure is the empirical counterpart of the distribution of the same name, and the proof is a bookkeeping of how far apart the two stay.

One stage does three things to π_{k-1}^N , and between them they contribute two terms to the recurrence. The *flow pushforward* adds no randomness and loses nothing, because G_k^{-1} is a deterministic map: it carries the whole empirical measure across,

$$\nu_k^N = (G_k^{-1})_{\#} \pi_{k-1}^N, \quad \nu_k = (G_k^{-1})_{\#} \pi_{k-1}, \quad (53)$$

the second identity being the definition of ν_k . The *importance sampling* contributes both terms of the recurrence: its weighting adds no randomness but distorts the measure, and the distortion is driven by the error already present, while its resampling injects fresh randomness. The *rejuvenation* injects randomness too, but together with the resampling it is one independent draw per particle, and on average the two return the weighted particles put through Q_k . So the pushforward contributes the identity that moves the error across a stage, the weighting contributes the inherited term, and the two randomizing draws contribute the single Monte Carlo term of the recurrence proved in Appendix C.2.

The pushforward identity is what replaces the change of variables one would otherwise perform, and it is what carries the whole burden of keeping the stage index. By the definition of $(G_k^{-1})_{\#}$, integrating h against ν_k is integrating $h \circ G_k^{-1}$ against π_{k-1} , and the same holds for the empirical measures because

$Y_k^i = G_k^{-1}(X_{k-1}^i)$; taking h to be an indicator matches the null sets of the two distributions, so h and $h \circ G_k^{-1}$ have the same essential range. Hence

$$\nu_k^N \langle h \rangle = \pi_{k-1}^N \langle h \circ G_k^{-1} \rangle, \quad \nu_k \langle h \rangle = \pi_{k-1} \langle h \circ G_k^{-1} \rangle, \quad \text{osc}(h \circ G_k^{-1}) = \text{osc}(h), \quad (54)$$

for every bounded measurable h , the two oscillations belonging to stages $k-1$ and k . An error committed before stage k is therefore handed to stage k at the same size and in the same units.

Two facts are used repeatedly and are worth recording once. The first is where the particles can be. Every X_0^i is drawn from π_0 ; if the law of every X_{k-1}^i is dominated by π_{k-1} , then the law of every $Y_k^i = G_k^{-1}(X_{k-1}^i)$ is dominated by ν_k , resampling only selects among the Y_k^i , and the rejuvenation preserves domination because invariance makes $Q_k(x, \cdot)$ ignore the null sets of π_k . By induction on k , almost surely no particle lies in a set null at its own stage, so essential suprema and essential infima serve for the particle averages exactly as they do for the exact ones. The second is what the kernel does, and that Q_k is a Markov kernel leaving π_k invariant is all of it. Invariance says the kernel does not move a π_k -average,

$$\pi_k \langle Q_k h \rangle = \pi_k \langle h \rangle \quad \text{for every bounded measurable } h, \quad (55)$$

and $Q_k h(x)$ is an average of values of h over a distribution charging no π_k -null set, so

$$\text{ess inf } h \leq Q_k h(x) \leq \text{ess sup } h \quad \text{for } \pi_k\text{-almost every } x, \quad \text{hence} \quad \text{osc}(Q_k h) \leq \text{osc}(h). \quad (56)$$

The first makes the error decomposition of Appendix C.2 exact; the second keeps the inherited term from growing, by keeping the unit in which every error is measured fixed.

The weighted distribution from which stage k resamples, with $W_k^i = w_k(Y_k^i)$, averages a bounded h to

$$\mathcal{A}_k(h) := \sum_{i=1}^N \frac{W_k^i}{\sum_{l=1}^N W_k^l} h(Y_k^i) = \frac{\nu_k^N \langle w_k h \rangle}{\nu_k^N \langle w_k \rangle}, \quad (57)$$

so that $\mathcal{A}_k(h)$ is the average of h under the weighted particles, a random number once h is fixed. It is the hinge of the proof: applied to $Q_k h$ it is the mean, given the pushforward particles, of what the resampling and rejuvenation draw, so it stands between the error inherited from earlier stages and the error those draws inject. The same ratio in (57) taken at ν_k instead of ν_k^N returns $\pi_k \langle h \rangle$, since $\nu_k \langle w_k h \rangle = \pi_k \langle h \rangle$ and $\nu_k \langle w_k \rangle = 1$; so the denominator is one for the exact sequence and random for the particle one.

C.2 Proof of Theorem 2

The proof is an induction on the stage index. Two facts about the importance weight w_k are used throughout. First, $\nu_k \langle w_k \rangle = 1$, because $w_k = \pi_k / \nu_k$ and π_k is a probability distribution. Second, $\nu_k^N \langle w_k \rangle$ lies almost surely in $(0, \|w_k\|_\infty]$, by the null sets recorded in Appendix C.1, so no normalization vanishes or diverges.

The next lemma bounds the error of the weighted average (57) by the errors of ν_k^N against ν_k on two fixed functions, and so accounts for the error the weighting propagates. The difficulty it removes is that $\mathcal{A}_k(h)$ is a ratio, and a ratio of two noisy averages is not an average of any fixed function against ν_k^N , which is the one form the pushforward particles are known to control. The device is to center the observable at its exact stage average and weight it, giving the function g_k below, whose exact average vanishes; the numerator error and the denominator error can then be read off separately as ordinary deviations of ν_k^N from ν_k .

Lemma 3 *Let h be bounded and measurable, and define $g_k := w_k(h - \pi_k \langle h \rangle)$. Then $\nu_k \langle g_k \rangle = 0$, $\text{osc}(g_k) \leq 2\|w_k\|_\infty \text{osc}(h)$, and, almost surely,*

$$\left| \frac{\nu_k^N \langle w_k h \rangle}{\nu_k^N \langle w_k \rangle} - \pi_k \langle h \rangle \right| \leq |\nu_k^N \langle g_k \rangle - \nu_k \langle g_k \rangle| + \text{osc}(h) |\nu_k^N \langle w_k \rangle - \nu_k \langle w_k \rangle|. \quad (58)$$

Proof. Since $w_k = \pi_k / \nu_k$, weighting by w_k converts a ν_k -average into a π_k -average, so $\nu_k \langle w_k h \rangle = \pi_k \langle h \rangle$, and the same identity holds with any other bounded function in place of h . Using $\nu_k \langle w_k \rangle = 1$, we obtain

$$\nu_k \langle g_k \rangle = \pi_k \langle h \rangle - \pi_k \langle h \rangle \nu_k \langle w_k \rangle = 0.$$

Write $D_k := \nu_k^N \langle w_k h \rangle / \nu_k^N \langle w_k \rangle - \pi_k \langle h \rangle$ and $q_k := \nu_k^N \langle w_k \rangle - 1$. Then

$$\nu_k^N \langle g_k \rangle = \nu_k^N \langle w_k h \rangle - \pi_k \langle h \rangle \nu_k^N \langle w_k \rangle = \nu_k^N \langle w_k \rangle D_k = D_k(1 + q_k),$$

hence $D_k = \nu_k^N \langle g_k \rangle - D_k q_k$, and the triangle inequality gives

$$|D_k| \leq |\nu_k^N \langle g_k \rangle| + |D_k| |q_k|.$$

Both $\nu_k^N \langle w_k h \rangle / \nu_k^N \langle w_k \rangle$ and $\pi_k \langle h \rangle$ are weighted averages of h and so lie in $[\text{ess inf } h, \text{ess sup } h]$, whence $|D_k| \leq \text{osc}(h)$. Using that bound on the right-hand side only, and keeping $|D_k|$ itself on the left, leaves

$$|D_k| \leq |\nu_k^N \langle g_k \rangle| + \text{osc}(h) |q_k|,$$

and inserting $\nu_k \langle g_k \rangle = 0$ and $\nu_k \langle w_k \rangle = 1$ into the two absolute values gives (58).

For the oscillations, $\pi_k \langle h \rangle \in [\text{ess inf } h, \text{ess sup } h]$ gives $\|h - \pi_k \langle h \rangle\|_\infty \leq \text{osc}(h)$ and therefore $\|g_k\|_\infty \leq \|w_k\|_\infty \text{osc}(h)$, so $\text{osc}(g_k) \leq 2\|g_k\|_\infty \leq 2\|w_k\|_\infty \text{osc}(h)$. $\square$

We are now ready to prove Theorem 2.

Proof. The argument reduces the whole schedule to a single scalar per stage. We measure how wrong the stage- k particles are by one number ε_k , show that ε_k obeys a linear recurrence in which the two randomizing draws together contribute a fixed amount and the previous stage enters multiplied by $3\|w_k\|_\infty$, and then unroll that recurrence. Taking a worst case over observables is what makes a single number possible: the recurrence has to apply at stage k to the functions g_k and w_k , which are not the observable the theorem asks about, so the quantity being recursed on must already cover every bounded function at once. Oscillation is the right scale for it, because that is the unit in which both the resampling bound and the lemma are stated.

Define ε_k by the worst-case L^2 error of the stage- k particles over observables of unit oscillation,

$$\varepsilon_k := \sup\{\|\pi_k^N \langle h \rangle - \pi_k \langle h \rangle\|_2 : h \text{ bounded measurable, } \text{osc}(h) \leq 1\}. \quad (59)$$

Then $\varepsilon_k \leq 1$, since $\pi_k^N \langle h \rangle$ and $\pi_k \langle h \rangle$ both lie in $[\text{ess inf } h, \text{ess sup } h]$. The definition of ε_k implies

$$\|\pi_k^N \langle h \rangle - \pi_k \langle h \rangle\|_2 \leq \text{osc}(h) \varepsilon_k \quad \text{for every bounded } h. \quad (60)$$

Combining (60) at stage $k-1$ with the pushforward identities (54) transfers the same bound to the pushforward particles: for every bounded h and every $k \geq 1$,

$$\|\nu_k^N \langle h \rangle - \nu_k \langle h \rangle\|_2 = \|\pi_{k-1}^N \langle h \circ G_k^{-1} \rangle - \pi_{k-1} \langle h \circ G_k^{-1} \rangle\|_2 \leq \text{osc}(h) \varepsilon_{k-1}. \quad (61)$$

The remaining proof rests on the recurrence

$$\varepsilon_k \leq \frac{1}{2\sqrt{N}} + 3\|w_k\|_\infty \varepsilon_{k-1}, \quad \text{with the second term absent at } k=0, \quad (62)$$

which we establish for every $k \geq 0$ and then unroll; Theorem 2 is a consequence of (62).

At $k=0$, (62) reads $\varepsilon_0 \leq 1/(2\sqrt{N})$. Here π_0^N is the empirical measure of N independent draws from π_0 ,

$$\mathbb{E}[(\pi_0^N \langle h \rangle - \pi_0 \langle h \rangle)^2] = \frac{\text{Var}_{\pi_0}(h)}{N}.$$

Then Popoviciu's inequality on variances (Popoviciu, 1935) bounds the variance $\text{Var}_{\pi_0}(h)$ by $\frac{1}{4} \text{osc}(h)^2 \leq \frac{1}{4}$.

For $k \geq 1$, let h be bounded with $\text{osc}(h) \leq 1$ and recall $\mathcal{A}_k$ from (57). Applying it to $Q_k h$ and using the invariance (55) on the right, we obtain the exact decomposition

$$\pi_k^N \langle h \rangle - \pi_k \langle h \rangle = \underbrace{[\pi_k^N \langle h \rangle - \mathcal{A}_k(Q_k h)]}_{\text{(I) resampling and rejuvenation error } (N^{-1/2})} + \underbrace{[\mathcal{A}_k(Q_k h) - \pi_k \langle Q_k h \rangle]}_{\text{(II) pushforward sampling error (Lemma 3)}}. \quad (63)$$

(I) is the error injected by the resampling and rejuvenation at stage k , and (II) the error inherited from before them; invariance is what lets (II) compare like with like, through the replacement of $\pi_k \langle h \rangle$ by $\pi_k \langle Q_k h \rangle$.

Since (63) is an identity, taking $\|\cdot\|_2$ and applying Minkowski's inequality bounds the stage- k error by $\|(I)\|_2 + \|(II)\|_2$, and the two are bounded separately below.

We condition on the pushforward particles $Y_k^{1:N} = (Y_k^1, \dots, Y_k^N)$, which determine the weights $W_k^i = w_k(Y_k^i)$ and leave the resampling draw and the kernel draw as the only randomness. Given $Y_k^{1:N}$, the resampled particles $Z_k^1, \dots, Z_k^N$ are independent draws from the weighted particles, and each X_k^i is then drawn from $Q_k(Z_k^i, \cdot)$ independently of everything else, so the pairs (Z_k^i, X_k^i) are independent across i and identically distributed. The stage- k particles are therefore N independent draws from the single distribution $\sum_j (W_k^j / \sum_l W_k^l) Q_k(Y_k^j, \cdot)$, whose h -average is $\mathcal{A}_k(Q_k h)$. This is where rejuvenation is absorbed: composing the kernel with the resampling leaves merely one independent draw per particle.

Hence $\mathbb{E}[\pi_k^N \langle h \rangle \mid Y_k^{1:N}] = \mathcal{A}_k(Q_k h)$, and, since each $h(X_k^i)$ lies almost surely in $[\text{ess inf } h, \text{ess sup } h]$, Popoviciu's inequality on variances gives

$$\mathbb{E}[(\pi_k^N \langle h \rangle - \mathcal{A}_k(Q_k h))^2 \mid Y_k^{1:N}] \leq \frac{\text{osc}(h)^2}{4N}.$$

This bound is almost surely uniform in $Y_k^{1:N}$, so the tower property gives

$$(I) \quad \|\pi_k^N \langle h \rangle - \mathcal{A}_k(Q_k h)\|_2 \leq \frac{\text{osc}(h)}{2\sqrt{N}}. \quad (64)$$

The collapse rests on the order of the two draws, which is the order the Boltzmann generator of Algorithm 5 uses: the resampling reads a weighted set fixed by $Y_k^{1:N}$, and the kernel then acts on each drawn particle alone. Were the kernel applied first, none of this would be available. The particles would be resampled from one realized post-kernel set, so given $Y_k^{1:N}$ they would be exchangeable but not independent, and a second Monte Carlo term would be genuine; worse, w_k would no longer be the weight that corrects them, since the kernel would have moved the particles before they were weighted.

For (II), we employ Lemma 3 with $Q_k h$ in place of the observable. It is bounded and measurable, so with $g_k := w_k(Q_k h - \pi_k \langle Q_k h \rangle)$ we obtain the L^2 error bound

$$\|\mathcal{A}_k(Q_k h) - \pi_k \langle Q_k h \rangle\|_2 \leq \|\nu_k^N \langle g_k \rangle - \nu_k \langle g_k \rangle\|_2 + \text{osc}(Q_k h) \|\nu_k^N \langle w_k \rangle - \nu_k \langle w_k \rangle\|_2,$$

where the two norms come from taking $\|\cdot\|_2$ in (58) and applying the triangle inequality. Neither g_k nor w_k is random: Q_k and π_k come from the schedule and not from the particles, so both are fixed functions and (61) applies to each, giving

$$\|\mathcal{A}_k(Q_k h) - \pi_k \langle Q_k h \rangle\|_2 \leq [\text{osc}(g_k) + \text{osc}(Q_k h) \text{osc}(w_k)] \varepsilon_{k-1}.$$

Finally $\text{osc}(g_k) \leq 2\|w_k\|_\infty \text{osc}(Q_k h)$ by Lemma 3, and $\text{osc}(w_k) \leq \|w_k\|_\infty$ because $w_k \geq 0$, so the two oscillations contribute $2\|w_k\|_\infty \text{osc}(Q_k h)$ and $\|w_k\|_\infty \text{osc}(Q_k h)$ respectively. The contraction (56) then replaces $\text{osc}(Q_k h)$ by $\text{osc}(h)$, and

$$(II) \quad \|\mathcal{A}_k(Q_k h) - \pi_k \langle Q_k h \rangle\|_2 \leq 3\|w_k\|_\infty \text{osc}(h) \varepsilon_{k-1}. \quad (65)$$

This is the second place rejuvenation could have cost something and does not: the kernel enters the inherited error only through $\text{osc}(Q_k h)$, which (56) bounds by $\text{osc}(h)$. Adding (64) and (65) as promised, by Minkowski's inequality, and taking the supremum over $\text{osc}(h) \leq 1$ yields (62).

It remains to unroll the recurrence. Write $a := 1/(2\sqrt{N})$ and $b_j := 3\|w_j\|_\infty$, and induct on k to prove $\varepsilon_k \leq a \sum_{i=0}^k \prod_{j=i+1}^k b_j$, the empty product being one. At $k=0$ the right-hand side is a , which is (62) at $k=0$. Assuming it at $k-1$, (62) gives

$$\varepsilon_k \leq a + b_k a \sum_{i=0}^{k-1} \prod_{j=i+1}^{k-1} b_j = a \left[1 + \sum_{i=0}^{k-1} \prod_{j=i+1}^k b_j \right] = a \sum_{i=0}^k \prod_{j=i+1}^k b_j,$$

the last equality because the $i=k$ term is the empty product. Taking $k=K$ and applying (60) with $h = \varphi$,

$$\|\pi_K^N \langle \varphi \rangle - \pi_K \langle \varphi \rangle\|_2 \leq \text{osc}(\varphi) \varepsilon_K \leq \frac{\text{osc}(\varphi)}{2\sqrt{N}} \sum_{k=0}^K \prod_{j=k+1}^K 3\|w_j\|_\infty.$$

Squaring gives (28). $\square$

C.3 The role of the propagation factor F_Σ

The experiments report $\hat{F}_\Sigma$, the sample counterpart of F_Σ , and the direction of the ordering $F_\Sigma \leq F_{\Sigma+}$ established in Section 4.4 is the first thing to be clear about: F_Σ is a lower bound on the factor of Theorem 2, so substituting it into (28) gives a statement the theorem does not support. F_Σ is not an error bound.

The gap can moreover be made arbitrarily large, because an ESS is built from a second moment and a supremum is not. Reading the ESS of (30) in measure form, take one stage and drop its index, writing ν for the pushforward distribution and w for the stage weight. Split $\mathbb{R}^d$ into two Borel sets of ν -probabilities $1-p$ and p , and let w take the values $(1-\Lambda p)/(1-p)$ and Λ on them, with $1 < \Lambda < p^{-1}$, so that $\int w d\nu = 1$ and $w > 0$. Then $\|w\|_\infty = \Lambda$, whereas

$$\frac{1}{\text{ESS}(w)} = \int_{\mathbb{R}^d} w^2 d\nu = \frac{(1-\Lambda p)^2}{1-p} + p\Lambda^2. \quad (66)$$

Choosing $p = \Lambda^{-3}$ gives $\text{ESS}(w) \rightarrow 1$ as $\Lambda \rightarrow \infty$ while $\|w\|_\infty = \Lambda \rightarrow \infty$, so a stage schedule whose stages behave this way would send F_Σ to $K+1$, the least value it can take, while $F_{\Sigma+} \geq (3\Lambda)^K$ diverges without limit. An ESS weights a large weight by the probability of the set carrying it, and a weight of size Λ on a set of proposal probability Λ^{-3} barely moves the ESS; a supremum is fixed by that weight alone.

For a singular potential the supremum need not be finite. A stage weight is an exponential of a difference of potentials and a log-Jacobian, and the soft cap of a molecular potential slows the growth of the interaction core without bounding that difference, so $\|w_k\|_\infty$ is not controlled by the regularization and may be infinite. Should it be infinite at some stage, $F_{\Sigma+} = \infty$ and the bound of Theorem 2 is unavailable, while a square-integrable weight still has $\text{ESS}(w_k) > 0$ and leaves F_Σ finite. The two factors are governed by different norms of the same weight, the L^2 norm and the L^∞ norm, and the second is the fragile one.

The size of the constant is a third limitation, and the recurrence of Appendix C.2 is where it comes from: each application of (62) re-enters ε_{k-1} , a supremum over all observables of unit oscillation, and the same worst case is taken again at the next stage. Taking $\pi_k \equiv \pi$, $G_k = \text{id}$ and $Q_k(x, \cdot) = \delta_x$, the identity kernel, which is π -invariant, makes every stage weight one, so the scheme is independent sampling followed by K rounds of uniform resampling, and yet $F_{\Sigma+} = (3^{K+1} - 1)/2$ and the constant of (28) grows like 9^K , so the bound says nothing until N reaches that order, while $\text{osc}(\varphi)$ bounds the error at every N . Theorem 2 should therefore be read as establishing the rate $N^{-1/2}$ and the finiteness of the constant, not its size.

F_Σ is nevertheless the reasonable quantity to report. The ESS is the natural second-moment substitute for the supremum, and the ordering of Section 4.4 is exactly the statement that the L^2 measurement never exceeds the L^∞ one. It is also the classical measurement for importance sampling, since $\int w_k d\nu_k = 1$ gives

$$\frac{1}{\text{ESS}(w_k)} = \int_{\mathbb{R}^d} w_k^2 d\nu_k = 1 + \chi^2(\pi_k \| \nu_k), \quad (67)$$

so F_Σ is assembled from the per-stage chi-squared divergences, and a stage contributes nothing beyond the floor exactly when its proposal equals its target. What it keeps from Theorem 2 is the structure of the constant rather than its value: the same sum over the stage at which an error enters, the same product over the stages it must still traverse, with the problem-independent factors 3 removed.

It is also the more stable of the two to [estimate from a sample, being a second moment of the whole stage weight vector rather than its largest entry](#). Neither factor can be computed directly, and [only \$F_\Sigma\$ is given a sample counterpart](#): (32) reads $\hat{F}_\Sigma$ off the stage weight vectors $W_j = (W_j^1, \dots, W_j^N)$ that a run already evaluates. The reported $\hat{F}_\Sigma$ decreases when a stage ESS improves and increases when the schedule lengthens, combining the two things that make one accepted schedule harder than another.

Section 4.4 already limits the reading of $\hat{F}_\Sigma$ twice, in that $\hat{F}_\Sigma$ is not a bound on F_Σ at finite N and that the factor does not measure the difficulty of the target. One consequence for reporting follows. Two runs that traverse the same stage schedule are the most directly comparable, since they share the same least value $K+1$ and their per-stage ESS values are matched one to one. Across schedules of different length the factor still orders runs by accumulated stagewise weight degeneracy, but the comparison also carries the difference in stage count, and a ratio of two factors is in neither case a prediction of the ratio of two errors.